Investigation of the Liquefaction of Partially Dried and Oxidized Coals

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ABSTRACT

This project was undertaken to determine the effect of predrying and partial oxidation on the liquefaction behavior of subbituminous coal. It was found from both micro- and bench-scale experiments that the drying of Belle Ayr subbituminous coal, even in nitrogen, results in a significant reduction of product yields when a comparison is made to similar runs using as-received coal. A further reduction occurs when the drying gas contains oxygen. As observed from a microscopic study of the reaction product residues, the yield reduction is due in part to a repolymerization of coal liquid products resulting in a sizable portion of coke-like solids.

Samples of both Belle Ayr subbituminous and Powhatan bituminous coals were dried in various gases in three scales of equipment, namely, TGA and laboratory and bench-scale fluid bed units. The rates and maximum levels of coal oxidation are discussed in this report. It is noted that the lower rank coal oxidizes more readily, but the maximum up-take of oxygen is only about 3 wt% while that of the higher rank coal is about 8 wt%.

EPRI PERSPECTIVE

PROJECT DESCRIPTION

Under this screening project (RP779-25), the nature and effects of drying and partial oxidation of subbituminous coal upon its liquefaction behavior was studied. A detailed literature search was first performed in order to obtain background on the coal drying/oxidation/aging process and the safety considerations in drying powdered coal.

The experimental project was carried out in two steps and at various scales. The first step, the coal-treating experiments, used three sizes of equipment. Microscale runs (made under subcontract to Pennsylvania State University) used thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) units to study the kinetics of oxygen sorption and the heats involved in low-temperature oxidation. At Gulf, both bench- and pilot-scale fluidized-bed dryers were employed to treat the coal samples over a wide range of temperatures, drying gas oxygen concentration, and coal particle sizes. These dried and oxidized coal samples were characterized and then used in subsequent liquefaction studies. The second step, liquefaction of the treated coal samples, was performed in both microscale and bench-scale units.

PROJECT OBJECTIVE

The drying and partial oxidation of coals can significantly change their physical and chemical characteristics. These changes, in turn, can affect the liquefaction behavior of the coal. Whether these changes are a result of natural causes, such as aging and weathering, or induced, as in the case of drying (with some attendant level of oxidation), an understanding of the effects on liquefaction behavior is needed in order to be able to predict and circumvent adverse changes.

Because subbituminous coals typically have extremely high moisture contents, the cost of transporting and/or processing the raw undried coal is higher than if the coal were dried. On the surface, it would appear that drying is desirable. In the past, drying in an inert environment had not been thought to have a detrimental effect.

The objective of this project, therefore, was to determine how drying and/or partial oxidation affect the liquefaction behavior of western subbituminous coals.

PROJECT RESULTS

This project produced a number of important conclusions and added extensively to the understanding of the oxidation of western subbituminous coals. Of key significance, however, was the result on the impact of coal drying in an inert gas. The drying of subbituminous coal, even in pure nitrogen, resulted in a significant reduction of liquefaction product yield when compared with liquefaction of as-received coal. This result should be seriously considered in the future before predrying liquefaction feedstocks. Alternate methods of removing the moisture from the subbituminous coal should be pursued. One such method is drying the coal after it is in slurry with the recycle solvent. (Both Exxon, RP778, and Mobil, RP1655, dry coal in this manner.)

Equally important, though less startling, results include the further reduction of liquefaction yield when the drying gas contains oxygen.

The Belle Ayr subbituminous coal was more rapidly oxidized than the Powhatan No. 5 bituminous coal. The maximum oxygen uptake of the lower-rank coal, however, was considerably less than the higher-rank coal.

Overall, it can be recommended that, based on the reactivity and nature of western subbituminous coals, great care should be taken in the collection, handling, transportation, storage, and preparation of the coal if it is to be used in the production of coal liquids.

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Section 1

SUMMARY

Liquefaction experiments were undertaken to determine the effect of predrying and partial oxidation on the liquefaction behavior of Belle Ayr subbituminous coal. Because subbituminous coals typically have moisture contents in excess of 25%, it would appear useful to dry these coals prior to liquefaction. During essentially any drying process some oxidation of the coal also occurs.

Coal pretreatment was carried out in micro-, laboratory, and pilot-scale units with typical drying temperatures from ambient to 200°C and with gases including nitrogen, air, nitrogen/air mixtures, and oxygen (only in TGA runs). The extent of oxygen up-take by the subbituminous coal generally increased with time and temperature to a maximum of about 3 wt%. The rate of oxygen consumption during drying appeared to fall in two zones, namely an initial high-rate period followed by one of low rate. The rate of coal drying in the fluid-bed units was not significantly affected by the presence of oxygen in the gas stream.

The drying of subbituminous coal, either in nitrogen or oxygen-containing gases, resulted in a significant decrease in yields of liquefied coal products. This was observed using SRC-II heavy distillate as solvent in both micro- and bench--scale liquefaction units. The residues recovered from the latter runs were examined microscopically and were found to contain high levels of coke. This coke appeared to have formed by repolymerization of coal liquid products.

Additional drying/oxidation runs were also made with Powhatan (Pittsburgh seam) bituminous coal. Similar trends were observed; this higher-rank coal had the capability of undergoing a greater degree of oxidation. No liquefaction experiments have been undertaken with these coal samples.

Section 2

INTRODUCTION

The overall objective of this project was to determine in a scoping study the effect of predrying and partial oxidation on the liquefaction behavior of subbituminous coal. Specifically, a sample of subbituminous coal from the Belle Ayr, Wyoming mine of Amax Coal Company was used for the majority of the work. The liquefaction experiments were performed using as solvent a heavy distillate (nominal 550°-850°F) derived from SRC-II liquefaction runs at the Fort Lewis (Tacoma) facility.

A detailed literature search was first undertaken to obtain background on the coal drying/oxidation/aging process, the effect of its oxidation on liquefaction, and safety considerations in the drying of coal powders. While an extensive amount of work has been reported on the oxidation process, little has been reported on its effect on liquefaction. The literature search is included herein in the Appendix.

The coal treating experiments were planned to be part of a scoping study using three sizes of equipment. Micro-scale runs were made at Pennsylvania State University (PSU) using TGA and DSC units along with a laboratory-scale fluid bed dryer. Experimentation was also undertaken at Gulf Research & Development Company (GR&DC) using both bench- and pilot-scale fluid bed dryers. A wide range of temperatures and drying gas oxygen concentrations were used at both PSU and GR&DC.

Coal samples prepared in the above units were liquefied in both micro- and bench-scale units. The micro unit runs were primarily of the screening type with subsequent bench-scale runs being made at conditions consistent with previously reported work (EPRI Report No. AF-913) concerning, among other things, liquefaction kinetics.

Residue samples recovered from the liquefaction runs were subsequently characterized by Professor A. Davis and D. Hoover at PSU. The primary goal was to correlate residue characteristics against coal pre-treatment procedures.

Another interesting aspect of this project was the collaborative work that was undertaken with Dr. David Schlyer of Brookhaven National Laboratory. He has been studying coal surface oxidation using sample activation in a cyclotron to measure oxygen level and concentration profiles. An interchange of samples and results has given both laboratories an opportunity to expand upon our knowledge.

Section 3

CONCLUSIONS

The objective of this project was primarily to determine the effect of predrying and partial oxidation on the liquefaction behavior of subbituminous coal. Experiments were also done to develop background on the characteristics of oxidation that occurs during coal drying in various gases. This work was undertaken as a scoping study, but some areas were expanded in depth as is indicated by the following conclusions:

- From both the micro- and bench-scale liquefaction experiments, the drying of subbituminous coal even in nitrogen alone results in a significant reduction of product yields when a comparison is made to similar runs made with as-received coal. Specifically, after Belle Ayr coal had been dried in nitrogen at 100°C for 1.0 hours, the conversion of coal to pyridine or toluene solubles was reduced by 10 to 12%.

- When the drying gas contains oxygen a further reduction in yield occurs.

- From a microscopic study of the reaction product residues, a portion of the yield reduction is due to the repolymerization of coal liquid products.

- Retrogressive reactions were observed in the micro-scale liquefaction experiments using SRC-II solvent and reaction conditions of 450°C, and a nitrogen atmosphere.

- Belle Ayr subbituminous coal is more rapidly oxidized than Powhatan No. 5 bituminous coal. However, the maximum up-take of the lower rank coal is about 3 wt%, while that of the higher rank coal is about 8 wt%.

- The rate of oxygen consumption in a fluid bed unit appears to fall into two zones, namely an initial high-rate period (particularly at high temperatures) followed by one of low rate.

- The rate of oxidation of the subbituminous coal is influenced by the remaining moisture content of the coal. During the drying step the flow of moisture within the particle reduces oxidation.

- Visual observations of the dried and oxidized coal particles did not show the presence of oxidation rims, discoloration at particle boundaries, and alteration of pyrite. In addition, oxygen analysis (CPAA) did not observe any oxygen gradient through the particles.

- FTIR analysis of coal particles was undertaken with a reasonable degree of success. It was shown that oxygen was lost during coal heating, presumably as a results of decarboxylation; however, oxygen was also taken up at other sites, apparently through the formation of carbonyl and other groups. (Further work will be done in this area.)

Section 4

SIMULTANEOUS DRYING AND OXIDATION OF PULVERIZED COAL SAMPLES

The purpose of the work described in this section was that of examining in a wide range of equipment the kinetics of the oxidation (and simultaneous drying) of coal samples. Specifically, the influence of temperature, particle size, oxygen partial pressure (P_{O_2}) and exposure time was observed for two coals over a range of moisture contents. Experimentation was carried out at both Pennsylvania State University using TGA, DSC and a laboratory scale fluidized bed unit, and at GR&DC using a laboratory fluid bed dryer (LFD) and a pilot scale fluid bed dryer (PFD).

Oxidation of coal in oxygen-containing gases has been reported in the literature (See Appendix) but certain features render the current study somewhat unique as listed below:

- Much of the previous work was with dry chars or carbons; here we used moist bituminous and subbituminous coals.
- Oxidation of pulverized coal is done in a fluid bed at a temperature range of 50-200°C. In the fluid bed, there is essentially constant P_{O_2}. Much previous work was done with packed beds or stagnant piles under conditions of changing P_{O_2} and at higher temperatures.
- Simultaneous oxidation and drying for exposure times of 5 to 120 min. were done.
- Determination was made of instantaneous oxygen consumption rate by a material balance from different gas analysis as well as coal oxidation as measured by PAA, FNAA, and Unterzaucher methods.
- Two fluid bed dryers of radically different sizes, varying gas/solid ratios, residence time of gas, and gas superficial velocity were used so that the effect of these variables on oxidation could be examined.
- FTIR was used to examine oxygen functionalities on coal before and after oxidation.

Experimentation in Micro-Scale Units (PSU)

Experiments made with Powhatan No. 5 (LR-26061) and Belle Ayr coal (LR-26104) were mainly aimed at gaining a better understanding of the relative roles of oxygen chemisorption (i.e., formation of oxygen-coal surface complexes) and gasification during low temperature coal oxidation. In addition, some techniques were applied to characterize coals and oxidized coals.

In order to investigate the relative roles of chemisorption and gasification, two techniques were employed; Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC). The former technique (TGA) was used to monitor continuously weight changes of a given sample under controlled heating conditions during oxidation. The major components of the TGA apparatus are a Cahn RG electrobalance, a 1 mv recorder to monitor the continuous output from the electrobalance, and a well-controlled furnace which surrounds the sample suspended from the balance (1). In a typical isothermal experiment, a small quantity (~40 mg) of powdered coal was placed in a small quartz bucket suspended from the electrobalance. It was then heated to a desired temperature in nitrogen; when this temperature was achieved, the oxidant gas (e.g., air) was introduced. The instrument then records changes in weight throughout the duration of the experiment. It should be noted that the system is also capable of being used under controlled heating rates. The output of the TGA is expressed as a continous weight (or weight change) as a function of time.

The other technique (DSC) is a thermal analytical technique which measures quantitatively the differential flow of heat between a sample and an inert reference material. Thus, in the presence of, say, flowing air, it is possible to measure exothermal heat evolved during chemisorption and/or gasification (1,2) as a function of time and temperature. The unit utilized in these studies consisted of a DuPont DSC system in conjunction with a DuPont 990 Thermal Analyzer. Under isothermal conditions the output of this unit is expressed as heat evolved per unit weight of coal as a function of time.

It is important at this stage to consider further the information gathered by these techiques. When an oxygen atom interacts with the surface of coal it can form an oxygen-coal complex. This would result in a weight gain. We describe this phenomenon as chemisorption. However, if this oxygen-coal surface complex decomposes to evolve gaseous species, e.g., CO, CO_2 or H_2O then there is a weight loss which we define as gasification. Ismail has shown previously (1) that for Saran chars this low temperature gasification can occur at temperatures as low as 125°C. In order to discriminate between chemisorption and gasification it is necessary to measure the heats involved as well as weight change.

In this light, certain aspects of the TGA/DSC data must be considered prior to the presentation of results. Firstly, the TGA unit measures net weight change. Thus, under the experimental conditions if a weight gain is observed one cannot say conclusively that it is due solely to chemisorption of oxygen by the coal sample. It does indicate, however, that chemisorption does dominate. On the other hand, if a net weight loss is observed it certainly indicates that gasification dominates. Secondly, the heats released during these processes can be expressed in several ways. One can describe the heat evolved in terms of calories per original gram of coal (Q). Alteratively, one can use the TGA-derived data to define the heat/mole O_2 from:

$$\Delta H = 32Q/W \tag{4-1}$$

where W is the weight change as determined from the TGA data.

It is assumed that W represents the amount of O_2 chemisorbed by the coal. However, because in this instance W represents the net weight change and is not always a direct measure of chemisorbed O_2, we have opted to express the data in terms of:

$$\Delta H = Q/W \tag{4-2}$$

which represents the heat evolved per unit mass of weight change. This equation does normalize the data in terms of the interaction of O_2 with the coal. But, it must be remembered that in the cases where gasification dominates the measured W values are an underestimation of the amount of O_2 which interacts with the coal surface. It has been shown (1) that if ΔH remains constant or decreases as a function of time then chemisorption dominates; however, if ΔH increases with time, gasification is the dominant process.

Initial Experiments

Screening TGA experiments were performed in which samples of the Powhatan and Belle Ayr coals were heated at 10°C/min in flowing air until complete burn-out was achieved to determine the temperature ranges that should be used for further investigation. It was found, not surprisingly, that the Belle Ayr coal was relatively much more reactive to O_2 than Powhatan. Generally, Belle Ayr coal was completely consumed by 595 to 625°C whereas Powhatan, under similar conditions, was not completely burned-out until 800 to 900°C. These TGA results also indicated that the temperature region of main concern for our low temperature oxidation studies was between 100 and 200°C for both coals, i.e., in this temperature range neither coal was grossly gasified under these non-isothermal conditions.

Belle Ayr Coal: Results and Discussion

Fresh samples of Belle Ayr coal (40x70 mesh) were heated in the presence of flowing (40 cc/min) air in the TGA at 120, 150 and 175°C. Results are shown in Figure 4-1. It can be seen that at 120°C and 150°C there is always a net increase in weight with time (even up to 25 hr). This observation indicates that, in terms of weight, chemisorption of O_2 dominates. It will be noted that most of the weight gain occurs in the first 8 hr. The maximum net weight increase, noted for the run at 150°C, was approximately 2.6%. The result obtained at 175°C shows a different trend. There is an initial relatively rapid uptake of O_2, but after 6 hr there is a weight loss. It is suggested that at this temperature the early stages are dominated by chemisorption, and after a period of time (6 hr) gasification processes begin to dominate. The maximum apparent weight increase at 175°C is just over 2.1 wt%. It should be remembered that even during the initial stages, gasification may well be occurring and it will lead to a reduced net weight gain.

The output from a typical DSC run is shown in Figure 4-2. The heat evolved Q continuously increases with time (note data are reported to 3 hr). This run was made on the 40x70 mesh fresh Belle Ayr coal at 150°C. If we combine the TGA/DSC data to yield ΔH from equation 2 the following (Table 4-1) results are obtained. At 120°C ΔH is essentially constant with time but overall does show

Table 4-1

Values of ΔH (kcal/g (wt change)) of
40x70 mesh Belle Ayr Coal in Air

	Time (hr)					
Temperature °C	0.5	1.0	1.5	2.0	2.5	3.0
120	9.5	10.7	9.3	8.6	7.9	-
150	11.9	12.5	12.8	13.1	13.1	13.3

a modest decrease, thus chemisorption is dominant. However, at 150°C ΔH values are larger than those found at 120°C and there is a slight increase in ΔH with time indicating that there is a gasification contribution but it is not dominant. A DSC run was not made at 175°C because of the obvious dominance of gasification from TGA data.

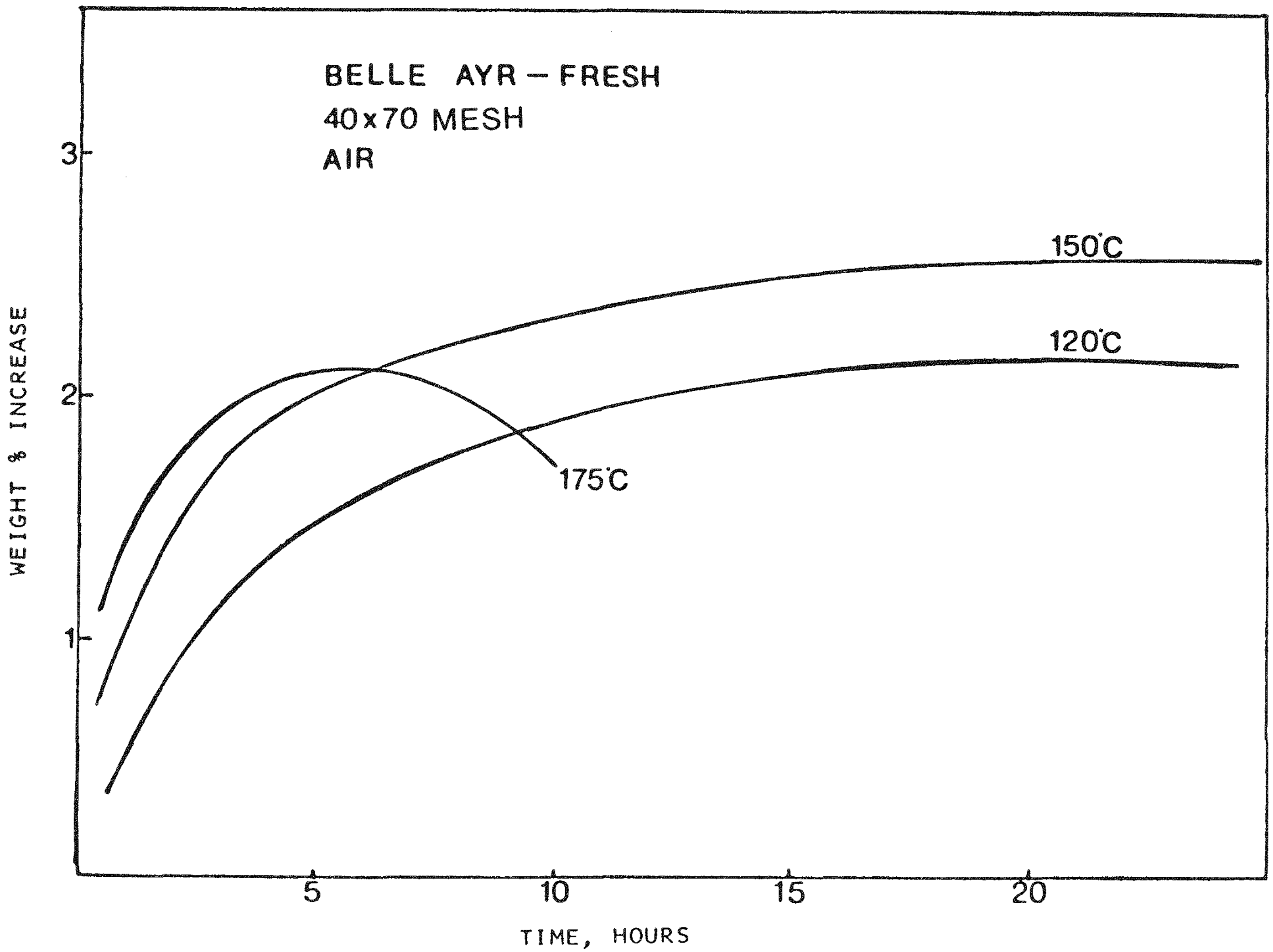

FIGURE 4-1 TGA TREATMENT OF FRESH BELLE AYR COAL IN AIR

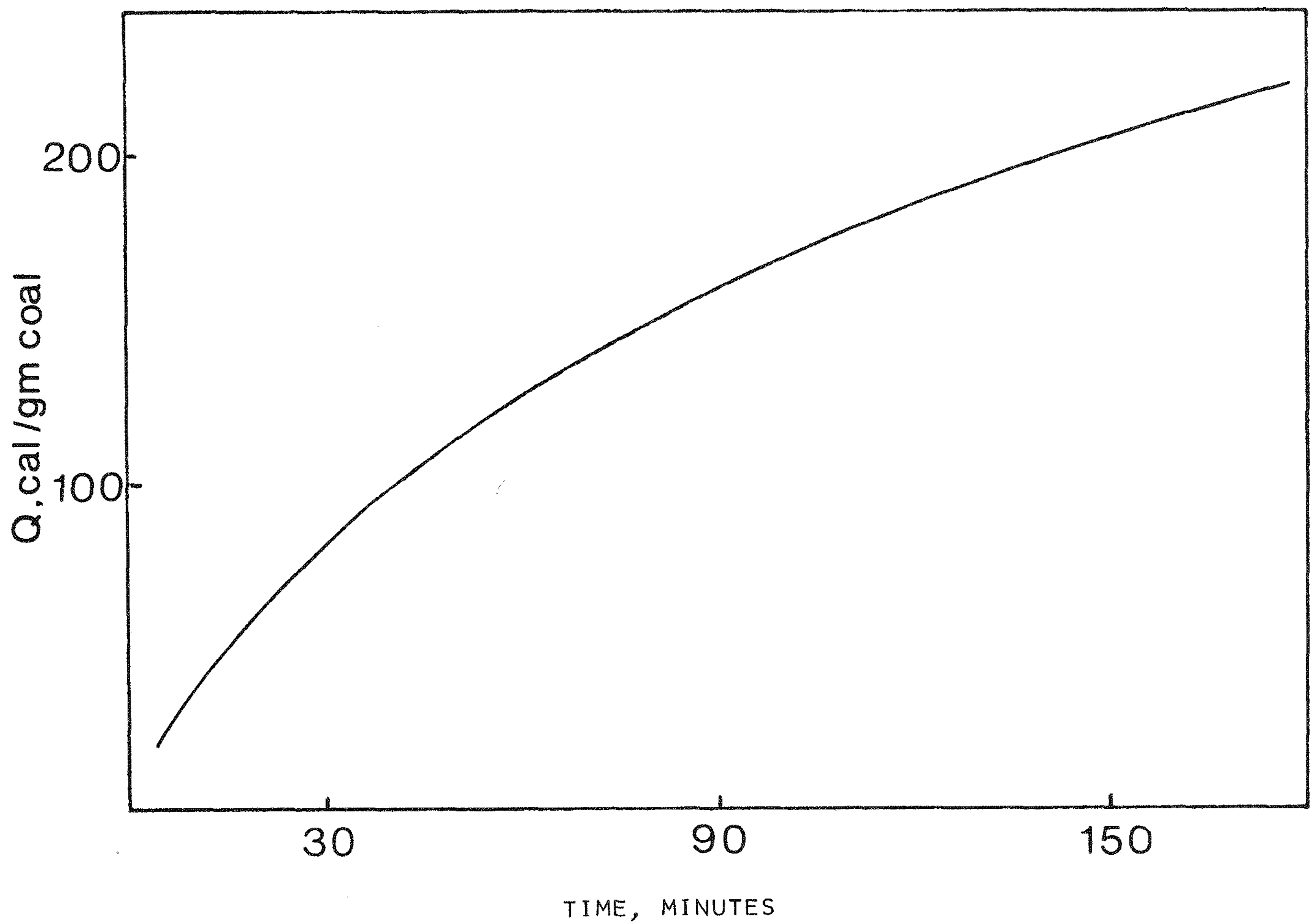

FIGURE 4-2 DSC PLOT FOR BELLE AYR COAL (40X70 MESH) AT 150^{O}C IN AIR

Another sample of Belle Ayr coal (40x70) was stored in dry air for 3 weeks prior to reaction in TGA and DSC. The TGA and DSC runs were made under the same conditions as just described for the fresh sample. Plots of the TGA data are shown in Figure 4-3. The overall trends are similar to those found for the fresh sample but there is a marked decrease at all temperatures in the apparent weight increase. Even at 150°C the maximum weight gain is only 1.7% (at 20 hr) compared to about 2.6% for the fresh coal. It is concluded that, as would be expected, the surface of this coal is already, to some degree, saturated with oxygen-coal complex. In Figure 4-3, we also note that the maximum observed for the run at 175°C occurs in a shorter period of time (~3 hr) than for fresh coal, indicating a more dominant role of gasification. Table 4-2 lists calculated ΔH values for these samples. In both instances ΔH increases with time, most markedly for the reaction of 175°C. Thus it is concluded that gasification plays a major role even at 120°C for this sample. These results give an indication of the marked effects which storage in dry air can bring about.

Table 4-2

Values of ΔH (kcal/g (wt change)) for
40x70 mesh Belle Ayr Coal (previously stored in air)

Temperature °C	Time (hr) 0.5	1.0	1.5	2.0	3.0
120	10.2	13.3	-	14.6	15.3
175	15.9	18.6	20.2	21.5	24.5

Another series of experiments was performed to determine the effects of oxygen partial pressure and particle size on the low-temperature oxidation of Belle Ayr coal. All these runs were performed at 150°C. The TGA data are reported in Figure 4-4. From the runs made on the two particle size fractions (40x70 and 200x270 mesh) there is apparently very little effect of particle size on the TGA data. If anything, the smaller size fraction increases in weight at a slightly lower rate than that for the larger particles. This observation would appear to be in conflict with accepted views that low temperature oxidation is very strongly dependent on particle size. However, comparable combined TGA/DSC (Table 4-3) indicate that for the 200x270 mesh size fraction at 150°C gasification plays a somewhat dominant role. Thus, the TGA data alone are misleading; the observation that there is little particle size effect on the TGA data is a reflection of gasification which apparently counteracts the chemisorption effects. Note again

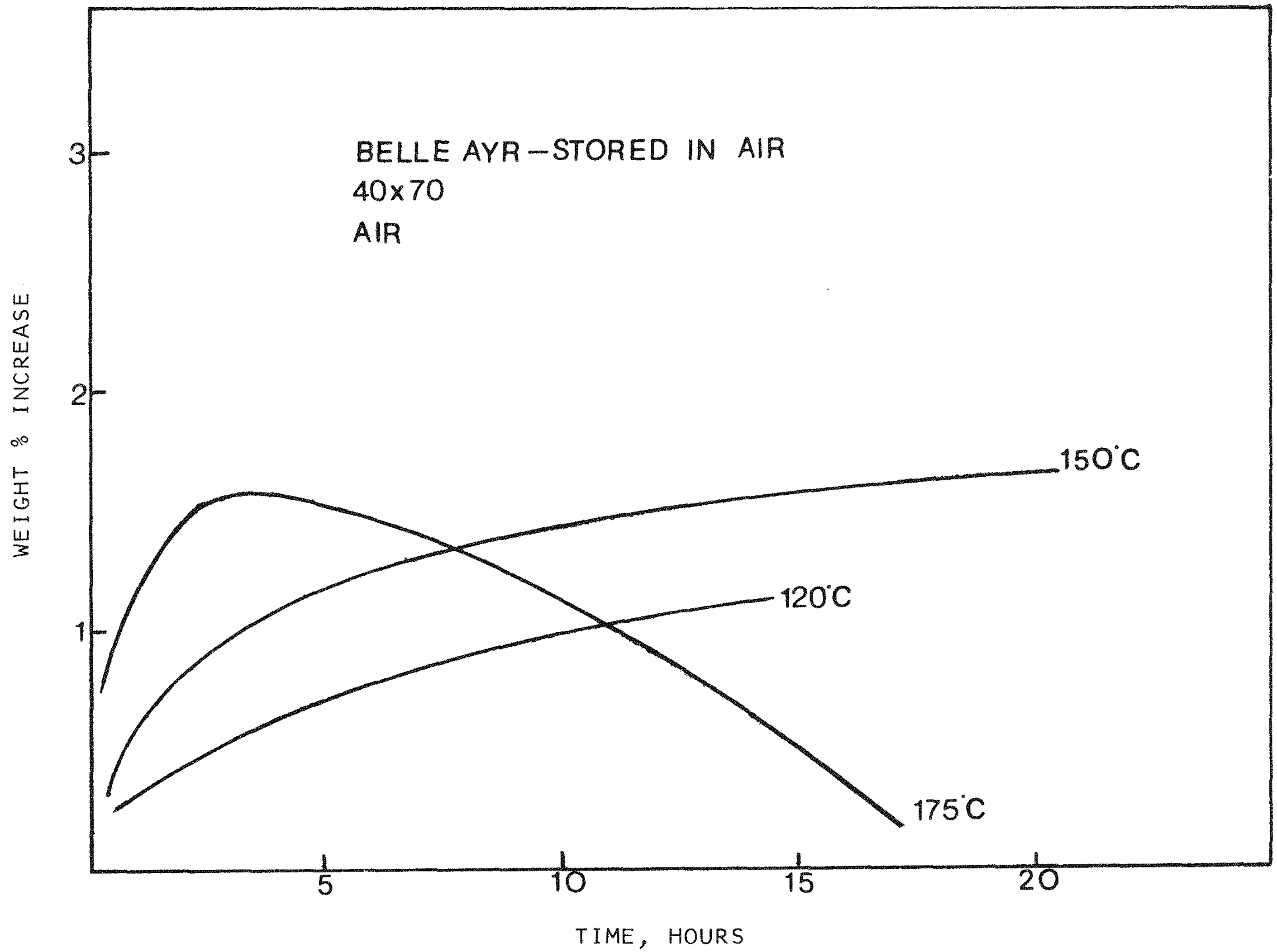

FIGURE 4-3 ISOTHERMAL TGA PLOTS OF 40X70 MESH BELLE AYR COAL (PREVIOUSLY DRIED IN AIR)

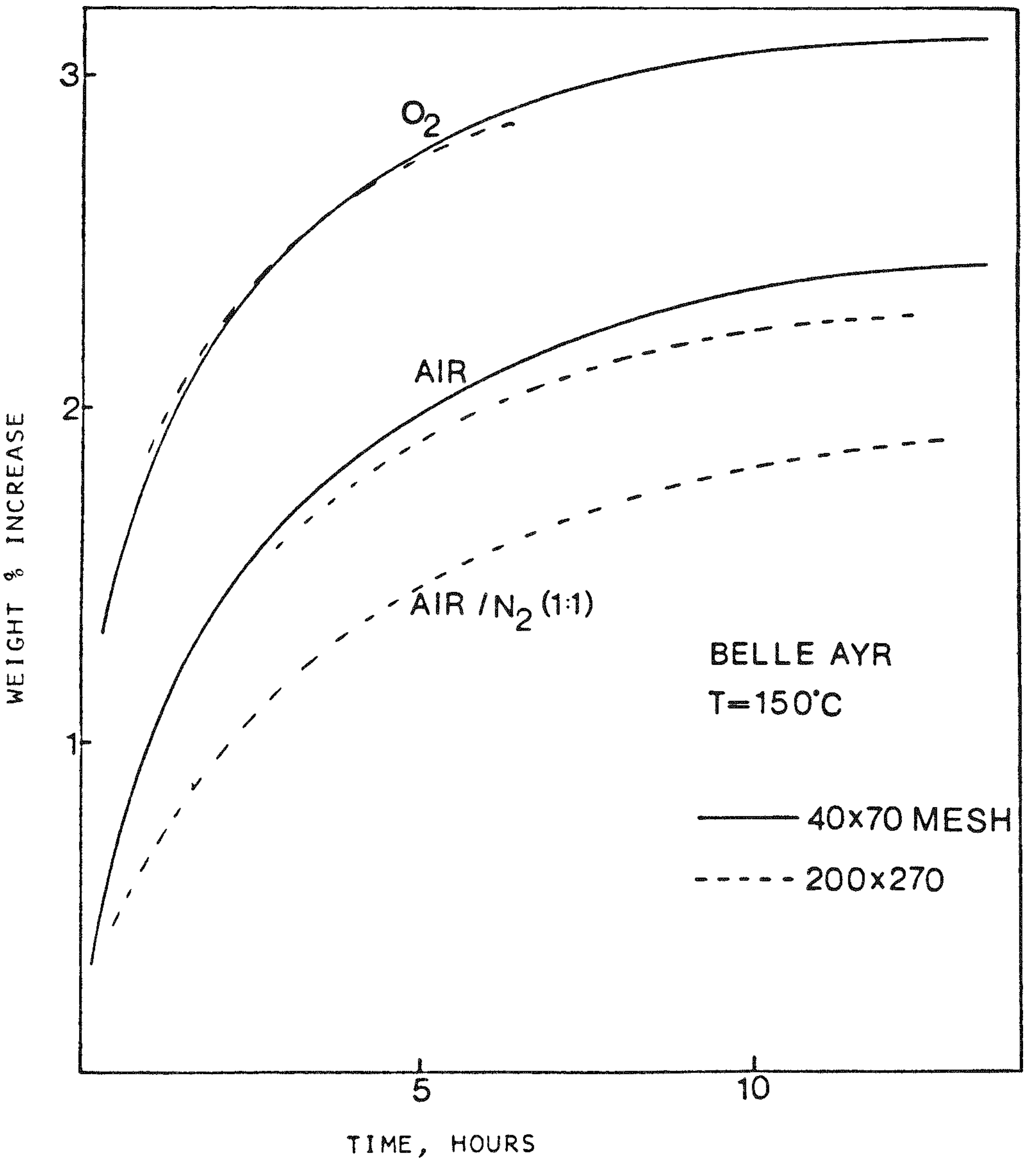

FIGURE 4-4 TGA TREATMENT OF BELLE AYR COAL IN VARIOUS GASES

from Table 4-3 that, at 175°C, gasification does play a major role.

Table 4-3

Values of ΔH (kcal/g (wt change)) for 200x270 mesh Belle Ayr Coal in Air

	Time (hr)					
Temperature °C	0.5	1.0	1.5	2.0	2.5	3.0
125	10.9	11.7	10.0	10.0	9.7	9.4
150	11.6	12.3	12.4	13.1	13.0	13.3
175	13.0	14.3	15.9	16.8	18.3	19.6

As also shown in Figure 4-4 there is an obvious dependency on the partial pressure of oxygen at 150°C. In the run made in oxygen on the 40x70 mesh size fraction over 3 wt% increase was noted after about 8 hr, while that run made in air under the same conditions resulted in an increase of 2.2 wt%. Two other TGA runs were performed in which moist air was used at 150°C on the two size fractions. In both cases no real difference in behavior was noted when compared to results obtained in dry air under these conditions.

Powhatan Coal: Results and Discussion

The other coal examined in this study was Powhatan No. 5 bituminous (LR 26061). TGA runs made in air over a range of temperatures (125°C to 225°C) for the 40x70 mesh size fraction are shown in Figure 4-5. It can be seen that the rate of uptake at 125°C and 150°C is slower than that noted for the same fraction of the Belle Ayr coal. However, over the time range studied no asymptotic value is approached, i.e., the uptake does not show any signs of leveling off. Rates of uptake increase with increasing temperature; only at 225°C does the TGA plot show signs of passing through a maximum. It will be recalled from Figure 4-1 that the equivalent experiment with the Belle Ayr coal showed a maximum for the run made at 175°C. These observations reflect the inherent lower gasification reactivity of the higher rank Powhatan coal. Combined TGA/DSC data (ΔH) are reported for the 40x70 mesh Powhatan coal in Table 4-4. These results show that in all cases the value of ΔH is either constant or decreases with time, indicating that chemisorption is the dominant process. It is surprising that the absolute values for 150°C are somewhat higher than the other temperatures. It should be noted that a DSC run was not made at 225°C because of the obvious role of gasification at that temperature.

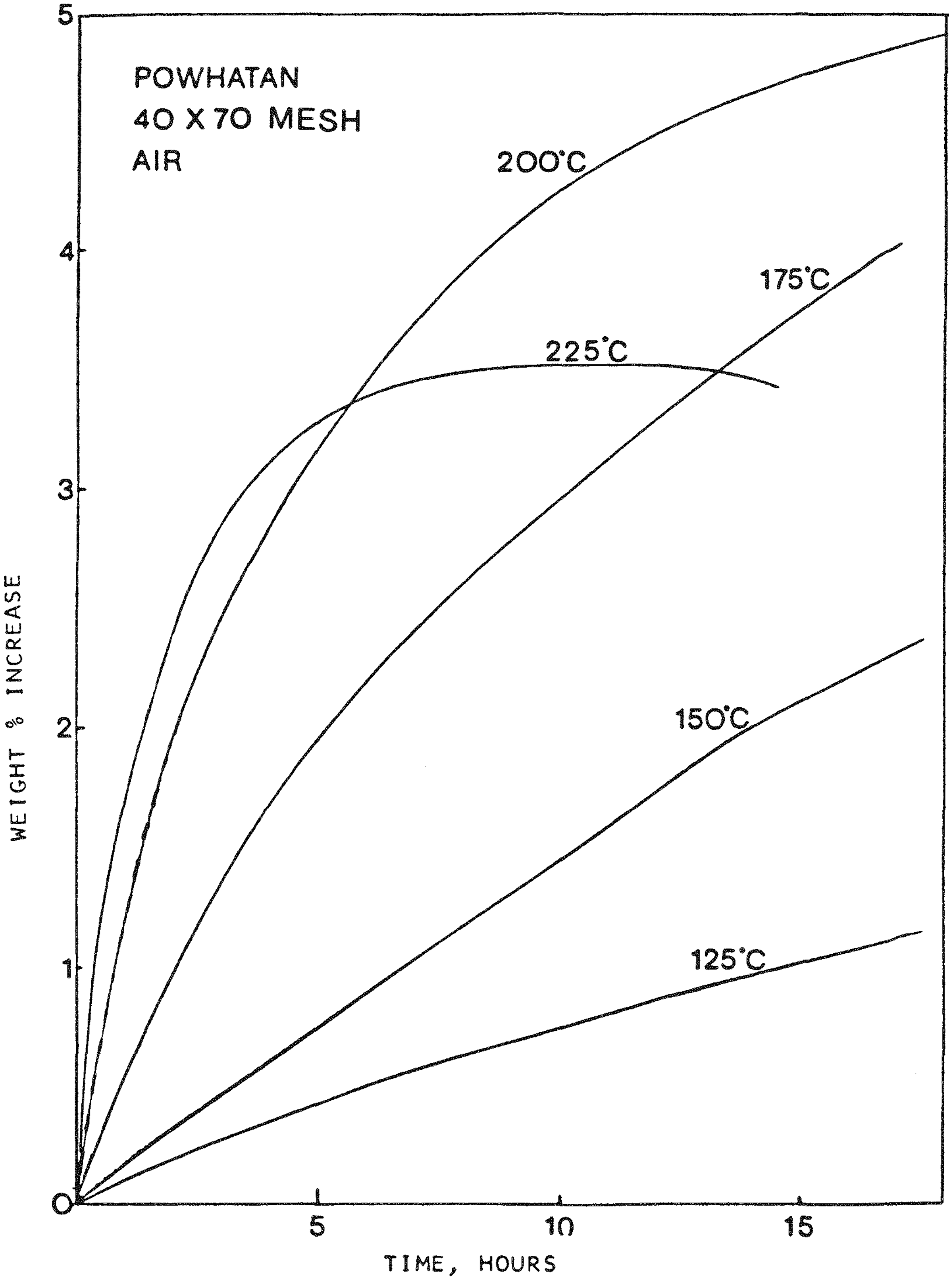

FIGURE 4-5 TGA TREATMENT OF POWHATAN COAL AT VARIOUS TEMPERATURES USING 40X70 MESH COAL

TGA results obtained on 200x270 mesh samples of Powhatan (Figure 4-6) reflect the marked particle size dependency of this coal. These runs were made in a 3-hr period in which time all uptake data exceed those of the 40x70 mesh coal at the same time (Figure 4-5). Thus, the 200x270 mesh Powhatan coal is much more reactive for oxygen chemisorption than the 40x70 mesh material. At 200°C this smaller range Powhatan coal shows a weight increase of 6.2%, compared to about 2.6% after 3 hr.

Table 4-4

Values of ΔH (kcal/g (wt change)) for
40x70 mesh Powhatan Coal in Air

Temperature °C	Time (hr) 0.5	1.0	1.5	2.0	2.5	3.0
125	11.9	10.9	9.9	8.9	8.4	8.3
150	14.2	13.4	13.9	13.9	14.7	14.6
175	12.7	12.3	12.6	13.0	12.8	-
200	12.4	12.3	12.9	13.2	13.1	-

Combined DSC/TGA obtained in air (Table 4-5) for the 200x270 size fraction show that ΔH is essentially constant for temperatures up to 175°C. Therefore, at these temperatures chemisorption dominates. However, the data obtained at 200°C show a market effect of gasification, i.e., ΔH increases with time. A small increase of ΔH with time is noted for the reactions at 175°C.

Table 4-5

Values of ΔH (kcal/g (wt change)) for
200x270 mesh Powhatan Coal in Air

Temperature °C	Time (hr) 0.5	1.0	1.5	2.0	2.5	3.0
125	4.5	4.4	4.3	4.6	4.7	4.6
150	6.6	6.6	6.7	6.9	6.9	6.9
175	5.9	6.3	6.8	6.9	6.8	7.2
200	9.2	9.5	10.2	10.6	11.2	11.6

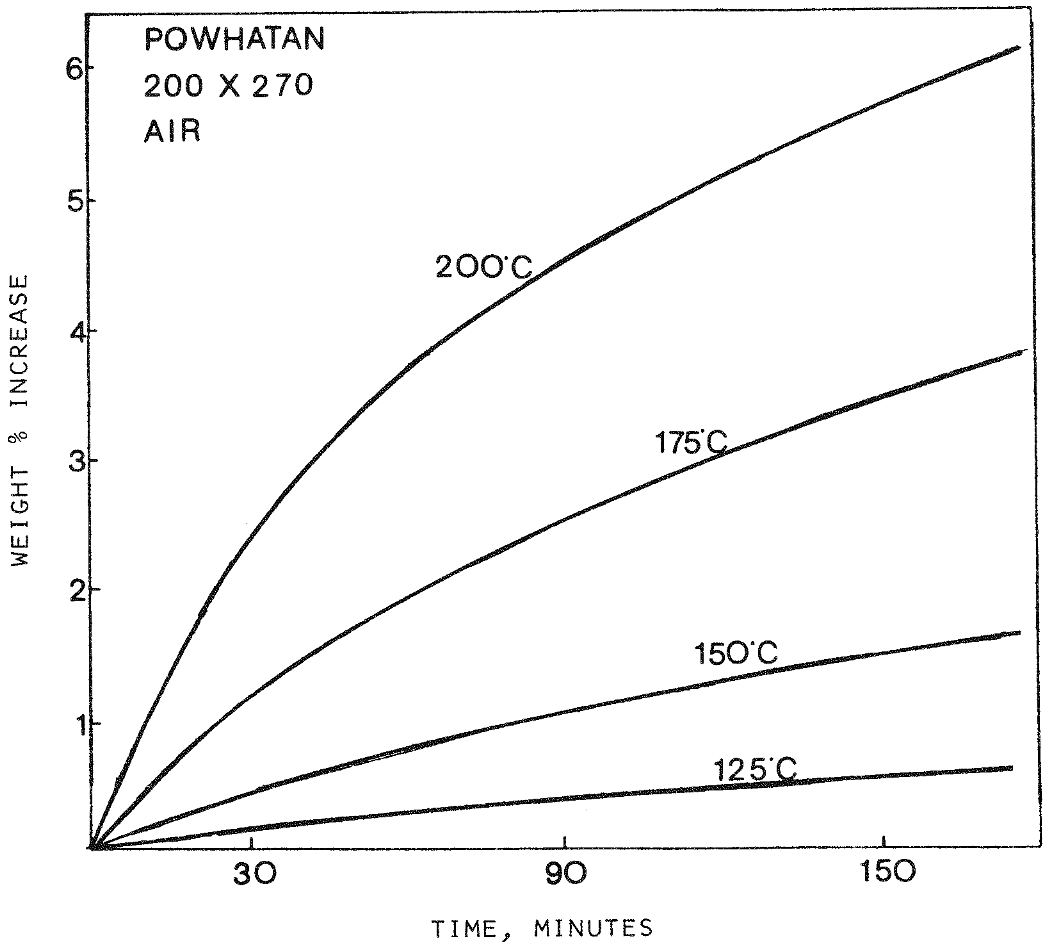

FIGURE 4-6 ISOTHERMAL TGA PLOTS OF 200X270 MESH POWHATAN COAL TREATED IN AIR

The effect of O_2 partial pressure was determined for 40x70 mesh Powhatan (Figure 4-7) at 200°C. Again, it is noted that there is a marked dependency on O_2 partial pressure. A similar set of data was obtained for the 200x270 size fraction (Figure 4-8) which again reflects the noticeable increased reactivity of this smaller size fraction. Figure 4-9 emphasizes the particle size dependency. In this plot, the weight increase at 200°C for the two size ranges is expressed as a function of time; these data were obtained in a pure oxygen atmosphere. Selected TGA runs were made with moist air at 200°C for both particle size ranges. As was the case for the Belle Ayr coal, for each particle size range no significant effect was noted.

Summary of TGA/DSC Results:

Most of the TGA/DSC data are summarized in Table 4-6 to emphasize the previous discussions. That is: under the conditions studied, Belle Ayr coal is more reactive for gasification than Powhatan; there is little apparent effect of particle size for Belle Ayr, whereas it is most marked for Powhatan.

Table 4-6

Low Temperature Air Oxidation of Belle Ayr and Powhatan Coals
Dominant Reaction: Chemisorption (C) or Gasification (G)

Belle Ayr	Temperature °C				
Size Range	125	150	175		
40x70	C	C/G	G		
200x270	C	C/G	G		
Powhatan					
Size Range	125	150	175	200	225
40x70	C	C	C	C	G
200x270	C	C	C	G	G

A series of oxidized samples of both coals in two size ranges (40x70 and 200x270) was prepared in a small fluidized bed. All runs were made at 175°C for varying periods of time in air. Subsamples of the material were set for optical and for other forms of analysis.

Surface Area Changes:

Surface area measurements were made by adsorption of CO_2 (room temperature) and N_2 (77°K). Some porosity data were also obtained on selected samples by mercury

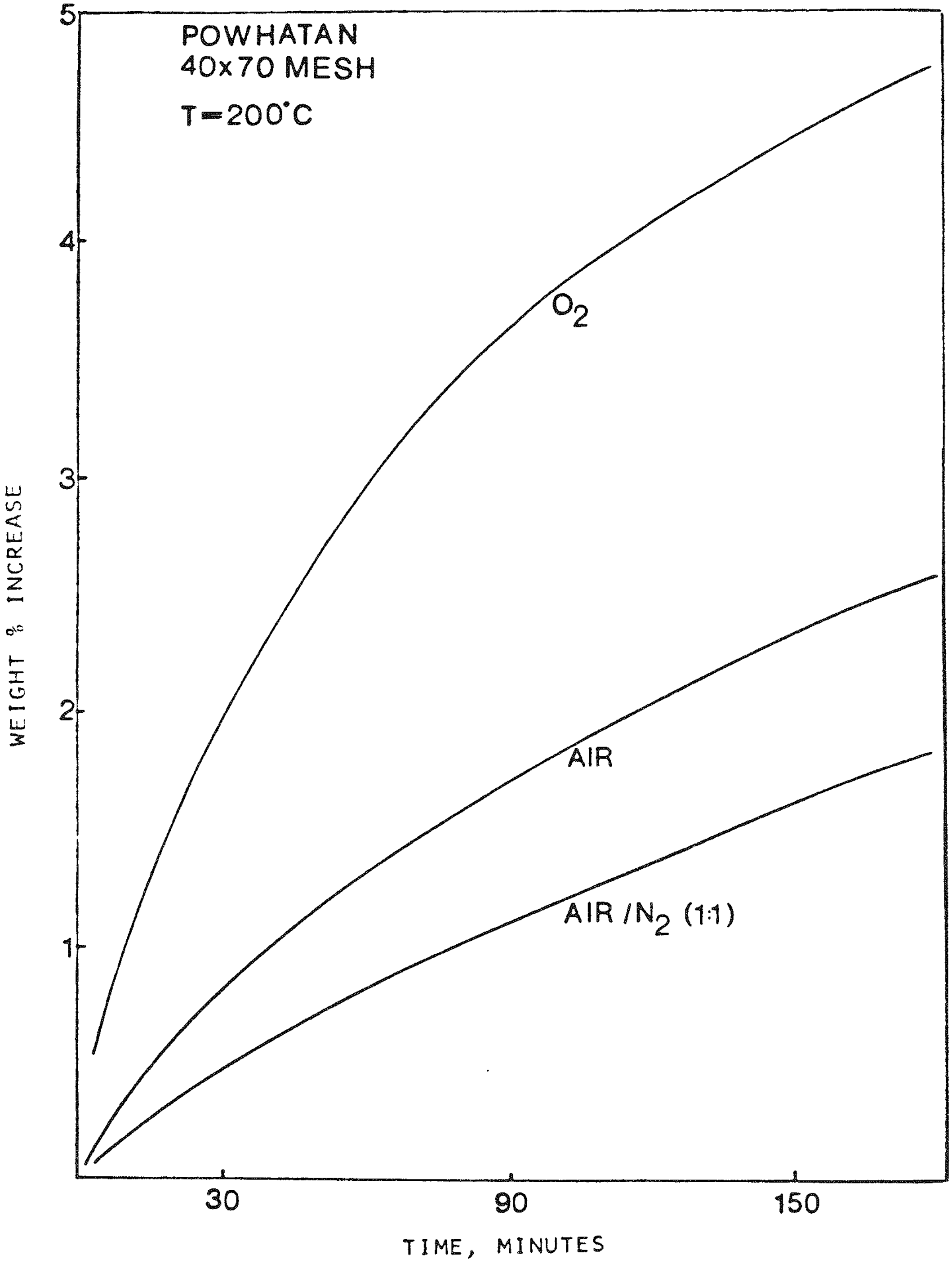

FIGURE 4-7 TGA TREATMENT OF POWHATAN COAL IN VARIOUS GASES AT 200°C

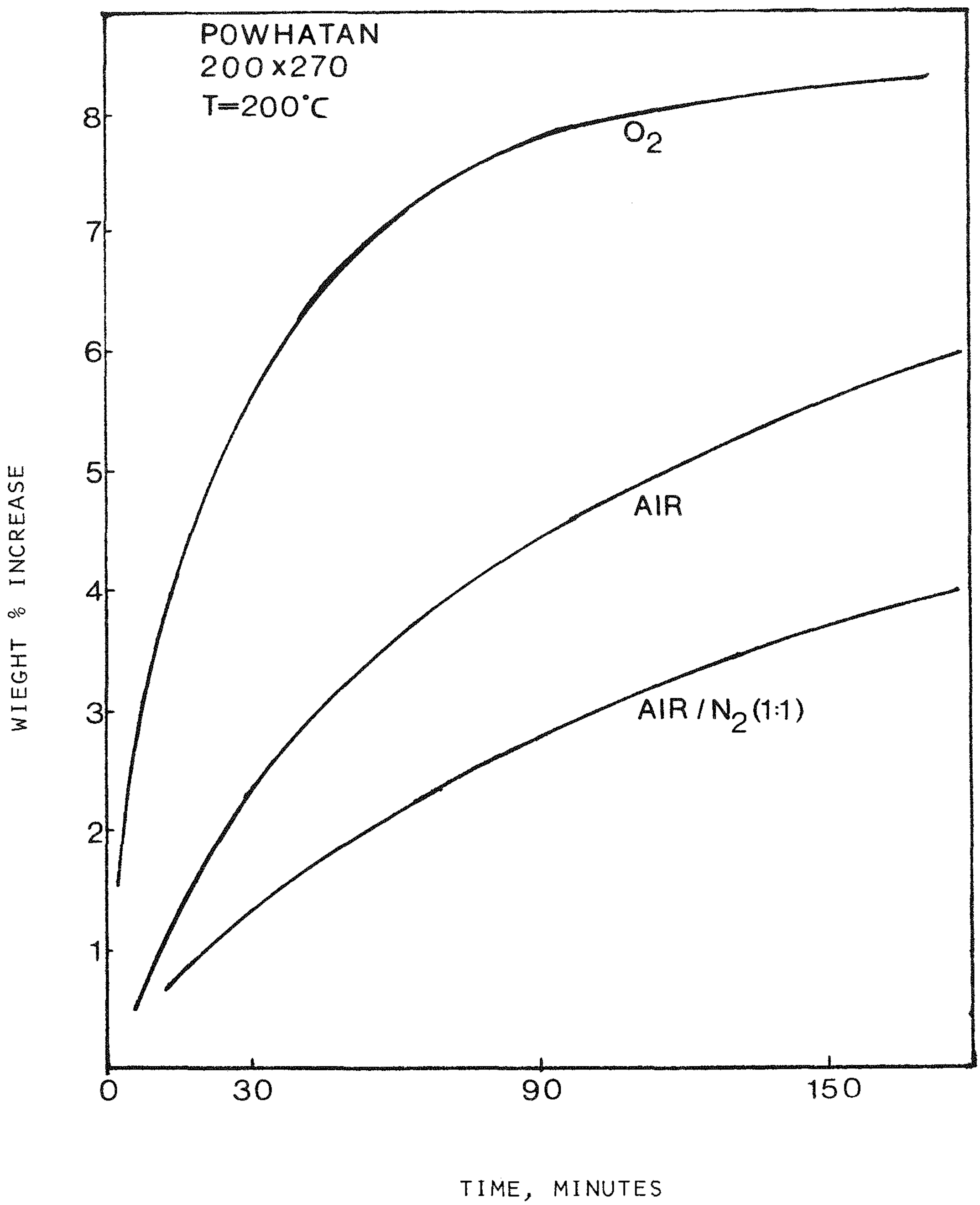

FIGURE 4-8 EFFECT OF OXYGEN PARTIAL PRESSURE: TGA PLOTS OF 200X270 MESH POWHATAN COAL

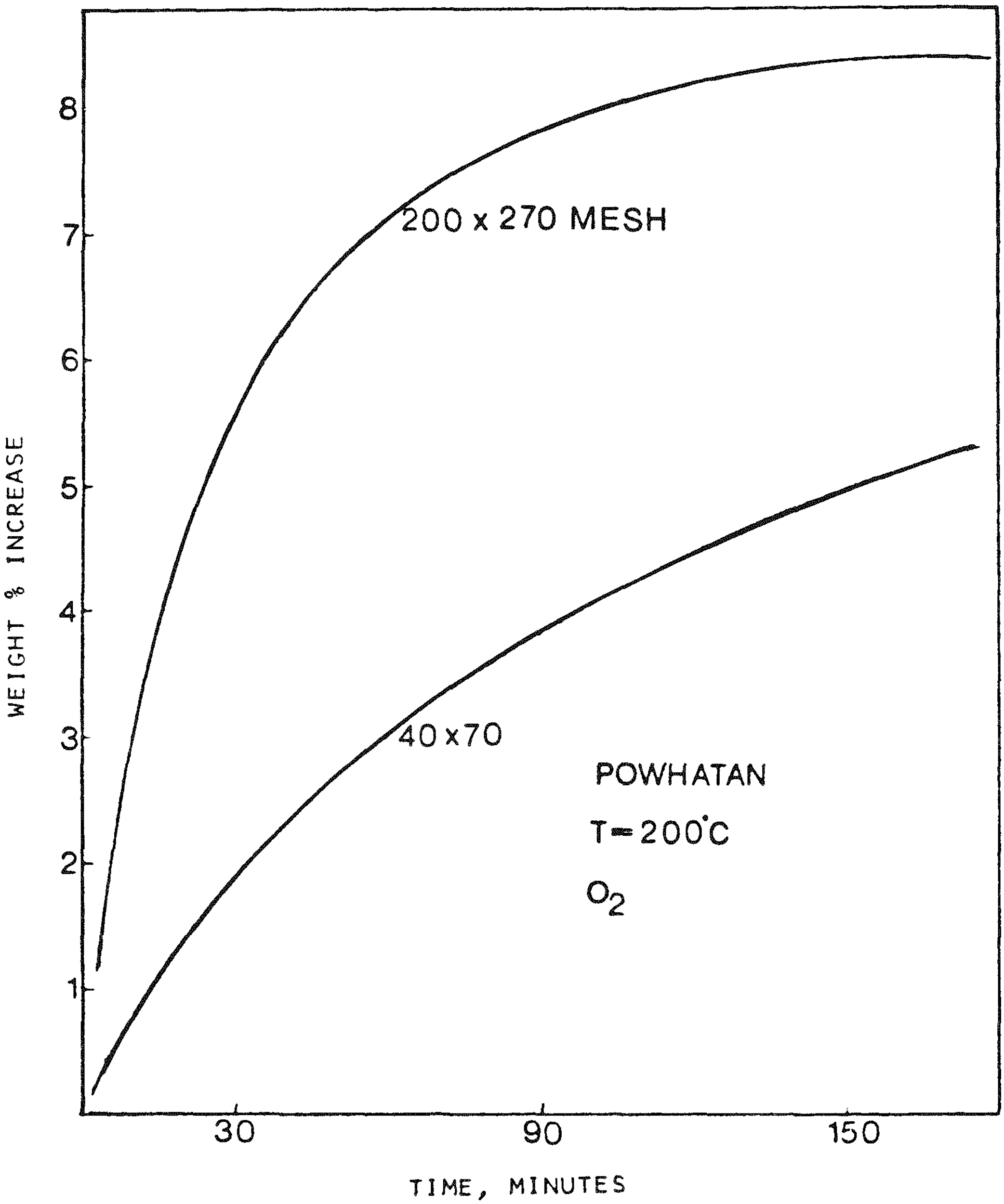

FIGURE 4-9 TGA TREATMENT OF POWHATAN COAL AT 200° C
(EFFECT OF PARTICLE SIZE)

porosimetry. Surface area data for the raw coal samples is given in Table 4-7. These results show (not unexpectedly) that the total surface area, as measured by CO_2 adsorption at 298°K, for the lower rank coal is considerably larger than for bituminous coal. The differences brought about by

Table 4-7

Surface Areas of Coals

	Size Range (mesh)	CO_2 (298°C) m^2/gm	N_2 (77°K) m^2/gm
Belle Ayr	40x70	171	24
	200x270	223	10
Powhatan	40x70	116	<1
	200x270	142	~2

grinding and sieving could, to a limited degree, be explained in terms of maceral differences. However, it is more likely that the increase in area for both coals is due to the opening up of previously occluded porosity. Increased area due to particle comminution would be comparatively small. Examination of the N_2 surface areas (which reflect the extent and distribution of transitional and macroporosity) reveals that pores in this larger size range are more prevalent in the subbituminous coal. This observation is in agreement with concept of structural disorder in the lower rank coals. The reduction of N_2 surface area with reduction of particle size for Belle Ayr coal is somewhat of a problem. It has been our experience that N_2 surface areas are very sensitive to the amount of chemisorbed oxygen on coals. This results in the data being somewhat a function of outgassing conditions. It is possible that the 200x270 size range took up more oxygen than the 40x70 range during sample handling, which was not subsequently removed sufficiently on outgassing at 110°C at 10^{-5} mmHg.

Experimentation in the Laboratory Fluid Bed Dryer (LFD)

Belle Ayr Coal - Qualitative Observations:

Belle Ayr coal was simultaneously dried and oxidized in the LFD located at GR&DC under isothermal conditions to determine the effect of time, particle size, temperature and oxygen partial pressure (P_{O_2}) on oxidation rate, as measured by vent gas analysis. Table 4-8 summarizes operating conditions for all runs.

An initial observation was that at a given temperature, the drying isotherm appeared independent of gas oxygen content, as shown in Figure 4-10 for nitrogen

Table 4-8

ISOTHERMAL RUNS IN THE LABORATORY, FLUID BED DRYER

Run	Coal[(a)]	Gas[(b)]	Avg. Bed Temp[(c)]	Minutes	Mois. Wt%
27	BA 84+	MG	120[c]	30	17.20
28	BA 84+	MG	120[c]	30	16.42
29	BA 84+	MG	95[c]	60	10.95
30	BA 84+	MG	110[c]	10	23.0
31	BA 84+	MG	120	5	24.0
32	BA 84+	MG	120[c]	5	25.0
33	BA 84+	MG	115[c]	10	22.80
34	BA 84+	N_2	105[c]	60	2.2
35	BA 84+	MG	120[c]	120	0.0
36	BA 84+	MG	135	35	5.0
37	BA 84+	A	120	120	2.0
38	BA 84+	A	100	35	3.9
39	BA 84+	A	70[c]	15	23.4
40	BA 84+	A	105	50	1.0
41	BA 84+	MG	70	14	19.8
42	BA 84+	MG	125	15	12.65
43	BA 84+	MG	119	60	0.8
44	BA 84+	x	x	x	x
45	BA 84+	MG	125[c]	15	8.0
46	BA 84+	MG	55[c]	15	24.0
47	BA 84+	MG	180[c]	20	8.0
48	BA 84+	MG	125	60	0.5
49	BA 84+	MG	125	15	11.0
50	BA 84+	MG	150	60	0.9
51	BA 84+	MG	150	15	n.a.
52	BA 84+	MG	150	15	3.6
53	BA 84+	MG	152	30	1.4
54	BA 84+	A	153	60	0.6
55	BA 140x230	MG	50	60	10.7
56	BA 140x230	N_2	50	60	12.8
57	BA 84+	MG	50	60	14.4
58	BA 140x230	N_2	85	15	3.6
59	BA 140x230	N_2	110[c]	30	1.8

(a) BA = Belle Ayr Wyodak Subbituminous, P = Powhatan Pitt Seam Bituminous

(b) A = Air; MG = mixed gas (10.6% vol% O_2 in 89.4% N_2)

(c) Indicates erratic temperature control

x = Run terminated.

Table 4-8 (continued)

Run	Coal(a)	Gas(b)	Avg. Bed Temp(c)	Minutes	Mois. Wt%
60	BA 140x230	N_2	110[c]	30	0.6
61	BA 140x230	N_2	100[x]	120	0.4
62	BA 140x230	N_2	100	15	3.2
63	BA 140x230	N_2	100	60	0.7
64	BA 140x230	MG	100	50	1.0
65	BA 140x230	MG	100	30	1.2
66	BA 140x230	MG	100	120	1.2
67	BA 140x230	MG	50	120	3.0
68	BA 140x230	MG	50	30	15.2
69	BA 140x230	MG	135[c]	15	1.3
70	BA 140x230	MG	150[c]	30	1.2
71	BA 140x230	A	150[c]	15	0.2
72	BA 140x230	A	150[c]	60	1.1
73	BA 140x230	A	150[c]	120	0.0
74	BA 140x230	MG	200[c]	30	1.1
75	BA 140x230	A	205[c]	30	1.4
76	BA 140x230	N_2	120	20	1.5
77	BA 140x230	N_2	125	60	0.1
78	BA 140x230	N_2	125	20	1.0
79	BA 140x230	N_2	125	10	1.15
80	BA 140x230	A	50	30	10.5
81	BA 140x230	A	50	30	16.8
82	BA 140x230	A	50	120	3.7
83	BA 140x230	A	50	60	6.4
84	BA 140x230	A	50	10	22.7
85	BA 84+	A	51	10	24.0
86	BA 84+	A	51	30	14.8
87	BA 84+	A	51	60	7.4
88	BA 84+	A	51	120	3.9
89	BA 84+	N_2	50	30	17.5
90	BA 140x230	A	160[c]	5	1.2
91	BA 140x230	MG	160[c]	5	1.0
92	BA 140x230	A	103	5	5.3
93	BA 140x230	MG	75	10	8.3
94	BA 140x230	A	75	10	13.6
95	BA 140x230	MG	75	60	2.8
96	BA 84+	MG	50	10	n.a.

(a) BA = Belle Ayr Wyodak Subbituminous, P = Powhatan Pitt Seam Bituminous

(b) A = Air; MG = mixed gas (10.6% vol% O_2 in 89.4% N_2)

(c) Indicates erratic temperature control

x = Run terminated.

Table 4-8 (continued)

Run	Coal[a]	Gas[b]	Avg. Bed Temp[c]	Minutes	Mois. Wt%
97	BA 84+	MG	50	60	n.a.
98	BA 84+	MG	50	20	n.a.
99	BA 84+	MG	50	30	11.5
100	P140x200	MG	75	60	0.1
101	P140x200	MG	75	30	0.4
102	P140x200	MG	90[c]	10	n.a.
103	P140x200	A	75	60	0.8
104	P140x200	A	75	30	0.8
105	P140x200	A	75	10	0.7
106	P140x200	A	100	15	n.a.
107	P140x200	A	50	10	n.a.
108	P140x200	MG	75	10	n.a.
109	BA 84+	A	150	15	n.a.
110	P140x200	A	150	15	n.a.
111	P140x200	MG	150	15	n.a.
112	P140x200	A	125	15	n.a.
113	P140x200	A	150	15	0.9
114	P140x200	A	125	15	n.a.
115	P30x70	A	100	10	n.a.
116	P30x70	A	50	10	1.7
119	P140x200	A	125	15	n.a.
120	BA 140x230	A	100	15	n.a.
126	BA 84+	N_2	100	75	5.4
127	BA 84x230	A	100	10	0.3

[a] BA = Belle Ayr Wyodak Subbituminous, P = Powhatan Pitt Seam Bituminous

[b] A = Air; MG = mixed gas (10.6% vol% O_2 in 89.4% N_2)

[c] Indicates erratic temperature control

x = Run terminated.

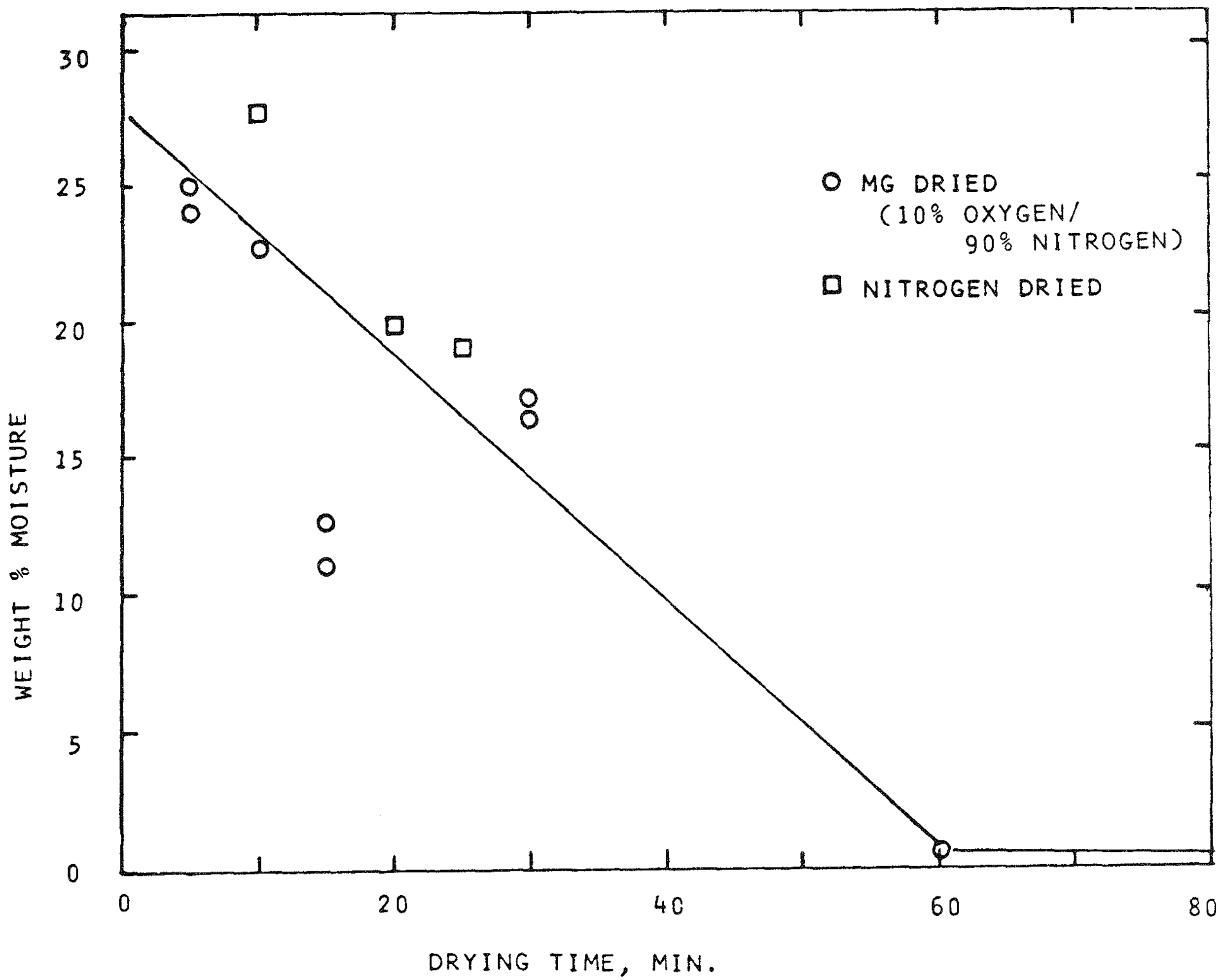

FIGURE 4-10 DRYING ISOTHERM (120 - 125°) OF BELLE AYR 84+MESH COAL

and mixed gas (MG) at 120°C. (Mixed gas consists of 10.6% oxgen in 89.4% nitrogen.)

A second major observation was that the oxidation reaction apparently proceeded in two stages. At the start of a run, nitrogen in the reactor is purged by the gas feed, then there is an initial period of high oxygen uptake followed by a second phase of lower and declining oxygen utilization. The initial phase is absent from runs below about 140°C. At higher temperatures, e.g., 200°C, a sustained combustion of the coal is observed in air but not in MG.

Belle Ayr Oxidation Kinetics:

Table 4-8 lists the oxygen consumption rate (OCR) data for all runs with Belle Ayr coal in the LFD. Obtaining reliable kinetic correlations from the isothermal Belle Ayr data was complicated by several factors, the primary of which was that small differences in handling or slight non-homogenity in coal samples resulted in only fair reproducibility. Better kinetic correlations were obtained from the non-isothermal runs discussed later, but the isothermal runs nevertheless provided valuable information.

With few exceptions, the average consumption rate in the declining rate phase was less than 1 g O_2/hr (0.52×10^3 millimole/min). Specifically, even if all the consumed oxygen was sorbed by the coal, the weight increase in a 1 h run would be less than 5%.

Since the initial high-rate oxygen consumption phase was only observed above about 140°C, relatively little data is available, as shown in Table 4-9. Figure 4-11 is an Arrhenius plot of the initial peak oxygen consumption rate versus the inverse Kelvin temperature. A line fitted to the points gives the following equation:

$$\ln r \left(\frac{\text{gmoles } O_2}{\text{min}}\right) = -5030\ (1/T^{\circ}K) + 5.6012 \qquad (4\text{-}3)$$

An activation energy for the initial high-rate oxidation of about 10.0 kcal/gm mol is indicated. No conclusions can be drawn from Figure 4-11 concerning the effect of oxygen partial pressure on the initial oxidation rate.

If the initial high-rate reaction is a surface phenomenon, the rate should be sensitive to particle size. The effect can be seen in the 150°C runs in Table 4-9, where the peak rate for the 140x230 mesh coal (runs 70, 71, 72, 73) was two to three times higher than that for the larger 84+ mesh coal particles. A particle size analysis Figure 4-12 of the 84+ mesh and 140x230 mesh sieve fractions show the mean particle size to be 203 and 83 microns respectively.

Table 4-9

OXYGEN CONSUMPTION CALCULATED FROM VENT GAS COMPOSITION

Run	BA Coal Size	Feed Gas	Temp. °C	Min.	Initial Phase Peak Height Millimoles O_2/Min	Dur. Min	Diffusion Phase Millimoles O_2/Min SOR	EOR
28	84+ MG	120	30	n	n	-	0.178	
29	84+ MG	95	60	n	n	0.297	0.178	
35	84+ MG	120	120	n	n	0.957	0.291	
36	84+ MG	135	35	a	n	0.26	0.208	
37	84+ AIR	120	120	n	n	0.208	0.208	
38	84+ AIR	100	35	a	n	0.624	0.52	
40	84+ AIR	105	60	n	n	77	0.478	
41	84+ MG	70	15	n	n	46	0.46	
46	84+ MG	55	15	n	n	0.46	0.46	
47	84+ MG	180	20	-	-	0.347	0.16	
48	84+ MG	125	60	-	-	-	1.02	
49	84+ MG	125	15	n	n	0.46	0.58	
50	84+ MG	150	60	n	n	0.347	0.392	
52	84+ MG	150	15	0.97	7	0.277	0.208	
53	84+ MG	150	30	0.76	-	0.53	0.116	
54	84+ AIR	153	60	0.81	2	0.647	0.58	
56	140x230	MG	50	60	n	n	0.208	0.208
57	84+ MG	50	60	n	n	0.116	0.116	
64	140x230	MG	100	60	n	n	0.416	0.07
65	140x230	MG	100	30	n	n	0.139	0.139
66	140x230	MG	100	120	n	n	0.392	0.116
67	140x230	MG	50	120	n	n	0.115	0.046
68	140x230	MG	50	30	n	n	0.116	0.116
69	140x230	MG	135	15	1.92	7	0.461	0.39
70	140x230	MG	150	30	1.22	3	0.55	0.485
71	84+ AIR	150	15	3.12	6	0.92	0.76	
72	84+ AIR	150	60	2.43	6	0.69	0.392	
73	140x230	MG	200	30	5.43	6	1.918	1.247
74	140x230	MG	200	30	5.43	6	1.918	1.247
75	140x230	AIR	205	30	c	c	n	n
80	140x230	AIR	50	30	n	n	0.23	0.23

n=not visible SOR=start of run

c=combusted EOR=end of run

Table 4-9 (continued)

Run	BA Coal Size	Feed Gas	Temp. °C	Min.	Initial Phase Peak Height Millimoles O_2/Min	Dur. Min	Diffusion Phase Millimoles O_2/Min SOR	EOR
82	140x230	AIR	50	120	n	n	0.393	0.30
83	140x230	AIR	50	60	n	n	1.09	0.393
84	140x230	AIR	50	10	n	n	0.393	0.370
85	84+ AIR	51	10	n	n	0.65	0.312	
86	84+ AIR	51	30	n	n	0.23	0.257	
87	84+ AIR	51	60	n	n	-	-	
88	84+ AIR	51	120	n	n	0.46	-	
90	140x230	AIR	160	5	2.22	>4	n	n
91	140x230	MG	160	5	2.54	>4	n	n
92	140x230	AIR	103	5	0.27	2.3	n	n
93	140x230	MG	75	10	n	n	0.166	0.166
94	140x230	AIR	75	10	n	n	0.333	0.333
95	140x230	MG	75	60	n	n	0.125	0.125
96	84+ MG	50	10	n	n	0.146	0.146	
97	84+ MG	50	60	n	n	0.208	0.208	
98	84+ MG	50	20	n	n	0.146	0.146	
99	84+ MG	50	30	n	n	0.291	0.291	

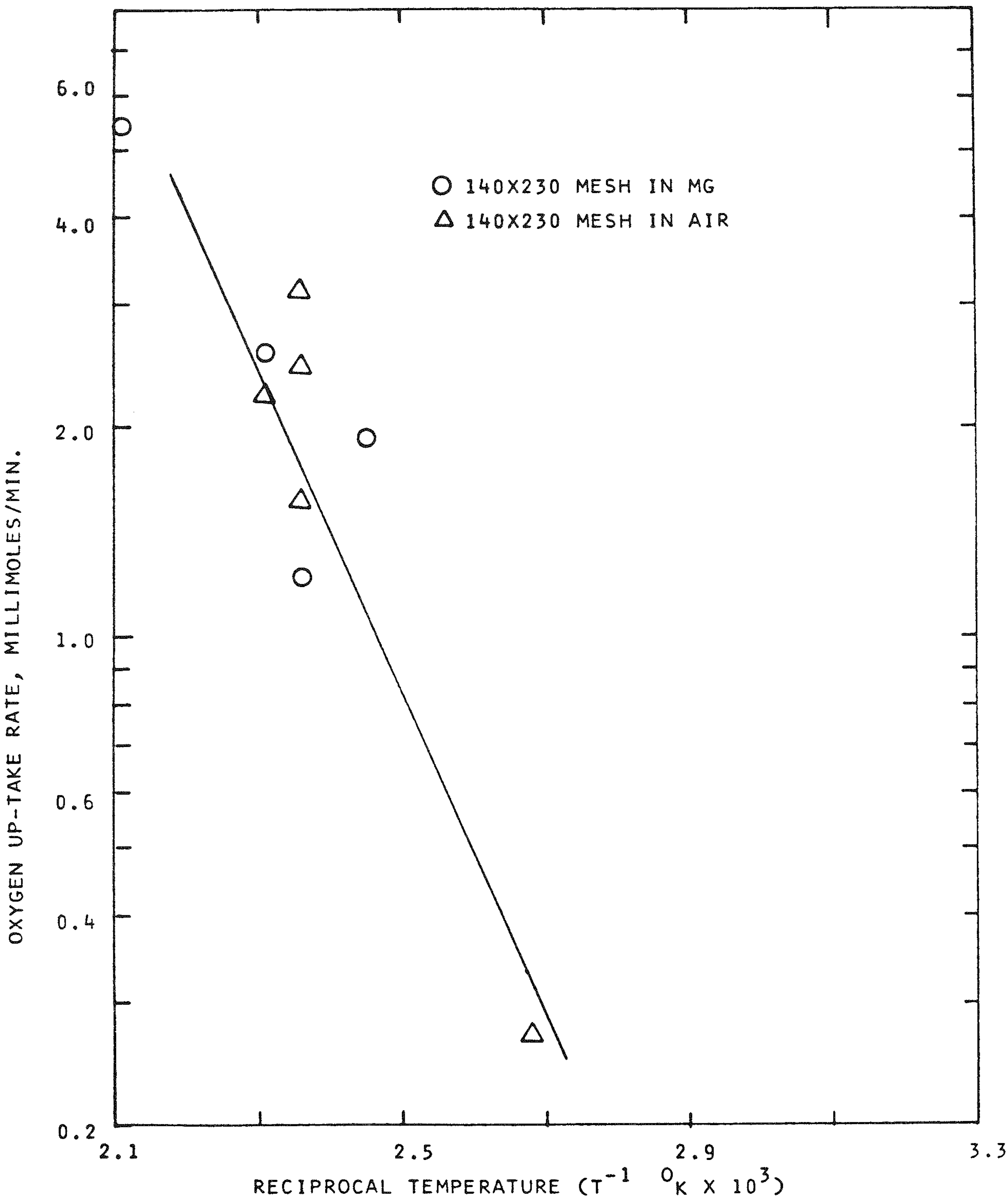

FIGURE 4-11 KINETICS OF HIGH RATE INITIAL OXIDATION PERIOD

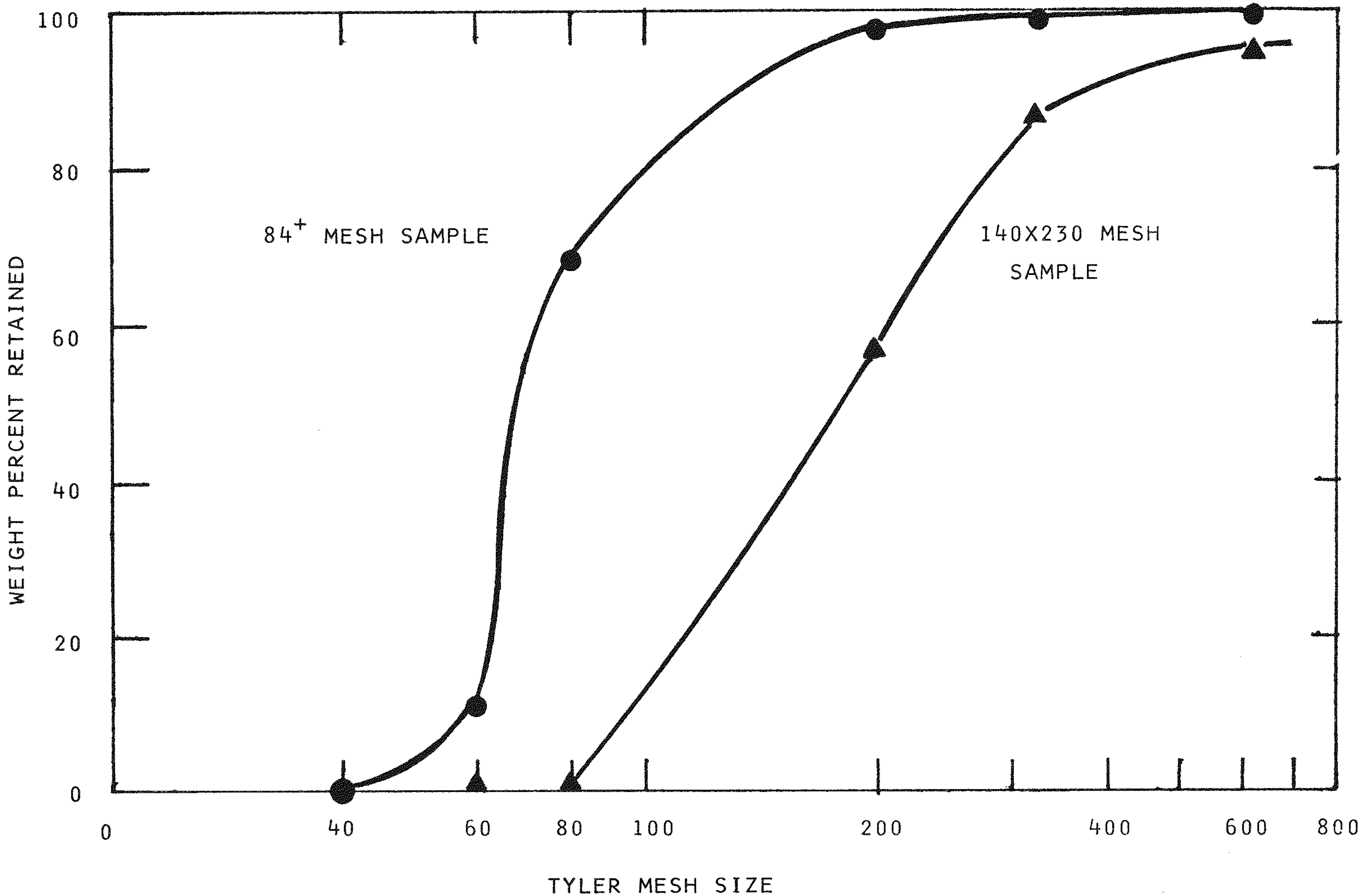

FIGURE 4-12 PARTICLE SIZE DISTRIBUTION OF BELLE AYR 84+ AND 140X230 MESH SAMPLES OF COAL

This results in a diameter ratio of 2.44:1 and a surface area ratio of 6:1. However, the mean particle size provides only a rough estimate of the actual surface area ratio.

After the initial high reaction rate phase, a period of declining reaction rate occurs. Figure 4-13 is an Arrhenius-type plot of the oxgyen consumption rate, OCR, (millimoles O_2/min) for the 140x230 mesh Belle Ayr coal fluidized in MG or air. There is scatter in the data, but the following lines can be fitted:

$$\text{AIR:} \quad \ln r \left(\frac{\text{gmoles } O_2}{\text{min}}\right) = -1567\ (1/T^{o}K)\ -3.5192 \qquad (4\text{-}4)$$

$$\text{MG:} \quad \ln r \left(\frac{\text{gmoles } O_2}{\text{min}}\right) = -1411\ (1/T^{o}K)\ -4.1038 \qquad (4\text{-}5)$$

The lines have similar slope with an activation energy of about 3.0 kcal/gmol O_2, about half that of the high-rate phase.

The data for 84+ mesh Belle Ayr coal as shown in Figure 4-14 are appreciably more scattered than for the 140x230 mesh coal. The line fitted for MG is:

$$\ln r \left(\frac{\text{gmol } O_2}{\text{min}}\right) = -928\ (1/T^{o}K) + 0.9883 \qquad (4\text{-}6)$$

It is clear that the larger particles give a lower rate at a given temperature.

Powhatan Oxidation Kinetics:

The literature documents that coals of different rank may diverge sharply in oxidation rate, mechanism and ultimate fixed oxygen. A limited number of runs were made with Powhatan (Pittsburgh seam) bituminous coal to compare with the Belle Ayr subbituminous coal. As described in Section 7, the composition of this coal is quite different from Belle Ayr, and the initial moisture content of the pulverized coal was 2.5% versus 27.5% for Belle Ayr.

The initial series of runs were isothermal, each made with a fresh 30 g (wet) charge of 140x200 or 30x70 mesh coal. The run conditions and observed oxidation rates are summarized in Table 4-10. In most runs there was little significant drop in oxidation rate from the start to end of run. As shown in Figure 4-15, the data for 140x200 mesh Powhatan coal show little scatter, unlike corresponding Belle Ayr runs. The lines for MG and air both indicate an activation energy of about 4 kcal/gmol O_2, but the rate in air is about double that in MG at the same temperature:

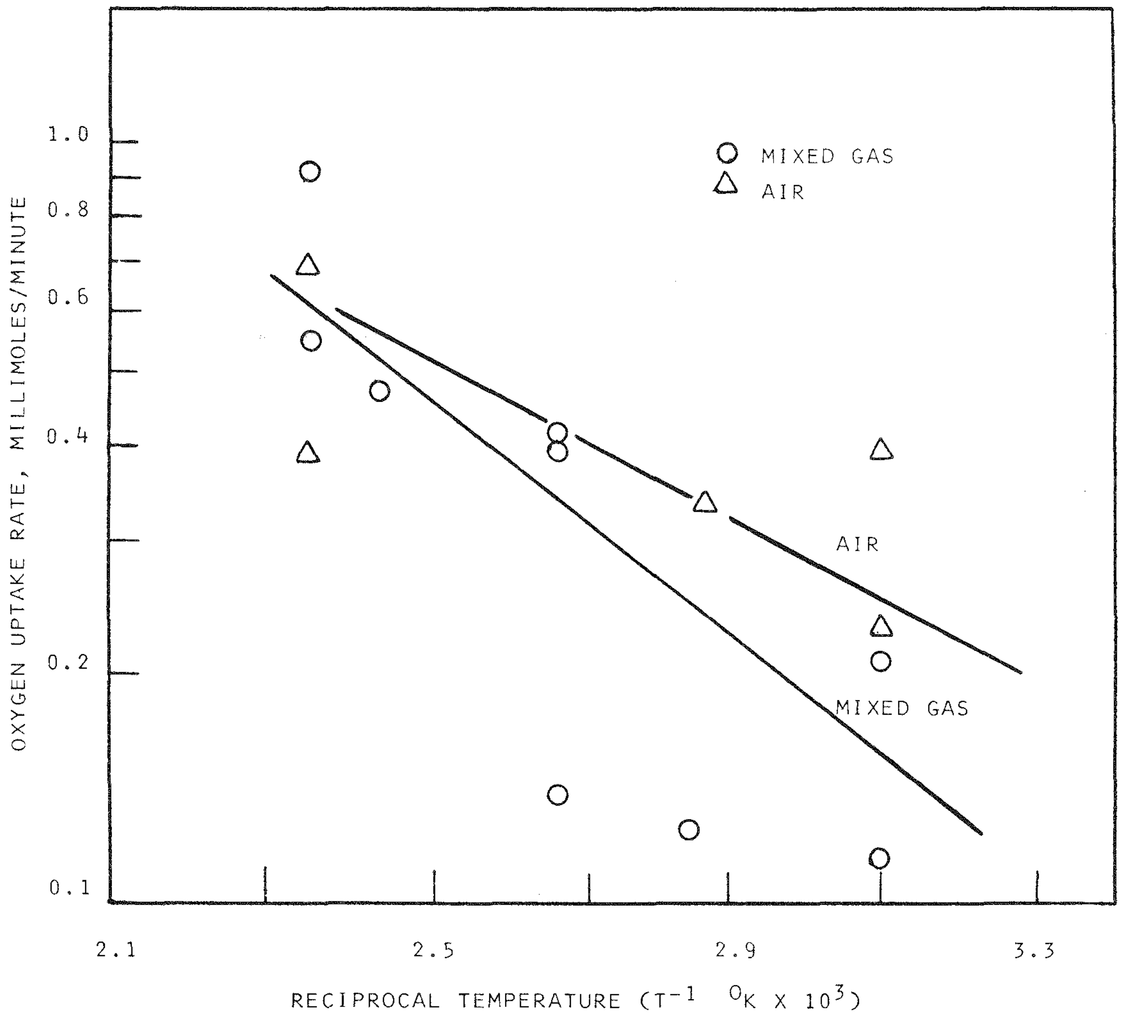

FIGURE 4-13 OXYGEN UPTAKE RATE IN DECLINING RATE PERIOD WITH 84^+ MESH BELLE AYR COAL

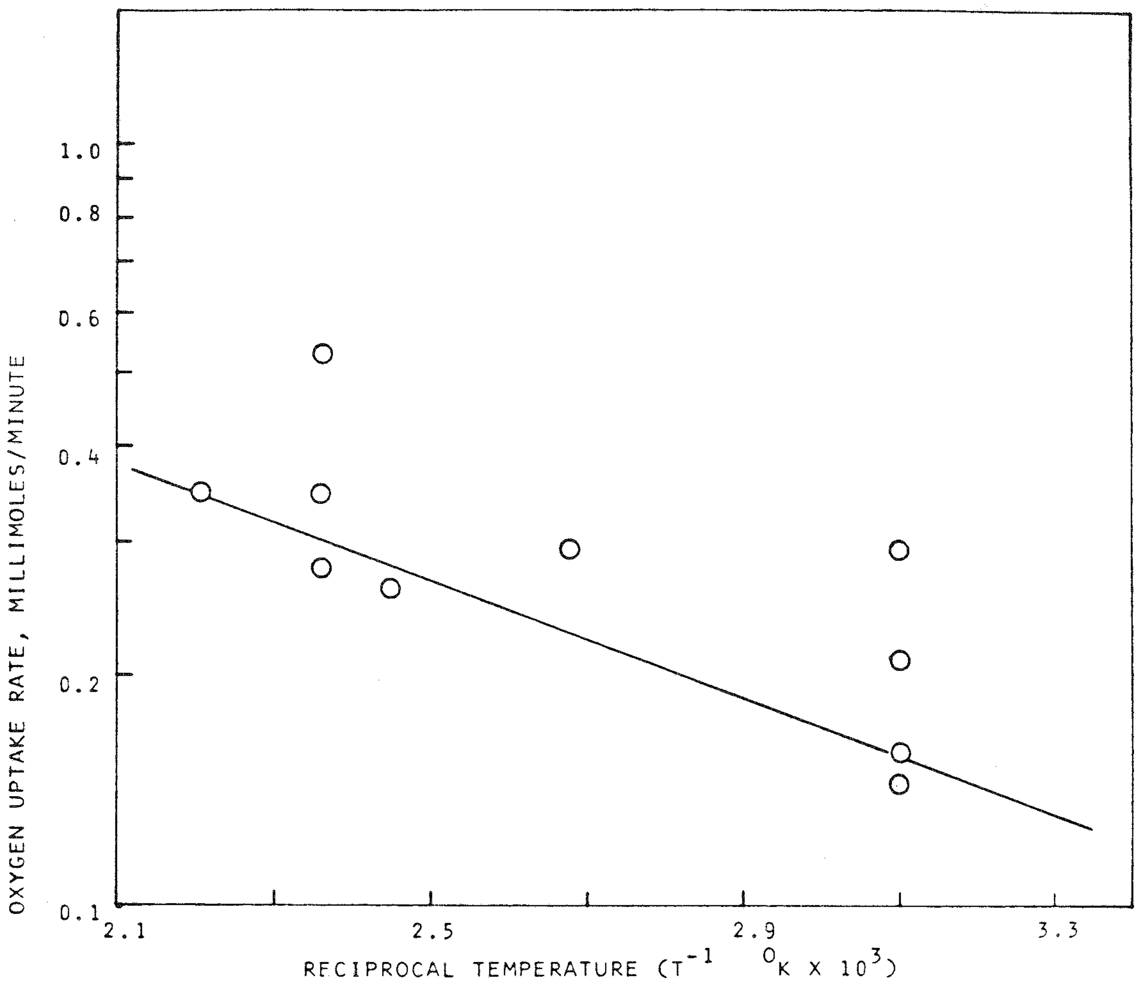

FIGURE 4-14 OXYGEN UPTAKE RATE IN DECLINING RATE PERIOD WITH 84^{+} MESH BELLE AYR COAL

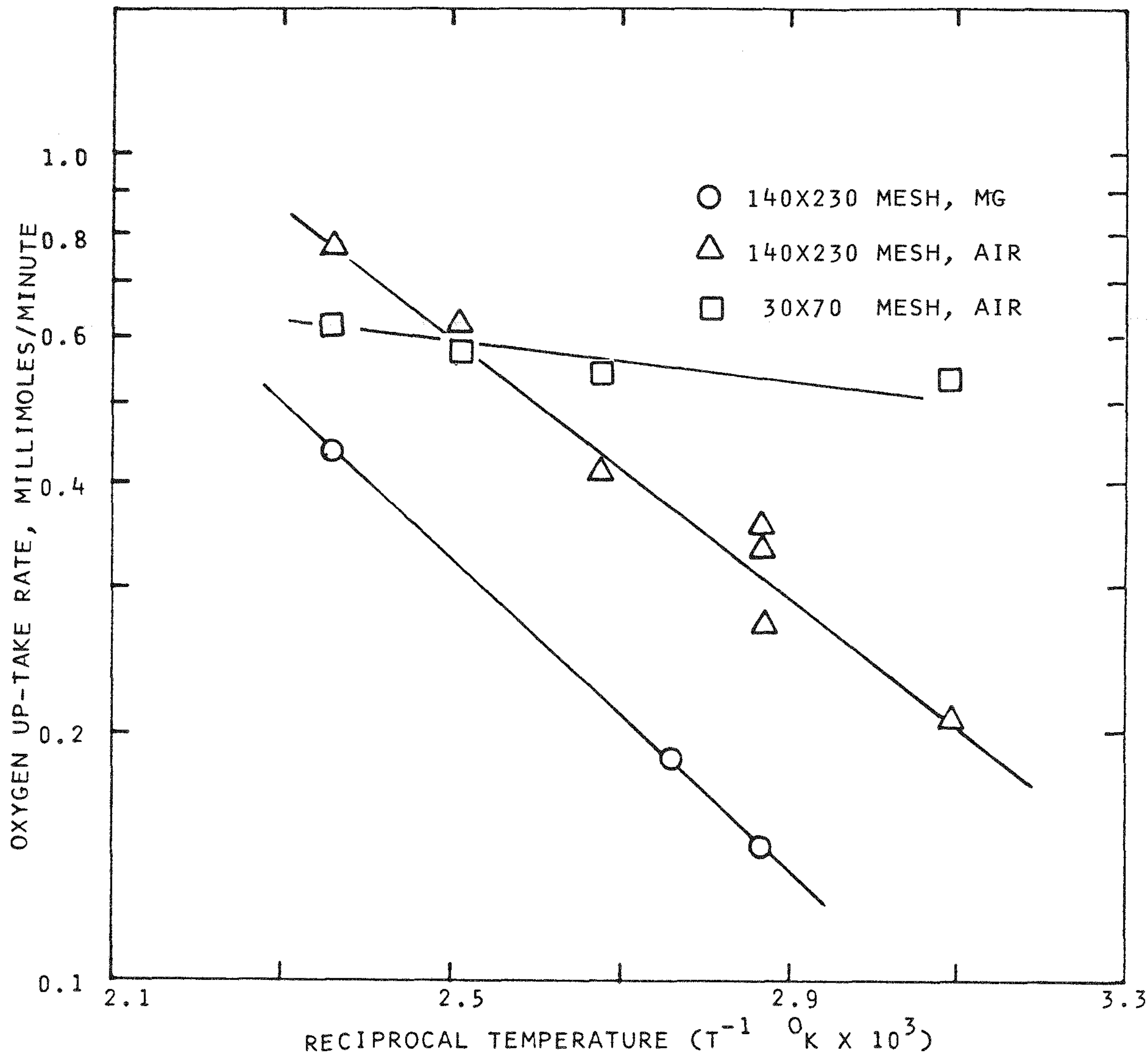

FIGURE 4-15 OXYGEN UP-TAKE RATE IN DECLINING RATE PERIOD WITH POWHATAN 140X200 COAL

Table 4-10

ISOTHERMAL DRYING/OXIDATION OF POWHATAN COAL OXYGEN CONSUMPTION CALCULATED FROM VENT GAS COMPOSITION

Run	Coal Mesh	Temp °C	Feed Gas	Dur. Min.	Fin. Mois%	Dec. Rate Phase (Gm Moles O_2/Min)x10^3 SOR	EOR
100	140x200	75	MG	60	0.1	0.146	0.125
101	140x200	75	MG	30	0.4	0.146	0.125
102	140x200	90	MG	10		0.187	0.146
103	140x200	75	Air	60	0.8	0.270	0.245
104	140x200	75	Air	30	0.8	0.333	0.333
105	140x200	75	Air	10	0.7	0.354	0.354
106	140x200	100	Air	15		0.416	0.416
107	140x200	50	Air	10		0.208	0.208
108	140x200	75	MG	10		NA	NA
110	140x200	150	Air	15		NA	NA
111	140x200	150	MG	15		0.437	0.437
112	140x200	125	Air	15		0.77	0.666
113	30x70	150	Air	115	0.9	0.624	0.603
114	30x70	125	Air	115		0.582	NA
115	30x70	100	Air	10		0.54	NA
116	30 70	50	Air	10	1.7	NA	NA
119	140x200	125	Air	15		0.416	0.354

$$\text{AIR:} \quad \ln r \left(\frac{\text{gmol } O_2}{\text{min}}\right) = -1798 \ (1/T^0K) \ -2.9334 \qquad (4\text{-}7)$$

$$\text{MG:} \quad \ln r = -2179 \ (2/T^0K) \ -2.5944 \qquad (4\text{-}8)$$

As with Belle Ayr coal, two phases were seen: an initial rapid period of high oxygen consumption (visible only above 125°C) followed by a phase of lower and slowly decreasing rate.

For the coarser 30x70 mesh coal oxidized in air, as also shown in Figure 4-15, the rate at 125°C was identical to the 140x200 mesh coal in air, but the limited data for this particle size indicates a lower temperature dependence, with an apparent activation energy of only about 0.8 kcal/gmol. Diffusion of moisture from the larger particles may be responsible for this effect.

Non-Isothermal Oxidation Runs:

The previous isothermal runs were required to prepare partially dried, oxidized coal samples for analysis and subsequent liquefaction studies, but the following non-isothermal runs provided a better kinetic picture of the oxidation process. Since the oxidiation rate for a single coal sample was determined over the entire temperature range, non-homogeneity between samples was avoided. Table 4-11 lists the conditions for the non-isothermal runs.

The first non-isothermal runs were made with Belle Ayr coal. Figures 4-16 and 4-17 show the temperature ascent and descent, respectively, for Run 118 where 84+ mesh coal was oxidized in MG. During the ascent the oxidation rate was temperature dependent over the entire range, but there was a break in slope at about 140°C. In contrast, during the descent there was a temperature dependence above 140°C with a slope similar to that of the ascent. However, the oxidation rate was constant at 0.125 millimoles/min below 140°C.

The results of Run 117, using air instead of MG, are seen in Figures 4-18 and 4-19. The results exhibit parallel behavior to Run 118, except that the rate in air is about double that in MG in the region below 140°C and 54% higher above 140°C (ascent).

Run 121 was made using 140x230 mesh coal in MG to provide a comparison of reaction rates for the two coal sizes. It was observed that, in the high temperature region of the descent, particle size had little effect on oxidation rate, while below 140°C the finer coal oxidized at 64% higher rate. It was also

Table 4-11

NON-ISOTHERMAL OXIDATION RUNS

	Slope Ascent		Slope Descent		Intercept Ascent		Intercept Descent			
Run	Coal[a]	Gas[b]	Lo Temp[e]	Hi Temp	Lo Temp	Hi Temp	Lo Temp	Hi Temp	Lo Temp	Hi Temp
117	BA 84+	A	-1403	-5524	0	-6182	2.8548	12.2150	0.270	13.3747
118	BA 84+	MG	-1399	-3596	0	-6610	2.2402	7.7340	0.125	14.0956
121	BA 140x230	MG	-1168	-3393	0	-6373	2.0070	7.6512	0.205	13.7400
122	BA 84+	MG[c]	0	-2186	n.r.	n.r.	0.48	5.0503	n.r.	n.r.
123	BA 84+	MG[d]	-3785	-3785	n.r.	n.r.	8.0131	8.0131	n.r.	n.r.
124	P 140x230	MG	-544	-4311	0	-3929	0.1066	9.3791	0.25	8.6309

n.r. = not run

[a]BA = Belle Ayr, P = Powhatan

[b]A = Air, MG = Mixed N_2 plus 10.4% O_2

[c]Prior to MG introduction, coal dried 65 min in N_2 at 100°C.

[d]Prior to MG introduction, coal dried 16 hr at 21°C in N_2.

[e]Lo Temp refers to below about 130°C, Hi Temp above this

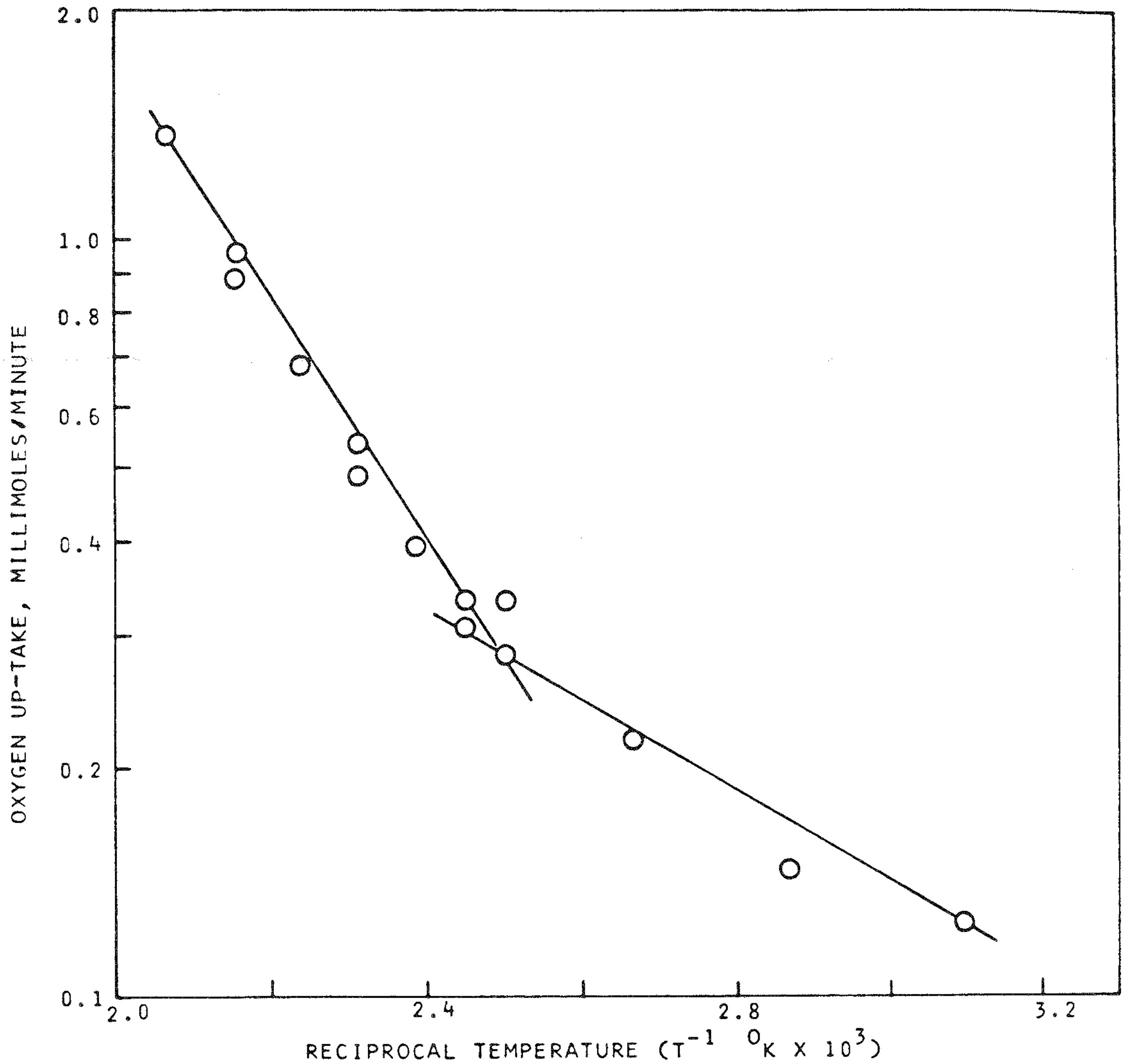

FIGURE 4-16 TEMPERATURE ASCENT FOR BELLE AYR COAL (84^{+}) IN MG

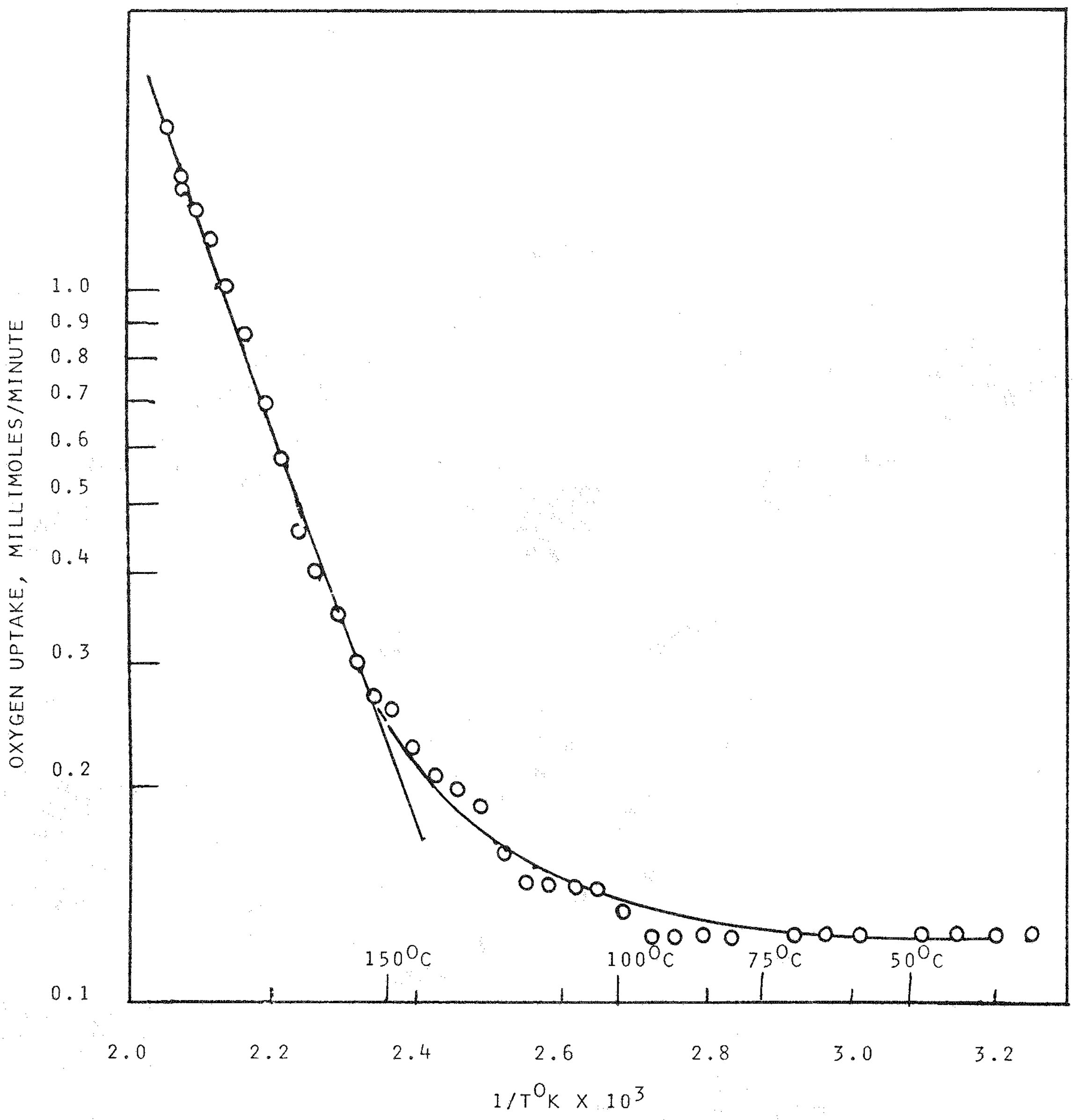

FIGURE 4-17 TEMPERATURE DESCENT FOR BELLE AYR 84$^+$ MESH COAL NON-ISOTHERMALLY OXIDIZED IN MIXED GAS

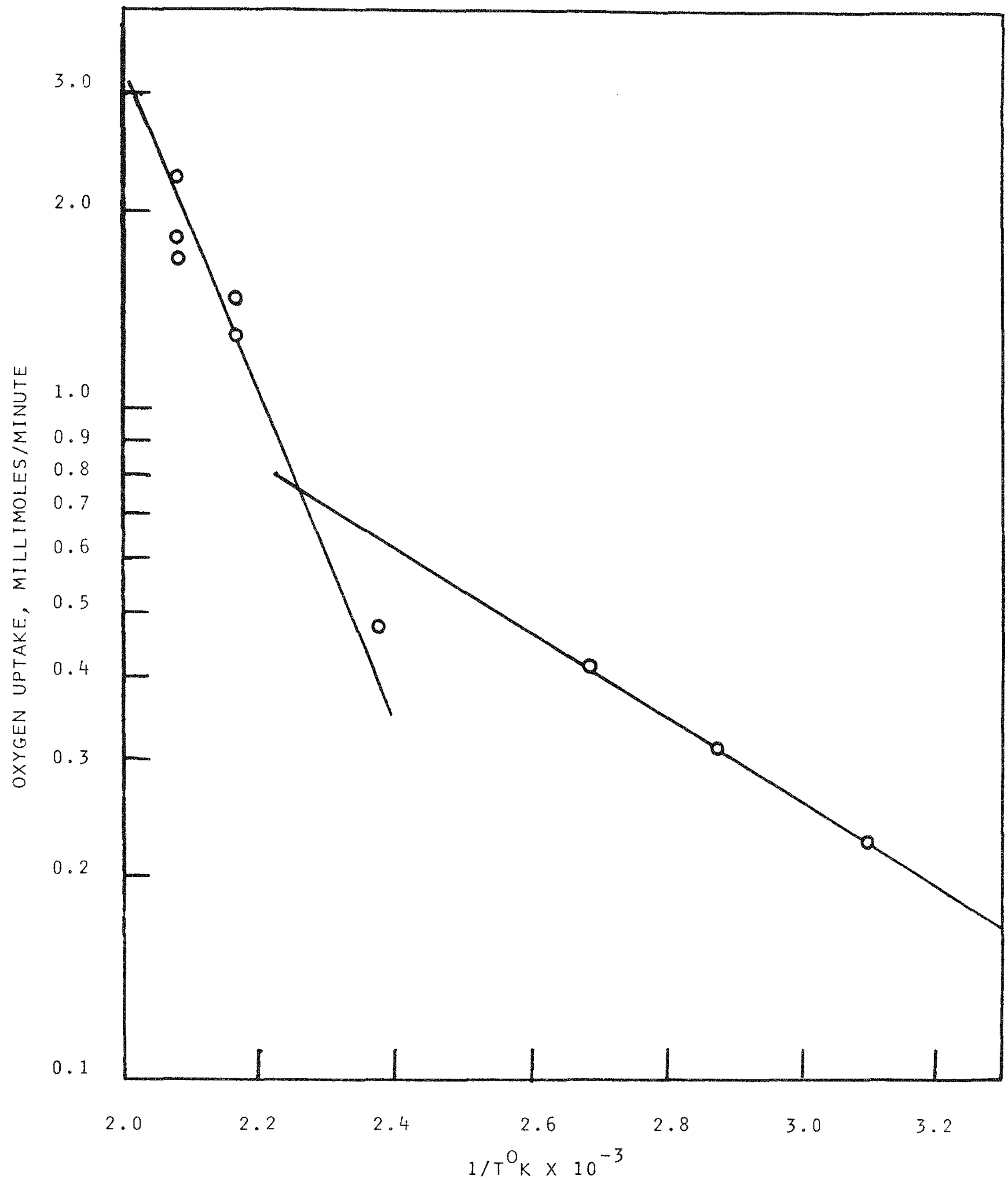

FIGURE 4-18 TEMPERATURE ASCENT FOR BELLE AYR 84+ MESH COAL NON-ISOTHERMALLY OXIDIZED IN AIR

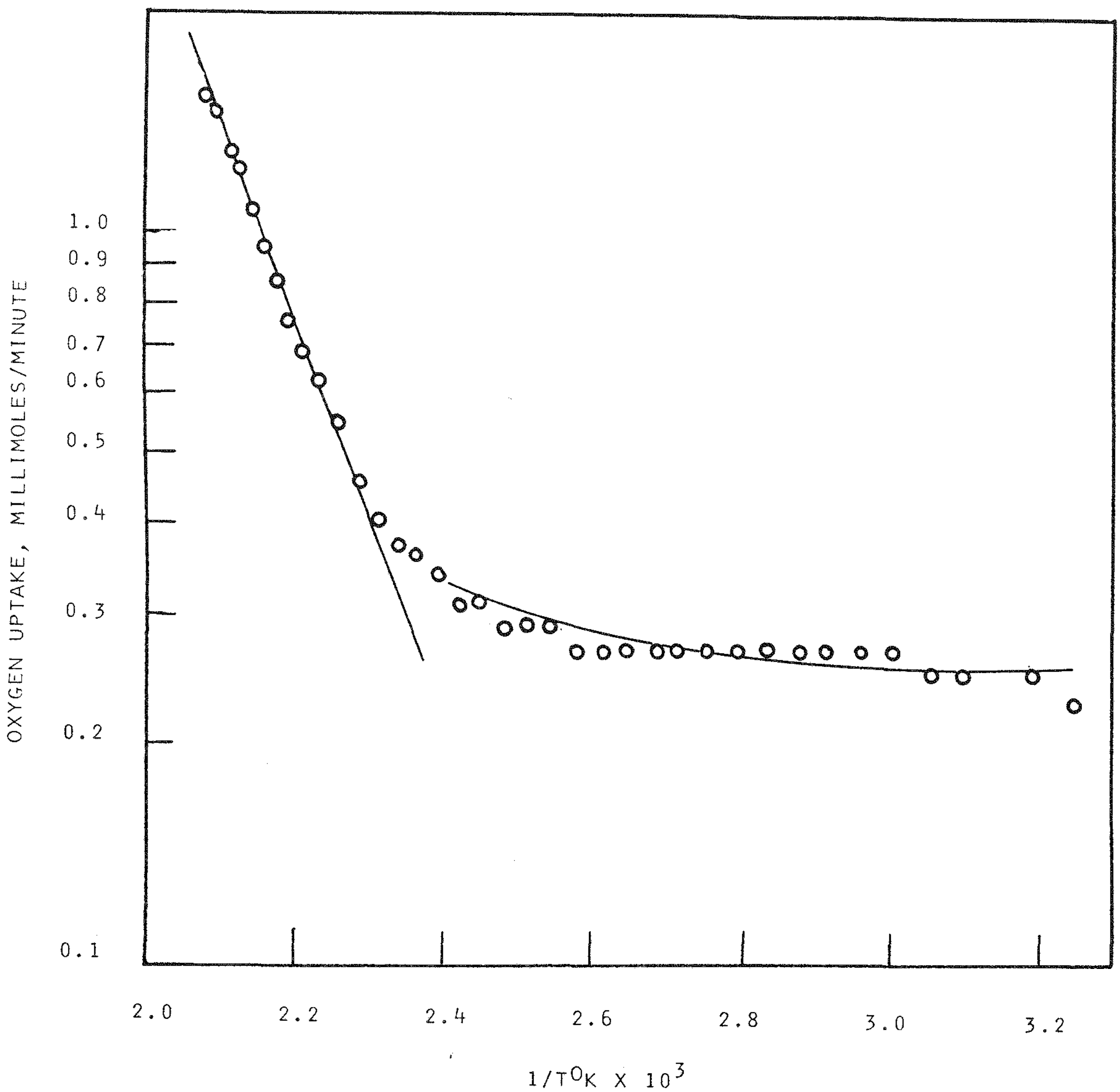

FIGURE 4-19 TEMPERATURE DESCENT FOR BELLE AYR 84^{+} MESH COAL NON-ISOTHERMALLY OXIDIZED IN AIR

observed that the finer coal oxidized 45-60% faster during the ascent over the entire range, although the slopes were similar.

Similar non-isothermal runs were made with Powhatan coal. The first non-isothermal run (124) was made with 140x230 mesh coal, as shown in Figures 4-18 and 4-19. As with Belle Ayr coal, during the temperature ascent there were two phases of temperature-dependent oxidation, with the second starting at about 130°C. On the descent, the slope and rate above 130°C was about equal to the ascent, but below 230°C there was essentially no temperature dependence. Comparing Powhatan to Belle Ayr in the high-temperature ascent region (Table 4-11), the apparent activation energy appears to be the same for both coals.

Discussion of Mechanisms

Observation of multiple "phases" during the oxidation of coal has been previously reported as discussed in the Literature Search (Appendix). The nature of oxidation phases varies with the rank and type of coal, ash content, temperature range, particle size and mode of oxidation. Considering our observations that there is a high reaction rate, short duration phase visible only above 140°C and a lower reaction rate declining phase, there are several possible mechanisms:

1. The coal contains two general types of reactive sites. The first type is more numerous but has less reactive (higher activation energy) "high temperature sites" (HTS) which do not become significant until the temperature exceeds 140°C (e.g., aromatic -C-H groups). The second type is more reactive, but has less numerous "low-temperature sites," LTS (e.g., methylene groups).

2. When the coal is initially exposed to oxygen-containing gas, a large surface area is exposed but thereafter oxygen must diffuse into a "shrinking" core of reactive coal.

3. When the coal is initially exposed at temperatures above 140°C, surface HTS react, giving a high oxygen consumption rate. However, these surface sites are consumed and the rate becomes limited by diffusion into the shrinking core. Eventually, as the "shell" of oxidized material penetrates to the center, the rate declines to zero.

4. When coal is simultaneously oxidized and dried, water vapor diffusing outward through the porous coal structure dilutes the oxygen partial pressure and effectively lowers reaction rate. As the coal becomes dried, the oxidation rate increases.

To provide further insight into the correct mechanism, two runs were made with 84+ mesh Belle Ayr coal which was pre-dried in nitrogen prior to oxidation in MG. In Run 123, the coal was pre-dried at 21°C overnight (16 hours), while in Run 122, the drying was at 100°C for 65 min. As shown in Figure 4-20, pre-drying at 100°C essentially removed temperature dependence in the region below 100°C, while above 100°C the temperature dependence was reduced to a level comparable to Run 118. In contrast, pre-drying in nitrogen at 21°C gave a single line (Figure 4-21) for the ascent with no break point in slope at 100°C.

These results suggest several interacting processes during the simultaneous oxidation and drying of Belle Ayr coal; supporting arguments are:

1. The "shrinking core" model is supported by somewhat higher oxidation rates at the same temperature during the ascent than during the descent when the radius of the reactive core is presumably smaller.

2. There appear to be three oxidation processes. At low temperatures, certain reactive "low temperature sites" with low activation energy, e.g., aliphatic methylene groups react. Also, at low temperatures there is another low rate, essentially temperature independent reaction, possibly physical adsorption. At higher temperatures, a second reaction of higher activation energy "high temperature sites," e.g., phenolic groups, becomes important. The low temperature sites may be destroyed or reacted when the coal is heated over 100°C. This is one possible explanation for the lack of temperature dependence in the low temperature region of the descent.

3. The water vapor dilution mechanism is supported by Run 123 where coal pre-dried at low temperature showed no slope break.

Pilot Fluid Bed Dryer (PFD) Experiments

Runs in the pilot fluid bed unit were primarily for preparation of large samples of partially oxidized, dried coals for subsequent continuous liquefaction studies. The gas/solid ratio and superficial gas velocity were considerably lower in the PFD than the laboratory unit (LFD). Also, the residence time of gas in the fluidized bed was much longer in the PFD, namely 43.4 seconds versus 1.3 seconds for the LFD. For these reasons the oxygen consumption in terms of oxygen reacted per unit volume of feed gas was much greater in the PFD. As seen in Table 4-8, generally less than 1 vol% oxygen was depleted in the LFD and constant P_{O_2} throughout the bed could be assumed. However, up to 50 vol% of the feed oxygen was consumed in the PFD and P_{O_2} varied greatly from the bottom to top of the fluid bed.

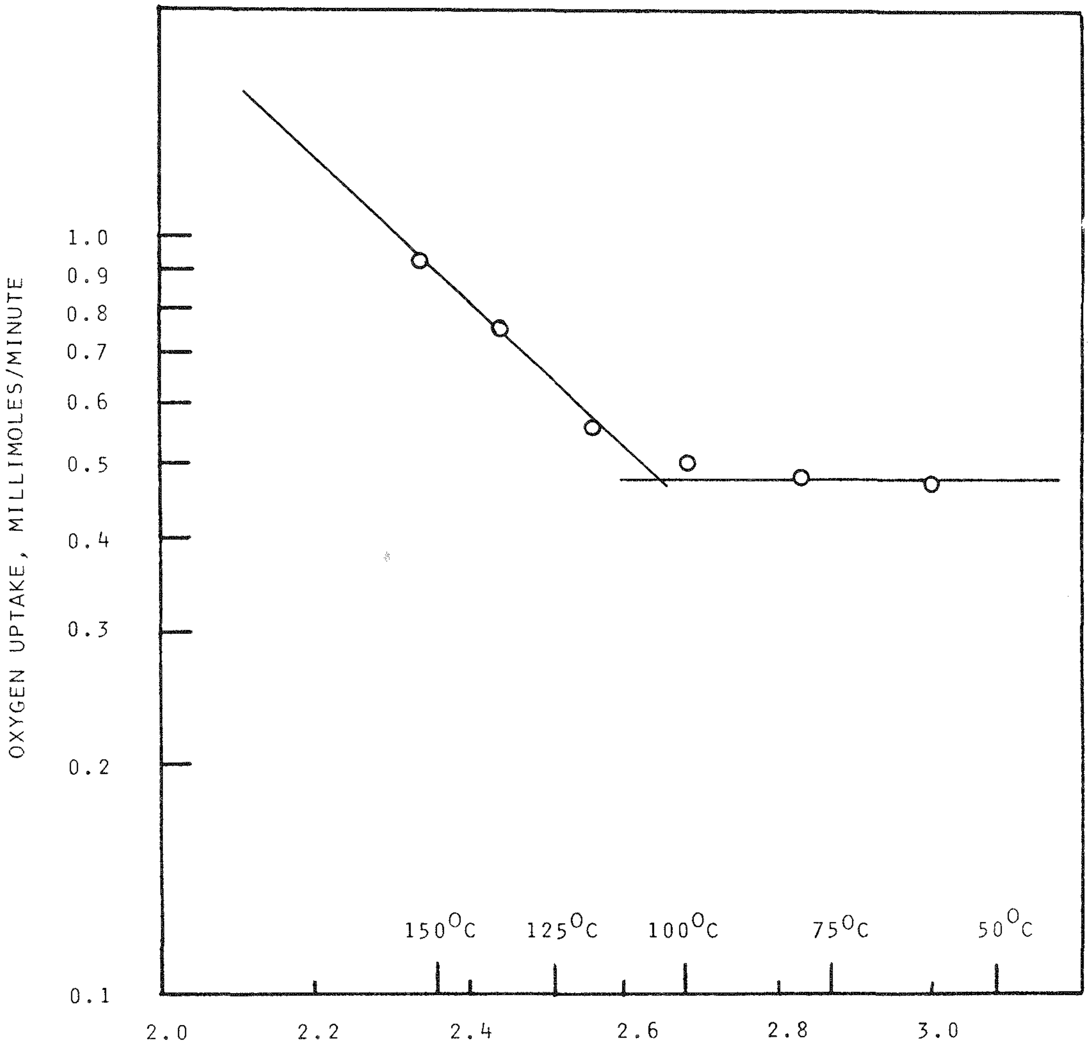

FIGURE 4-20 TEMPERATURE ASCENT FOR BELLE AYR 84+ COAL PREDRIED IN N_2 AT 100°C

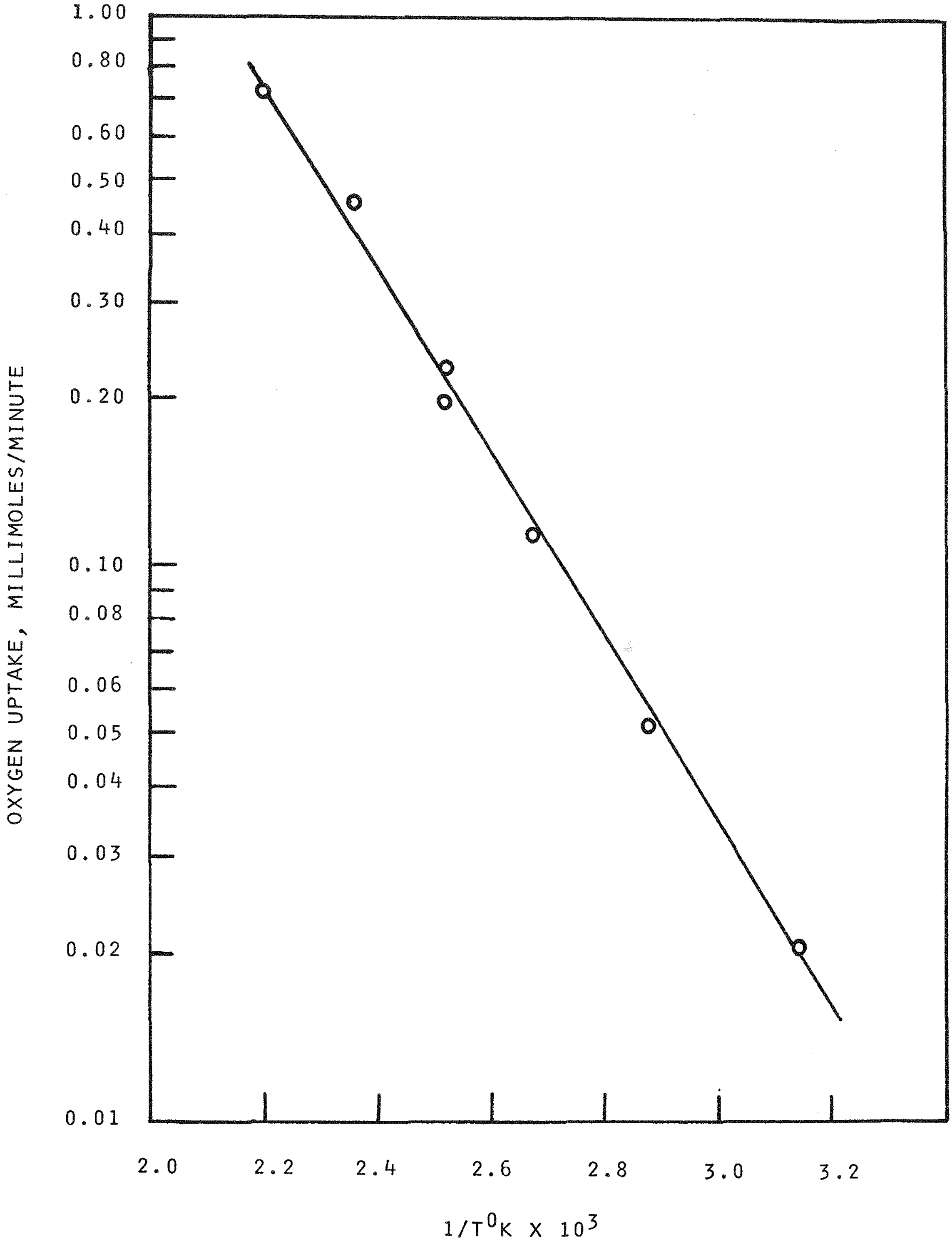

FIGURE 4-21 TEMPERATURE ASCENT FOR BELLE AYR 84+ MESH COAL OXIDIZED IN MIXED GAS AFTER 21°C N_2 PRE-DRYING

A total of 18 runs were made in the pilot unit, all with Belle Ayr coal and all but the first two with 84x230 mesh material, as shown in Table 4-12. (The coal used in Runs 11, 12 and 13 was later found to be contaminated with Powhatan coal, therefore oxidation results from these runs have been disregarded.)

Mixed gas (MG) was furnished by separately metering air and nitrogen at about a 1:1 ratio, then mixing prior to preheating. Therefore, the oxygen content of the MG varied somewhat between runs. Also, the oxygen consumption rate (OCR) in the PFD was determined by spot analysis, generally at 15 min intervals, rather than continuous analysis. Oxygen consumption rate for all PFD runs is shown in Table 4-13.

The following observations can be made concerning the drying/oxidation runs in the pilot unit.

1. The initial oxidation rate was high and then it decreased or reached some plateau value. (In those runs where oxygen consumption again rose near the end of the run, e.g., Runs 10 and 11, this was due to a temperature excursion above the desired temperature.)

2. All but one of the runs were at 100°C, and at this temperature the "steady state" OCR is about 30-35 g O_2/hr. At this rate, for a 7000 g coal charge (5075 g dry), if <u>all</u> the oxygen consumed was adsorbed by the coal, the weight gain would be about 0.6%. We may assume that oxygen removed from the gas but not reporting to CO_2 or CO remains adsorbed to the coal, but this neglects formation of H_2O, SO_2 or any other volatile oxygen compounds. Actual oxygen content of the coal cannot be verified by an analysis of oxygen on the coal, since the small percentage change is beyond the precision of the analysis.

3. Data for CO_2 production (Run 3, Table 4-13) show that about 2.5% of the total OCR is reporting to CO_2. Actually, heating in nitrogen alone at 100°C releases CO_2. As observed in Run 3 (Table 4-14), when the feed gas was switched from nitrogen (used during heatup) to MG, there was a <u>drop</u> in CO_2 production, and CO, which was small during nitrogen feed, disappeared entirely. Runs 11 and 12 show transient spikes with CO_2 production as high as 66 mg/min, but even this is less than 9% of oxygen uptake and the coal used in these runs inadvertently was contaminated and of unknown properties. We can conclude that at 100°C in MG, little or none of the oxygen removed from the gas feed is reporting to CO_2 or CO. At 50°C (Run 5), CO_2 production in MG was also about the same as at 50°C in nitrogen.

4. Pre-drying the coal in nitrogen at 100°C prior to exposure to MG (Runs 13, 17) appeared to increase considerably the initial (start-of-run) rate of oxygen consumption, but the limited data indicate that the steady-state rate is comparable to that

Table 4-12

COAL OXIDATION AND DRYING IN PILOT-SCALE FLUID-BED UNIT (PFD)

	BA Coal	Heatup	Isothermal				Kg Coal		Final
Run	Mesh	N_2,CFH	Temp°C	Min	Gas	CFH	Fed	Recov	Mois.wt%
1	84+	115.6	100+	55	MG[d]	56.5	7.0	4.0	1.4
2	84x145	105.9	35	540	N_2	37.0	7.0	3.758	17.6
3[a]	84x230	31.6	100	60	MG	21.8	7.0	(a)	(a)
4	84x230	31.6	100	60	MG	21.8	7.0	4.752	5.8
5	84x230	31.6	50	60	N_2	21.8	7.0	6.478	26.7
6	84x230	31.6	100	60	N_2	31.6	10.0	7.1140	7.7
7	84x230	31.6	100	60	N_2	31.6	7.0	4.544	3.5
8	84x230	31.6	100	60	N_2	31.6	7.0	3.598	n.a.
9	84x230	61.6	100	60	N_2	61.6	7.0	4.334	4.4
10	84x230	61.6	100	60	MG	43.1	7.0	5.005	4.8
11[b]	84x230	61.6	100	60	MG	46.2	7.0	4.698	1.8
12[b]	84x230	61.6	100	60	MG	36.5	7.0	n.a.	1.8
13[b]	84x230	61.6	100	45/15[c]	N_2/MG	45.7	7.0	n.a.	n.a.
14	84x230	61.6	100	60	MG	41.9	7.0	4.625	5.2
15	84x230	61.6	100	60	MG	41.9	7.0	4.595	5.3
16	84x230	61.6	100	60	MG	41.9	7.0	4.862	6.0
17	84x230	61.6	100	45/15	N_2/MG	41.9	7.0	5.088	5.2
18	84x230	61.6	100	30/30	N_2/MG	41.9	7.0	5.210	7.5

(a) Aborted due to cracked reactor

(b) Coal charge contaminated; not pure Belle Ayr

(c) Time exposed to N_2 (at 61.6 CFH) /MG (at CFH indicated)

(d) Mixed N_2 plus approx. 10.8% oxygen

Table 4-13

OXYGEN CONSUMPTION IN PILOT FLUID BED DRYER (PFD)

Run	Time: Min. (Isothermal MG Feed) 0	15	30	45	60
10					
PØD[a]	39.13	27.54	24.64	26.09	27.54
ØCR[b]	45.08	31.73	28.38	30.06	31.73
11[c]					
PØD	--	25.0	18.0	20.0	24.75
ØCR	--	44.18	31.8	35.34	44.17
12[c]					
PØD	--	--	--	--	15.15
ØCR	--	--	--	--	20.94
13[c]					
PØD	--	58.0	x	x	x
ØCR	--	90.29	x	x	x
14					
PØD	24.77	18.35	9.17	8.26	7.34
ØCR	4.27	32.05	16.02	14.43	12.83
15					
PØD	27.52	--	17.43	17.43	17.43
ØCR	48.07	--	30.45	30.45	30.45

Run	Time: Min. (Isothermal MG Feed) 0	15	30	45	60
16					
PØD	20.18	20.18	20.18	20.18	20.18
ØCR	35.26	35.26	35.26	35.26	35.26
17					
PØD	40.36	22.94	x	x	x
ØCR	70.5	40.04	x	x	x
18					
PØD	17.57	2.70	2.70	x	x
ØCR	20.83	3.2	3.2	x	x
3					
PØD	25.5	d	x	x	x
ØCR	21.25	d	x	x	x

Run	Time: Min. (Isothermal) 10	20	30	40	50	60
5						
PØD	38.25	8.61	3.28	0.41	0	0.68
ØCR	23.34	5.25	2.0	0.25	0	0.41

All runs with nominal 10-11 vol% O_2 in feed MG, but actual concentration used to calculate PØD and ØCR.
[a]PØD = Percent oxygen depletion [(Feed O_2 - Vent O_2)/Feed O_2] x 100; [b]ØCR = Oxygen consumption rate, g O_2/Hr; [c]Coal charge contaminated, not pure Belle Ayr; [d]Reactor cracked.

without pre-drying. The stimulus of pre-drying on oxygen consumption is the likely result of the removal of water vapor which dilutes oxygen partial pressure. Moisture analysis of the coal shows that about half of the original moisture content (27.5 wt%) is removed during the warm-up period, with the balance coming off as follows (data from Run 15):

Isothermal Time:	0*	10	20	30	40	60 min
% H_2O :	9.4	9.0	7.5	6.8	5.6	5.3

*At start of isothermal period, coal has been nitrogen-blown for 75 min during warm-up from 21° to 100°C.

5. The sole run at 50°C (No. 5) showed some initial oxygen uptake but this decayed to essentially zero within 30 minutes (Table 4-13). Integrating OCR over the run, about 7.4 g of O_2 were consumed, only 0.17 g going to CO_2. This resulted in an oxygen increase on the coal of less than 0.15 wt%.

In conclusion, behavior in the pilot unit was similar to that of the laboratory unit, but due to the lower gas to solid ratio and related factors, oxygen uptake by the coal at a given temperature was lower, e.g., at 100°C about 0.6% in the PFD versus 2.2 wt% in the LFD.

Table 4-14

CARBON OXIDE PRODUCTION IN PILOT FLUID BED DRYER (PFD)

Run 3

Gas	N_2	N_2	N_2	MG	MG
Temp°C	50	75	100	100	100
Time, min	--	--	--	0	15
CO_2, ppm	208	529	576	449	(a)
CO_2, mg/min	3.97	10.10	11.00	8.58	--
CO, ppm	22	31	35	n.d.	(a)
CO, mg/min	0.27	0.38	0.43	--	--

Run 7

Gas	N_2	N_2	N_2	N_2	N_2
Temp°C	51	75	100	100	100
Time, min	--	--	0	20	60
CO_2, ppm	312	290	277	165	238
CO_2, mg/min	8.64	8.03	7.67	4.57	6.59
CO, ppm	n.d.	n.d.	n.d.	n.d.	n.d.
CO, mg/min	--	--	--	--	--

Run 5

Gas	MG	MG	MG	MG	MG	MG
Temp°C	50	50	50	50	50	50
Time, min	10	20	30	40	50	60
CO_2, ppm	195	198	227	235	198	227
CO_2, mg/min	3.72	3.78	4.34	4.49	3.78	4.34
CO, ppm	n.d.	n.d.	28	55	44	36
CO, mg/min	--	--	0.34	0.67	0.53	0.44

Run 11

Gas	MG	MG	MG	MG
Temp°C	--	100	100	100
Time, min	CHG[b]	25	45	60
CO_2, ppm	235	238	1622	326
CO_2, mg/min	9.52	9.64	65.68	13.20
CO, ppm	n.d.	n.d.	334	n.d.
CO, mg/min	--	--	8.61	--

Run 6

Gas	N_2	N_2	N_2	N_2	N_2
Temp°C	53	75	100	100	100
Time, min	--	--	0	20	60
CO_2, ppm	267	364	310	290	260
CO_2, mg/min	7.40	10.08	8.59	8.03	7.20
CO, ppm	n.d.	n.d.	n.d.	n.d.	15
CO, mg/min	--	--	--	--	0.26

Run 12

Gas	MG	MG	MG
Temp°C	--	100	100
Time, min	CHG[b]	20	60
CO_2, ppm	170	170	1181
CO_2, mg/min	5.44	5.44	37.78
CO, ppm	n.d.	n.d.	323
CO, mg/min	--	--	6.58

(a) Cracked reactor; n.d. = not detected; (b) Charge gas, which should contain zero CO_2.

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Section 5

OPTICAL, PHYSICAL AND CHEMICAL CHARACTERIZATION OF SOLID SAMPLES

Effort was undertaken at both GR&DC and Pennsylvania State University to characterize the various solids handled in this project, namely: feed coals, dried and oxidized coal samples, and residues recovered from the bench-scale liquefaction experiments. In addition to optical characterization, chemical analyses of the samples were done with emphasis on oxygen analysis, including direct oxygen determination and FTIR analysis to observe functional group changes. In regard to direct oxygen analysis, Dr. D. Schlyer of Brookhaven National Laboratories aided by carrying out CPAA analysis of selected samples.

(It is noted that this EPRI project was intended to be a scanning type of effort. There are obvious areas in which questions remain, particularly with respect to oxygen analysis and FTIR work. Some of this work will continue with GR&DC support and will be reported in the literature.)

Physical and Optical Characterization of Oxidized Samples

Two size fractions (40x70 mesh and 200x270 mesh) of both Belle Ayr and Powhatan coal samples were selectively oxidized in air at 175°C for 2, 5, and 15 hour periods. Products of these experiments were prepared for examination in reflected light following procedures which have been previously reported.

All samples were examined for the detection of recognizable oxidation features including:

1. presence of oxidation rims
2. discoloration of particle boundaries and cracks
3. irregular and 'blanket-stitch' microfractures
4. irregular relief
5. discoloration of granular micrinite and clays
6. alteration of pyrite to sulfate minerals
7. overall increase in vitrinite reflectance

Samples of Powhatan coal, oxidized for 2, 5, and 15 hour periods showed no strong evidence of the first 5 characteristics noted above. In the 15 hour sample, there was some marginal discoloration of particle boundaries; however, it was so slight as to not generally be recognized as a result of oxidation. The 15 hour sample also showed some evidence of alteration of the pyrite, although once again it was not as clear as is generally seen in naturally oxidized coals.

A systematic reflectance study was carried out on these samples to determine the increase in reflectance of the vitrinite macerals. Standard reflectance analyses were performed on both size fractions of the progressively oxidized samples. The results, shown in Figure 5-1, indicate that significant increase in vitrinite reflectance occurred with only 2 hours of oxidation treatment. Further oxidation treatment resulted in a more gradual increase in reflectance. The change in reflectance for both size fractions was constant, within the error on the analysis.

Samples of Belle Ayr coal also exhibited only minor changes in their optical properties as a result of the oxidation treatment. Particles in the 15 hour treatment sample showed minor discolorations around the particle margins. The small amount of pyrite in this sample showed no signs of oxidation to sulfate minerals.

A reflectance study of the samples of Belle Ayr coal showed increase of ulminite reflectance with increasing time of treatment. Figure 5-1 shows this trend to be constant, with no major inflection points. It was also evident that the ulminite reflectance of the two size fractions increased in a similar manner. In summary, the artificially oxidized samples showed few or none of the features normally associated with oxidized coals. Both samples did, however, show a progressive increase in the vitrinite (or ulminite) reflectance with increasing time of oxidation. Although the increase was measureable, it was not of a magnitude that would be detected by visual inspection.

Three coal samples which were prepared at GR&DC were also examined for optical evidence of the oxidation treatment. Two of these samples were treated in a mixed gas of 90% nitrogen, 10% oxygen at 100°C for 1.0 hr (samples PFD-14 to 16 and PFD-17). The third (PFD-18) was first dried in nitrogen for 15 minutes and then in mixed gas for 45 minutes.

None of these samples showed clear evidence of the oxidation treatment. PFD-14 to 16 and PFD-17 showed some brownish rims around the margin of the particles, but these were not very pronounced. PFD-18 showed no evidence of the oxidation treatment.

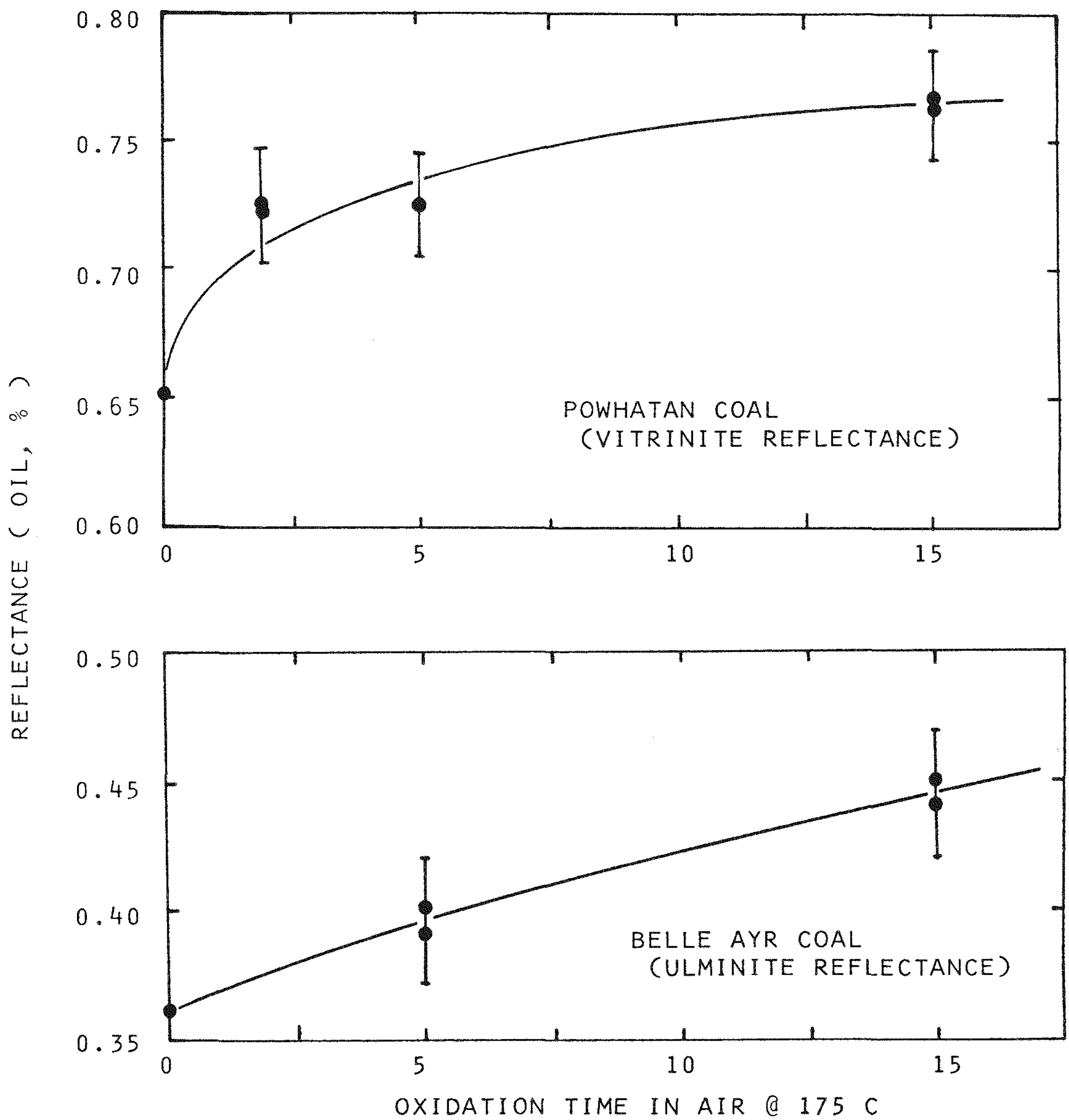


FIGURE 5-1 EFFECT OF OXIDATION TIME ON REFLECTANCE

Sample PFD-17 was subjected to reflectance analysis to test if its ulminite reflectance had increased due to the treatment procedures. The analysis yielded an ulminite reflectance of 0.38%, which is not significantly higher than the untreated coal (R_0 = 0.36%).

For reference, two microphotographs of as-received Belle Ayr coal are shown in Figure 5-2. Two photographs of highly oxidized Belle Ayr coal are shown in Figure 5-3 to demonstrate a high level of surface oxidation. These latter coal samples were recovered from air drying runs that had temperature exotherms in excess of 200°C.

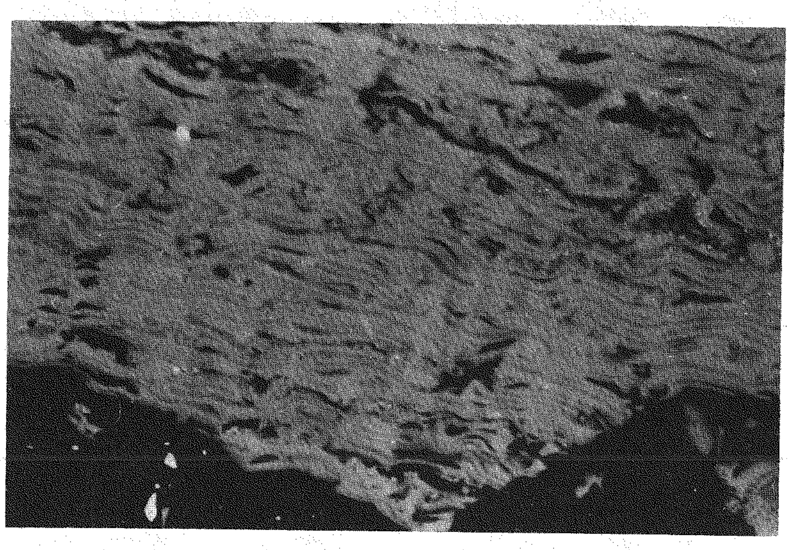
Analysis of Oxygen in the Treated Coal Samples

There are several points necessarily considered in measuring the level of oxygen concentration change during coal drying with or without partial oxidation. First, it is difficult to accurately measure the oxygen content of coal, particularly organically combined oxygen. It is sufficient here to state that oxygen is typically given by difference, namely 100% minus others. In addition, oxygen analysis is carried out using the "Unterzaucher method," "fast neutron activation analysis" (FNAA), and "charged particle activation analysis" (CPAA). These methods are discussed in Section 8.

Secondly, during the drying of coal, it is known that of the total oxygen removed from the feed gas stream, part will remain physically or chemically combined with the coal while part will be regasified as carbon oxides, water, sulfur dioxide, and other volatile oxygen species. The proportion of oxygen reporting to these two modes will vary with temperature, rank of coal and other reaction variables.

Finally, a complicating factor is that the coal loses moisture and the oxygen contained therein. Since the analytical techniques measure moisture oxygen as well as organic and often mineral oxygen, it is necessary to calculate oxygen on a dry basis, preferably a dry mineral matter free basis. To get an accurate value, it is necessary to know the precise moisture content of the sample and to ensure that no change in moisture content occurs up to the time of oxygen analysis. Considering that 100 grams of Belle Ayr raw coal contains 24.4 g oxygen as moisture while only 0.5 to 3 g were generally adsorbed during oxidation runs, the difficulty of precise analysis is obvious.

The maximum oxygen fixed in the coal can be set by the gas material balance, specifically, total oxygen consumed minus total oxygen released as carbon oxides.



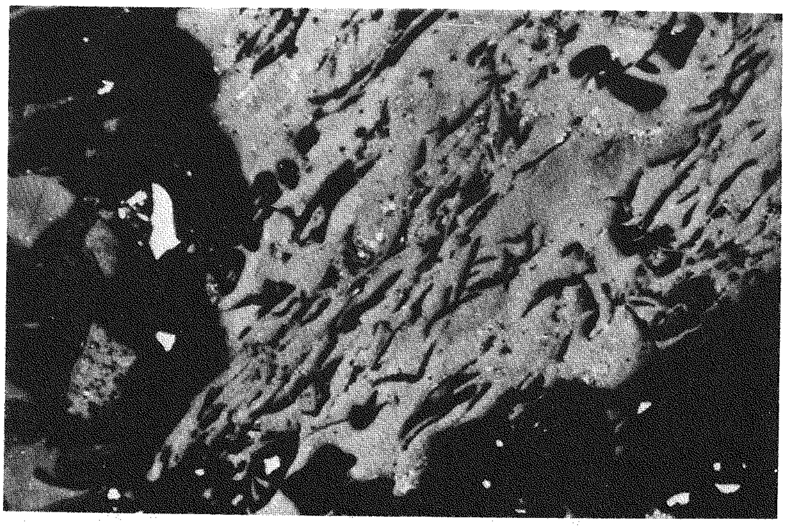


FIGURE 5-2 PHOTOMICROGRAPHS OF AS-RECEIVED BELLE AYR COAL

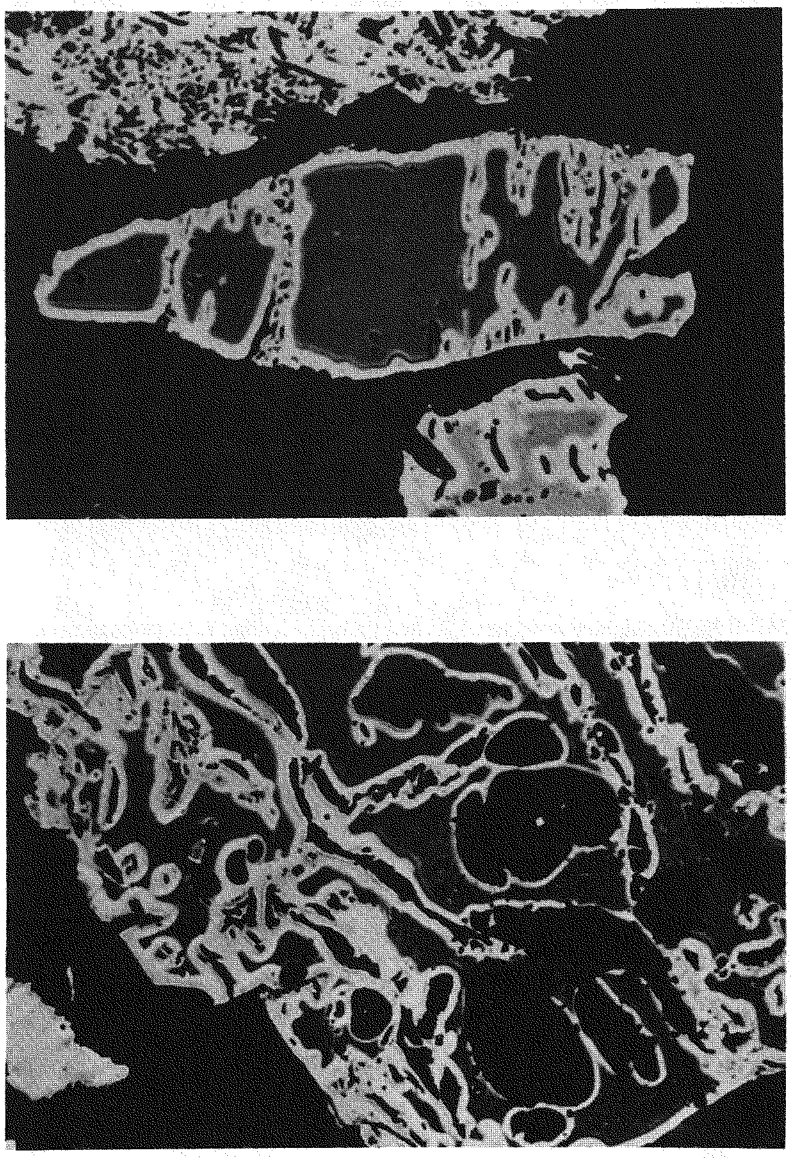


FIGURE 5-3 PHOTOMICROGRAPHS OF TWO SAMPLES OF HIGHLY OXIDIZED BELLE AYR COAL (200^{+} ^{o}C)

Change in coal oxygen content can exceed this value only if non-oxygenated compounds are pyrolyzed off the coal; this is highly unlikely at the temperatures of these runs (50-200°C).

Initially, selected samples from experiments using the laboratory fluid bed dryer (Runs 12-56) were analyzed for oxygen by the Unterzaucher method, with simultaneous determination of the carbon and hydrogen. A plot (Fig. 5-4) of total weight percent carbon (as-is basis) in the coal vs. wt% moisture gives a straight line with an intercept of 69.77% C, i.e., this is the carbon content of dry Belle Ayr coal. Likewise, a plot of hydrogen vs. moisture reveals the dry Belle Ayr hydrogen content to be 5.29 wt%.

However, a plot (Fig. 5-5) of total wt% oxygen vs. moisture gives considerable scatter. Even those samples dried with nitrogen were scattered, and no clear trend for those dried in mixed gas (MG) or air was obvious. A linear regression line gives a zero moisture intercept of 22%, and we can assume this approximates the dry coal oxygen content prior to any oxidation. It appears that some random moisture change occurred during sample handling or during the initial nitrogen purge of the Unterzaucher instrument.

In summary, it appears that, based upon the Unterzaucher results, the raw moist (27.5%) Belle Ayr coal contained 40.4 wt% oxygen while nitrogen-dried coal contained about 22%.

A second series of samples, selected from LFD Runs 65-92, were analyzed for oxygen by three different methods; the results are given in Table 5-1. As before, the Unterzaucher method gave considerable scatter with no apparent pattern. Since only three samples were run by FNAA, the reliability of this method is uncertain. The CPAA results appear to correlate best with the gas mass balance and oxidation conditions.

Considering that the raw coal contains approximately 22 wt% O_2 (dry basis), the most severe oxidation conditions gave a net increase of less than 2.5 wt%. As shown in Table 5-2, at a given exposure time, higher temperature gave more oxygen uptake and at a given temperature, increased time also gave more oxidation, but the differences were small.

Another group of samples was analyzed for oxygen by CPAA, including a repeat of ten previous samples. The results of the repeat runs indicated an oxygen content 1% lower than the original values, a possible consequence of moisture loss on storage. The trends were the same, namely increased oxygen content with

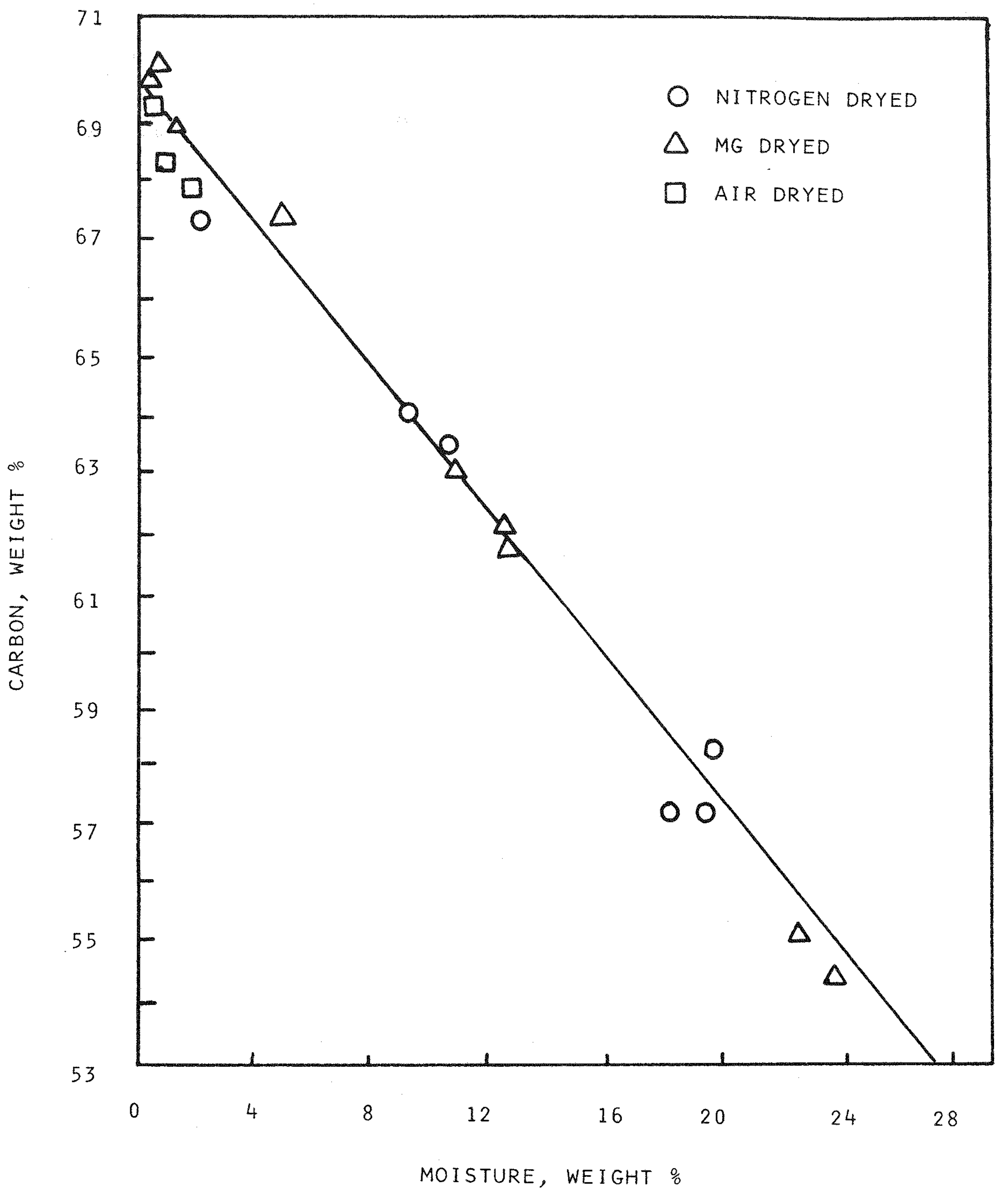


FIGURE 5-4 CARBON VERSUS MOISTURE CONTENT OF SELECTED SAMPLES (LFD RUNS 12 THROUGH 56)

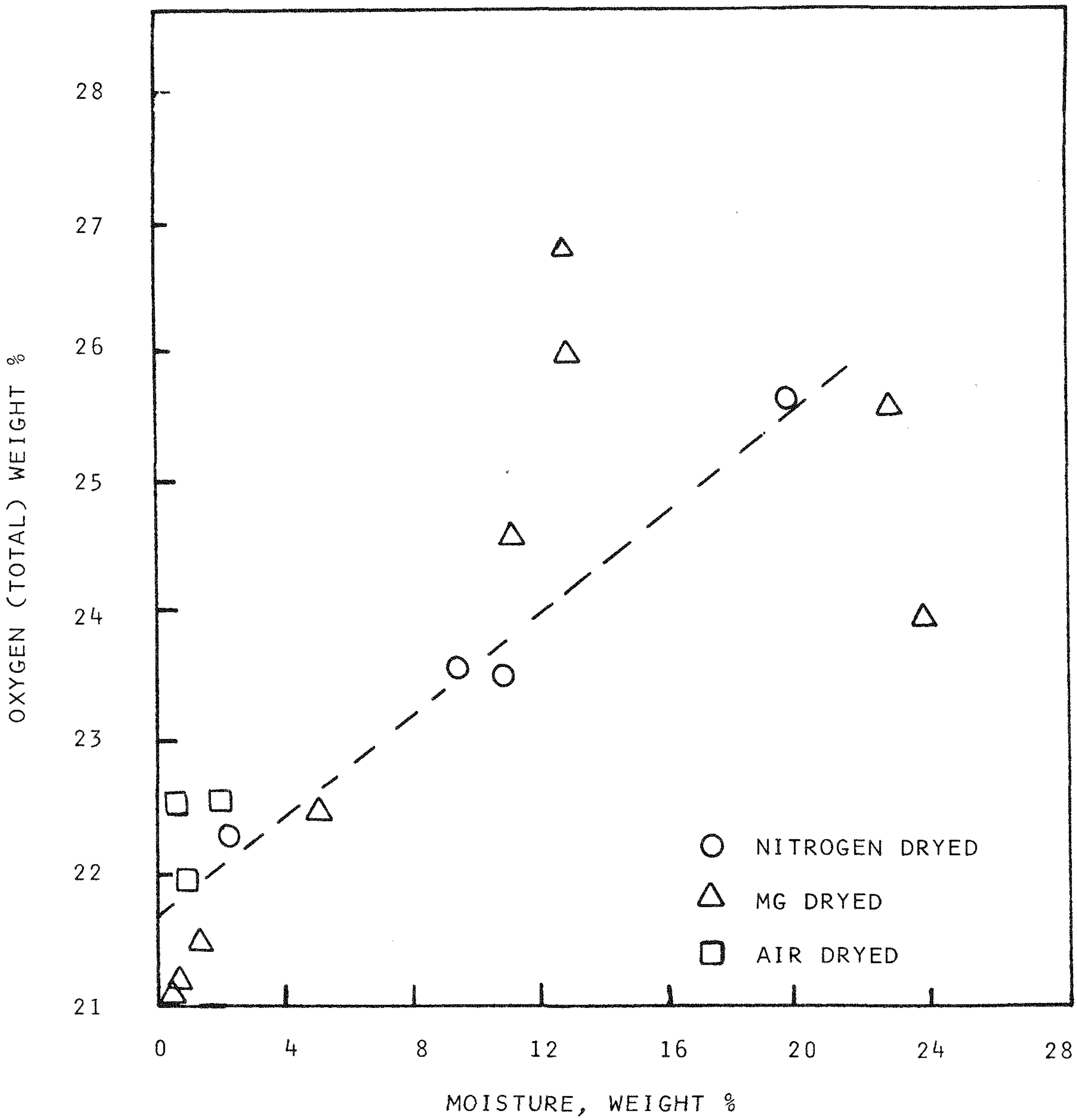


FIGURE 5-5 OXYGEN VERSUS MOISTURE CONTENT OF SELECTED SAMPLES (LFD RUNS 12 THROUGH 56)

Table 5-1

OXYGEN ANALYSIS OF SELECTED SAMPLES, LFD RUNS 65-92

Run		Moisture wt%	Unterzaucher AS-IS	Dry[c]	CPAA[d] AS-IS	Dry[c]	FNAA[e] AS-IS	Dry[c]
Ba-84+[a]		27.5	30.94	8.97	41.8	23.94	34.4	13.74
BA-VOD[b]		0.0	20.2	20.2	22.6	22.6	25.30	25.30
67	3.0	23.33	21.30	24.9	23.50			
86	14.8	26.6	15.78					
80	10.5	24.0	16.39	29.4	22.42			
81	16.8	26.6	14.02					
82	3.7	22.85	20.31					
83	6.4	23.13	18.63					
65	1.2	21.90	21.09	24.6	23.82			
66	1.2	22.06	21.25	24.3	23.52	24.1	23.31	
92	5.3	21.85	18.10	22.5	18.78			
69	1.3	20.95	20.06					
70	1.2	21.32	20.50					
91	1.0	21.40	20.72					
71	0.2	21.00	20.86	23.2	23.07			
72	1.1	20.90	20.14					
73	0.0	22.35	22.35	23.0	23.0			
90	1.2	20.72	19.89					
75	1.4	22.32	21.37	21.37	25.4	24.5		

(a) Raw, undried 84+ mesh Belle Ayr Coal

(b) Vacuum oven dried (N_2 purge) to zero moisture.

(c) Dry Basis Oxygen % = [Gm Non-moisture O_2/100 Gm Bone Dry Coal] x 100

(d) Charged particle activation analysis

(e) Fast-neutron activation analysis

Table 5-2

CORRELATION OF CPAA COAL OXYGEN ANALYSES WITH OXIDATION CONDITIONS

1. SERIES OF 100°C RUNS

Run	Gas	Time	Oxygen (wt%)(1)
92	A	15	18.78
65	MG	30	23.82
66	MG	120	23.52

2. SERIES OF 30 MIN. RUNS

Run	Gas	Temp.(°C)	Oxygen (wt%)(1)
80	A	50	22.42
65	MG	100	23.83
75	A	200	24.50

3. SERIES OF 120 MIN. RUNS

Run	Gas	Time	Oxygen (wt%)(1)
67	MG	50	22.92
66	MG	100	23.52
73	A	150	23.0

(1)Oxygen on a dry basis, correcting for moisture content.

increasing time and temperature, but the precision is not adequate for quantification.

It is obvious from the above results that additional work must be done to quantify oxygen uptake during the drying of subbituminous coal. We have scheduled such a project to be carried out in cooperation with Brookhaven National Laboratories.

Discussion of Material Balance

Oxygen removed from the gas can be fixed to the coal or can report to volative oxygen compounds in the vent gas (e.g., CO_2, CO, H_2O, SO_2). As shown in Table 5-3, selected samples of the vent gas from LFD runs indicate that CO is negligible except where combustion took place (200°C in MG, 150°C in air). Likewise, CO_2 is also low except in these runs.

Oxygen consumed from the gas can be calculated by integrating under the oxygen vs. time recorder trace. An approximation to this can be calculated by averaging SOR and EOR consumption rates at the start and end of the runs and multiplying by run time. This neglects oxygen consumed during the brief initial period in runs above 140°C. Table 5-3 indicates that in the low temperature runs very little (less than 10%) of the consumed oxygen is going to carbon oxides; while above 150°C, 25-65% of the consumed oxygen may be regasified. These latter percentages may be high since some carbon oxides are released even when the coal is heated at high temperatures in nitrogen. Considering that the original coal charge to the LFD contains 15.18 g carbon, even under the most severe conditions, 1% or less is gasified in a one-hour run.

FTIR Analysis

Selected oxidized coal samples from both the LFD and PFD units were analyzed by FTIR to determine oxygen functionality changes. The samples were pressed in KBr pellets (5 mg coal plus 50 mg KBr) and scanned in a Nicolet No. 7199 instrument. By internal computer calculation, difference scans were prepared by minimizing spectral features in either the aromatic (1570-1650) or C-H (2800-3000) stretching regions. The following table lists the product samples used for FTIR analysis:

Table 5-3

CARBON OXIDE EVOLUTION IN THE LABORATORY-SCALE FLUID BED DRYER

Run	Temp. (°C)	Gas	Min.	PPM Vol. CO	PPM Vol. CO_2	Mg/Hr CO	Mg/Hr CO_2	Mg/Run[2] CO	Mg/Run[2] CO_2	Total Mg[2] O2 In CO_x	Mg O_2[c] Uptake
70	150	MG	30	75	307	13	81	6	40	33	494
72	150	A	60	120	277	20	73	20	73	65	1033
73	150	A	120	79	1150	13	303	7	606	445	1018
74	200	MG	30	812	2195	136	578	68	289	249	1511
79	125	N_2	10	105	1795	18	473	3	79	59	-
81	50	A	30	<5	84	<1	22	<1	11	8	220
82	50	A	120	<5	53	<1	14	<1	28	20	1324
83	50	A	60	<5	322	<1	85	<1	85	62	751
91	150	MG	5	<5	15	<1	4	<1	0.3	<1	-
96	50	MG	10	<5	69	<1	18	<1	3	2	47
100	75	MG	60	<5	<5	<1	<1	<1	<1	<1	-
MG	-	-	-	<5	<5	<1	<1	<1	<1	-	-
MG	-	-	-	<5	<5	<1	<1	<1	-	-	

(1) All runs were made with 140x230 Belle Ayr, except 84+ mesh in Run 96 and 140x230 mesh Powhatan in Run 120

(2) Assuming constant rate over entire run.

(3) Total mg uptake over run, calculated from average rate in declining rate period.

Table 5-4

LIST OF FTIR SAMPLE SCANS

Run No.	Drying Conditions Gas	Temp (°C)	Time (min.)	Final Moisture (wt%)
LFD Series				
-55	N_2	50	60	10.7
-61	N_2	100	120	0.4
-91	MG	160	5	1.0
-84	Air	50	10	22.7
-75	Air	205	30	1.4
-88	Air	51	120	3.9
PFD Series				
	Raw	Coal		27.5
-9	N_2	100	60	4.4
-5	MG	50	60	26.7
-14 to 16	MG	100	60	5.5

A greater consumption of oxygen was observed in the LFD experiments; therefore these results will be discussed first. A typical FTIR spectra (Run LFD 61) is shown in Figure 5-6. (It is noted that Professor P. Painter of Pennsylvania State Univ. pointed out that the coal particles perhaps should have been ground finer; this is being taken into account in future works.) In addition, subsequent figures show differential spectra created by subtracting the spectrum of one run from another to aid in pinpointing areas of change.

The FTIR results indicate that the heating of Belle Ayr coal to high temperatures in nitrogen destroys certain oxygen functionality, probably by decarboxylation, as shown in Figure 5-7, where an absorbance decrease is seen in the 1750 nm region (Run 61 minus 55). Heating the coal in oxygen-containing gas adds oxygen functionality, but different from that removed. Reference is made to Figure 5-8a and b where Run 55 product is subtracted from products 91 and 84, respectively. There is a marked absorbance increase in the 1050-1150 nm region, possibly due to aliphatic ethers.

If we consider the mechanism postulated for coal oxidation, namely:

carbonyl

-CH- ---> peroxides ---> hydroxyl ---> carboxyl ---> CO_2

(methylene) (phenolic OH) (humic acids)

(aromatic rings)

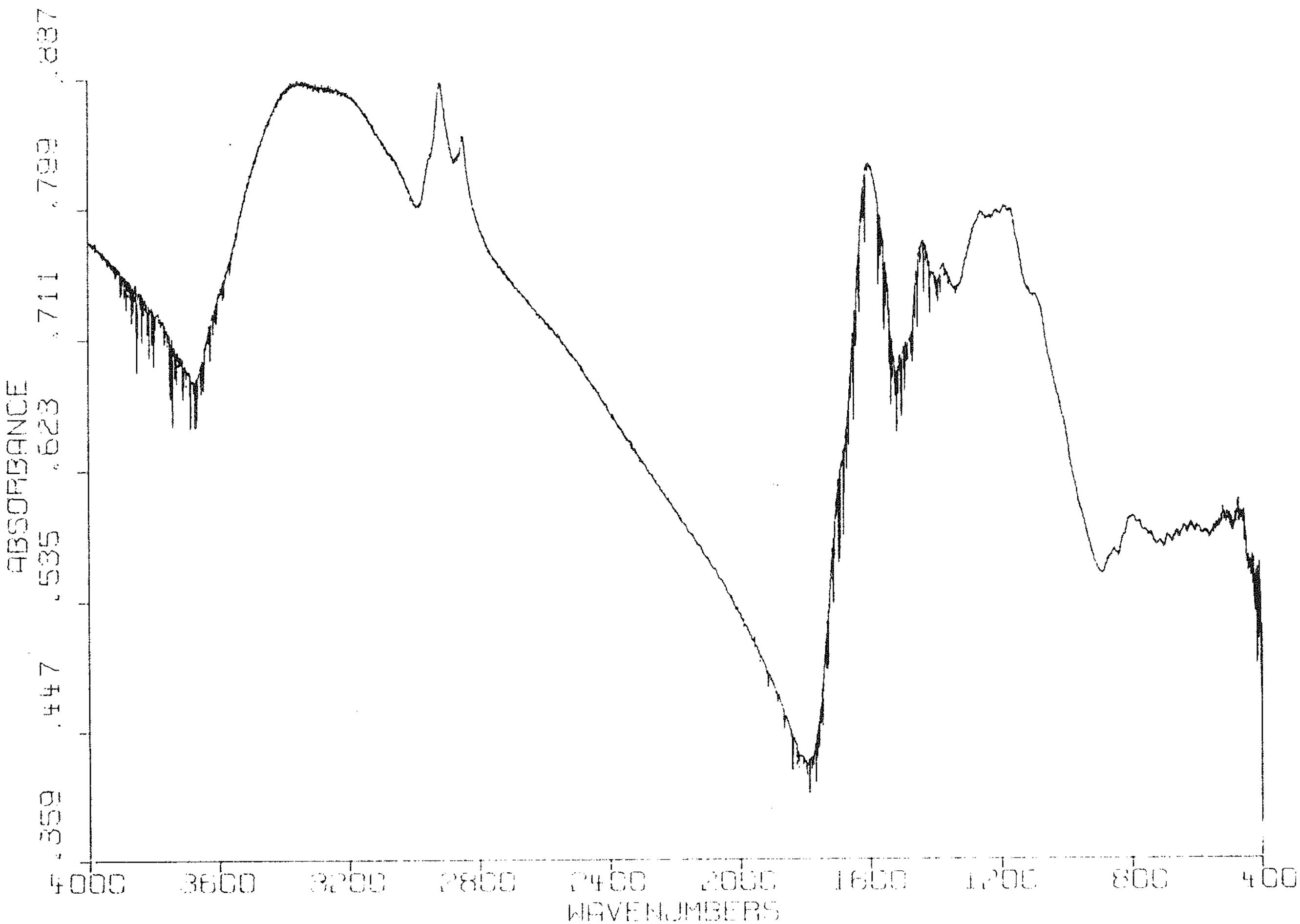


FIGURE 5-6 FTIR SPECTRA OF PRODUCT FROM LFD RUN 61 (N_2, $100^{O}C$, 120 MIN.)

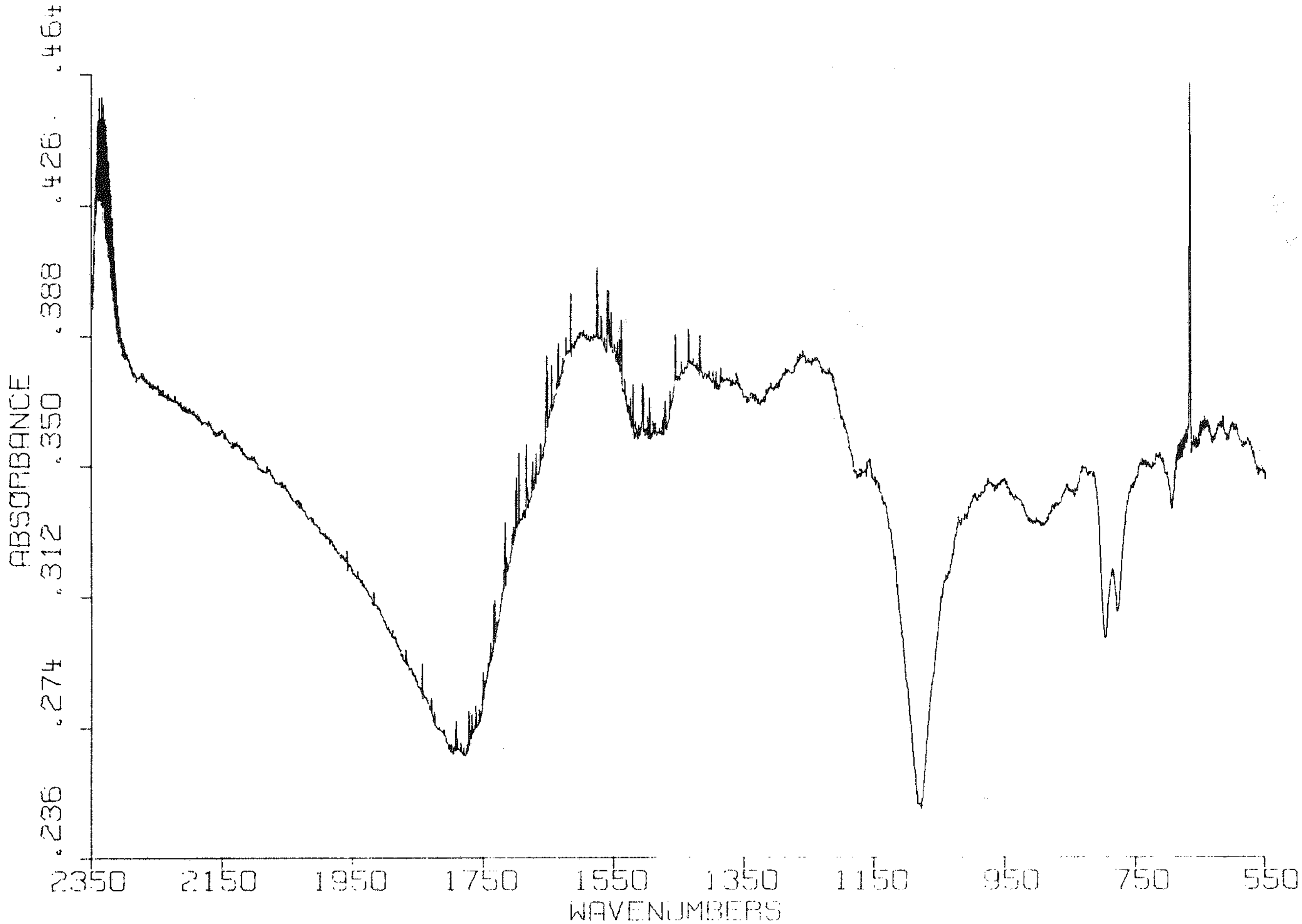


FIGURE 5-7 EFFECT OF DRYING SEVERITY IN NITROGEN; DIFFERENTIAL SPECTRUM RUNS 61 - 55

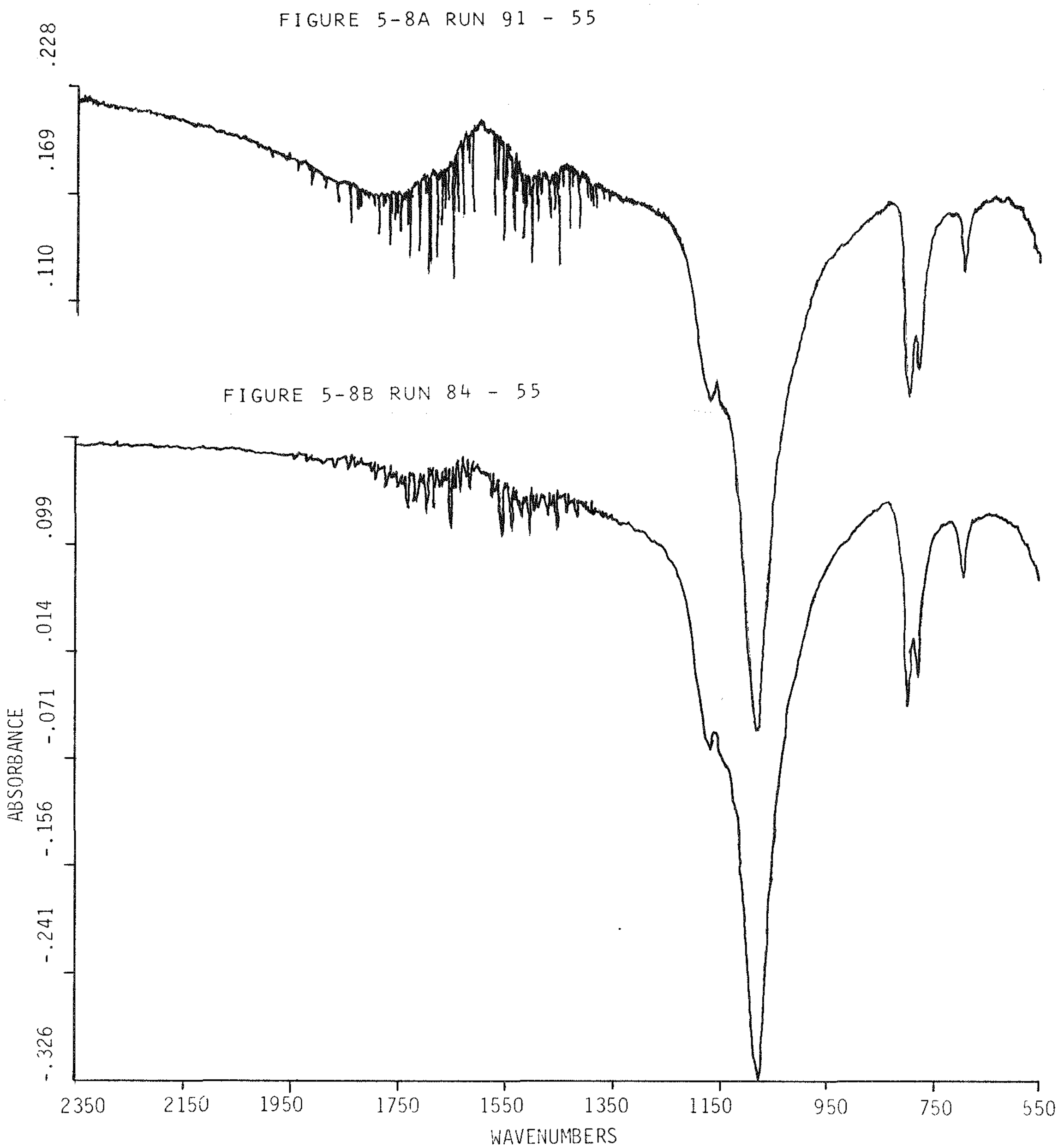


FIGURE 5-8 EFFECT OF DRYING IN OXYGEN CONTAINING GASES:;
DIFFERENTIAL SPECTRA RUNS 91 - 55 & 84 - 55

We would expect upon longer exposure time in the LFD to see a shifting in oxygen functionality along the reaction chain. Comparing runs 88 and 84 in Figure 5-9 (both 50°C in air but 84 exposed for 10 min, 88 for 120 min), it can be seen that the longer exposure increases the amplitude of various peaks (3320, 2910, 1600, 1420, 1200 nm) by a factor of 2.2-2.4. However, at higher temperatures, longer exposure may decrease oxygen functionality by decarboxylation, e.g., compare run 75 (30 min) vs 92 (5 min) where the former is reduced in carboxyl and other functionality (Figure 5-10).

The FTIR spectra of the coal samples prepared in the PFD were prepared in a similar manner to those discussed above. Typical spectra of the as-received coal and the most strongly oxidized coals are shown in Figures 5-11 and 5-12. The differential spectra are shown in Figure 5-13. As anticipated from the low level of oxidation, there was only a minor change of oxygen functionality, specifically less than 10% of that originally present. This primarily shows up as decarboxylation in the nitrogen-dried sample. The differential spectra of the MG-dried coal samples from run temperatures of 50 and 100°C are almost flat, indicating that any losses were nullified by subsequent reactions.

While the above spectra indicate that only minor structural changes occurred during drying in nitrogen or mixed gas, it is stressed that major changes in the liquefaction potential of these coals occurred. We plan to further study the changes of functionality because of this latter fact.

Optical Characterization of Liquefaction Residues

Ten liquefaction residues were recovered from bench-scale coal liquefaction runs in which the slurry product was mixed with pyridine with subsequent filtration and washing. The nitrogen dried solids were prepared for reflected light examination following procedures described previously.[3] A summary of the feed materials and the run conditions for these samples is included in Table 5-5. A summary of the results from the optical examinations is included in Table 5-6.

Examination of the sample set without regard to coal treatment or run conditions reveals that these samples can be grouped into one of three categories.

1. Good conversion - as indicated by large amounts of granular residue

2. Poor conversion - as indicated by large amounts of vitroplast plus uncoverted huminite.

3. Samples with significant semicoke formation (>5%).

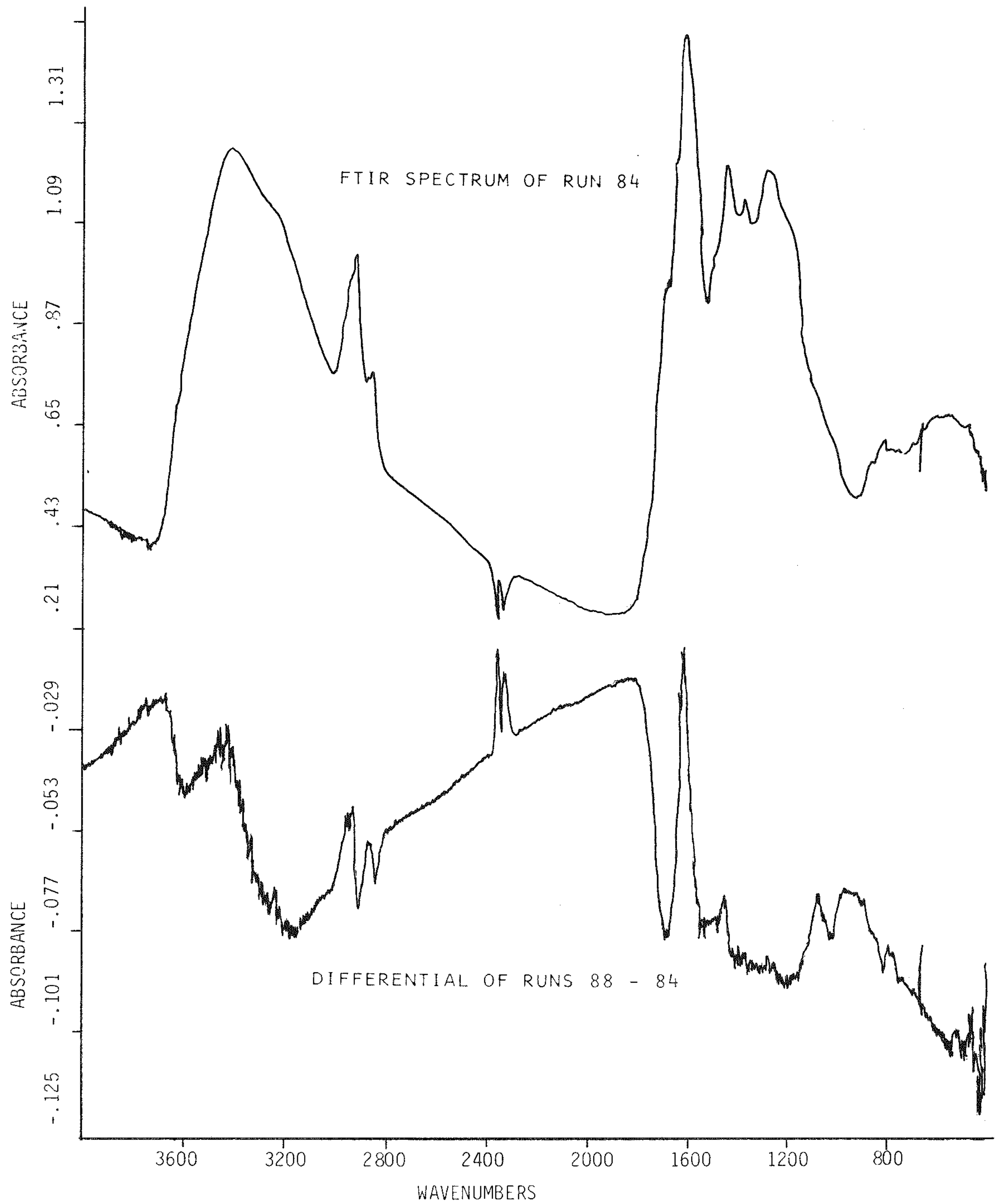


FIGURE 5-9 EFFECT OF EXPOSURE TIME: FTIR SPECTRUM OF RUN 84 WITH DIFFERENTIAL OF 88 - 84

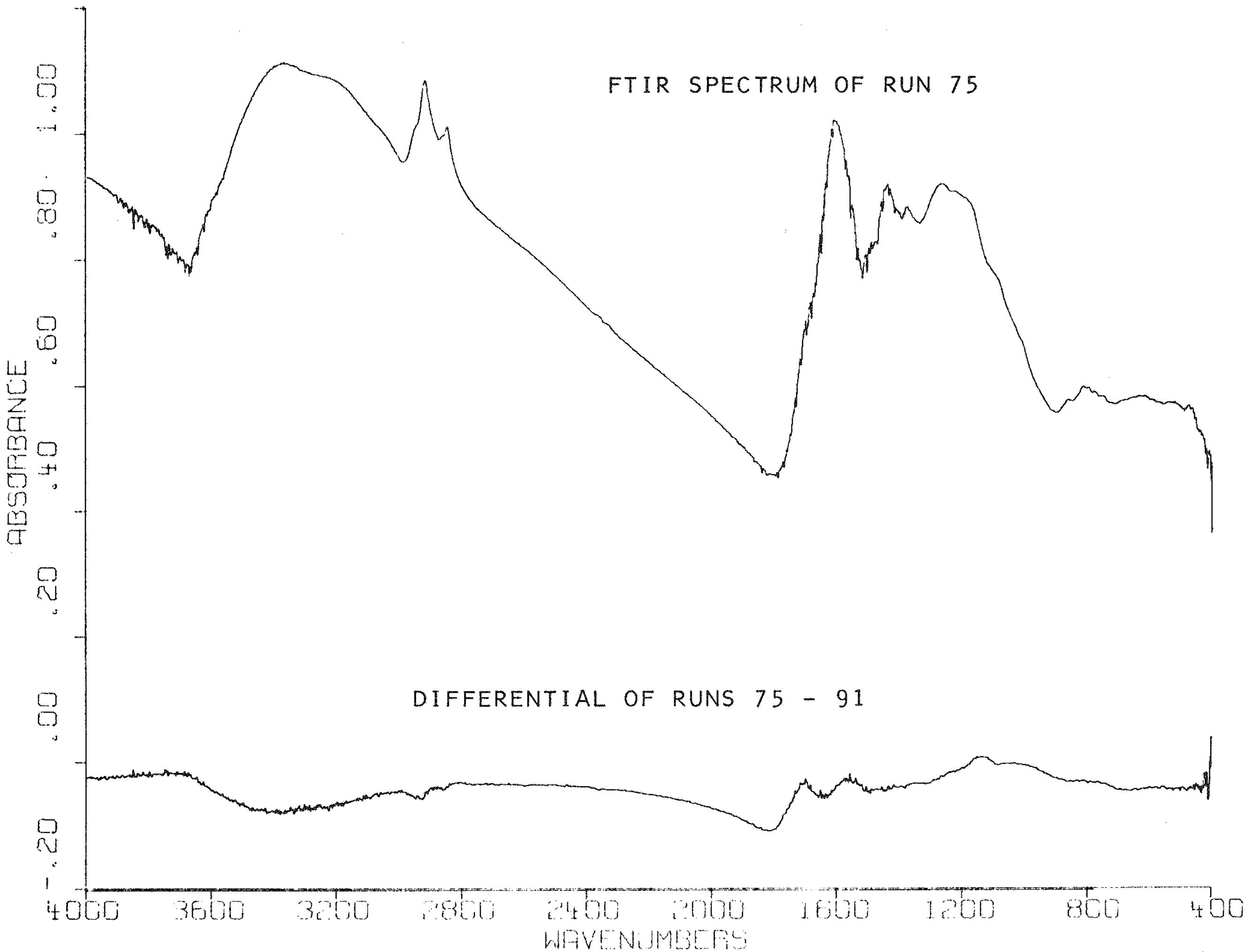


FIGURE 5-10 EFFECT OF TIME WITH OXYGEN CONTAINING GASES: FTIR SPECTRUM OF RUN 75 WITH DIFFERENTIAL OF 75 - 91

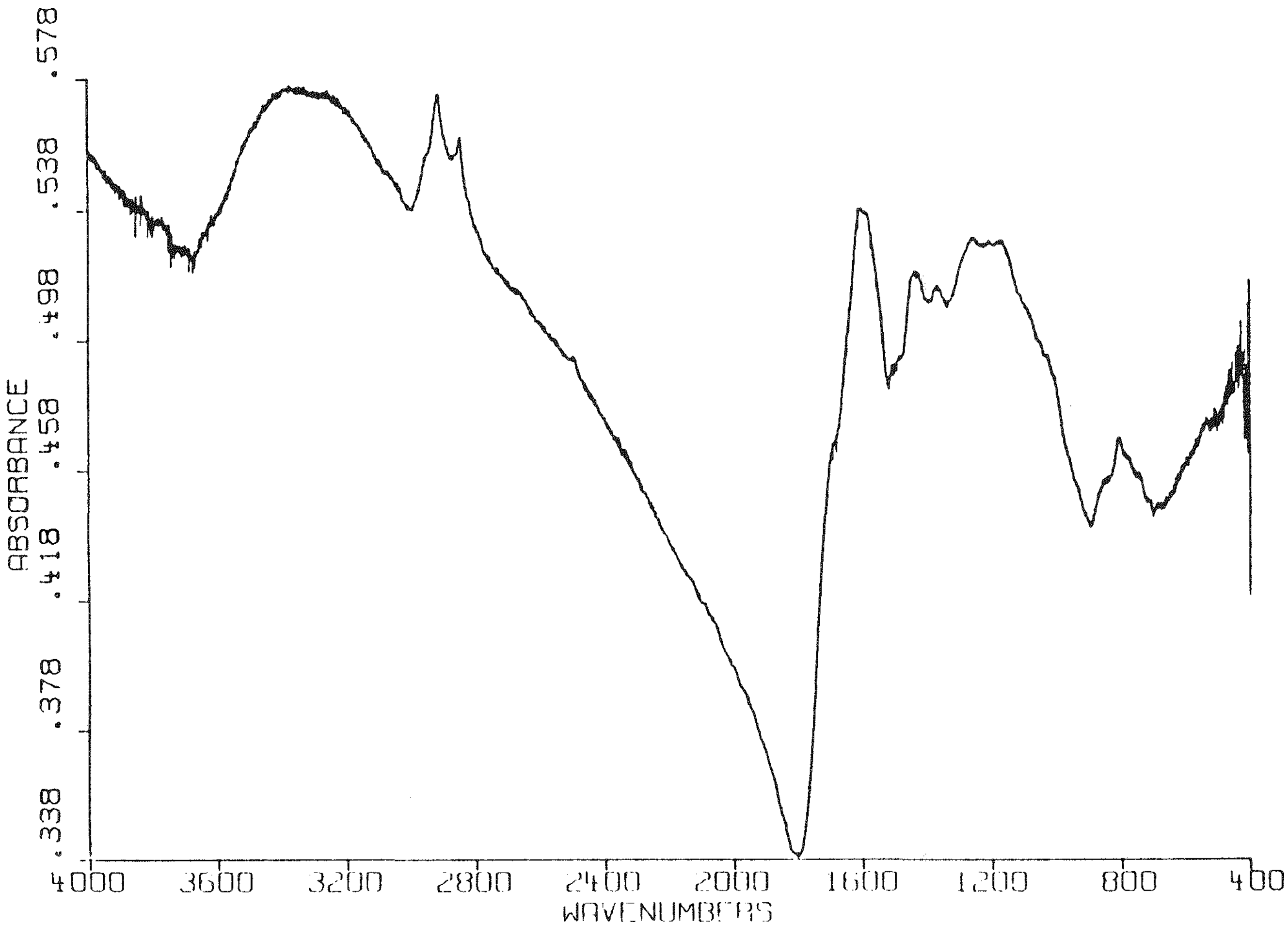


FIGURE 5-11 FTIR SCAN OF PULVERIZED BELLE AYR COAL

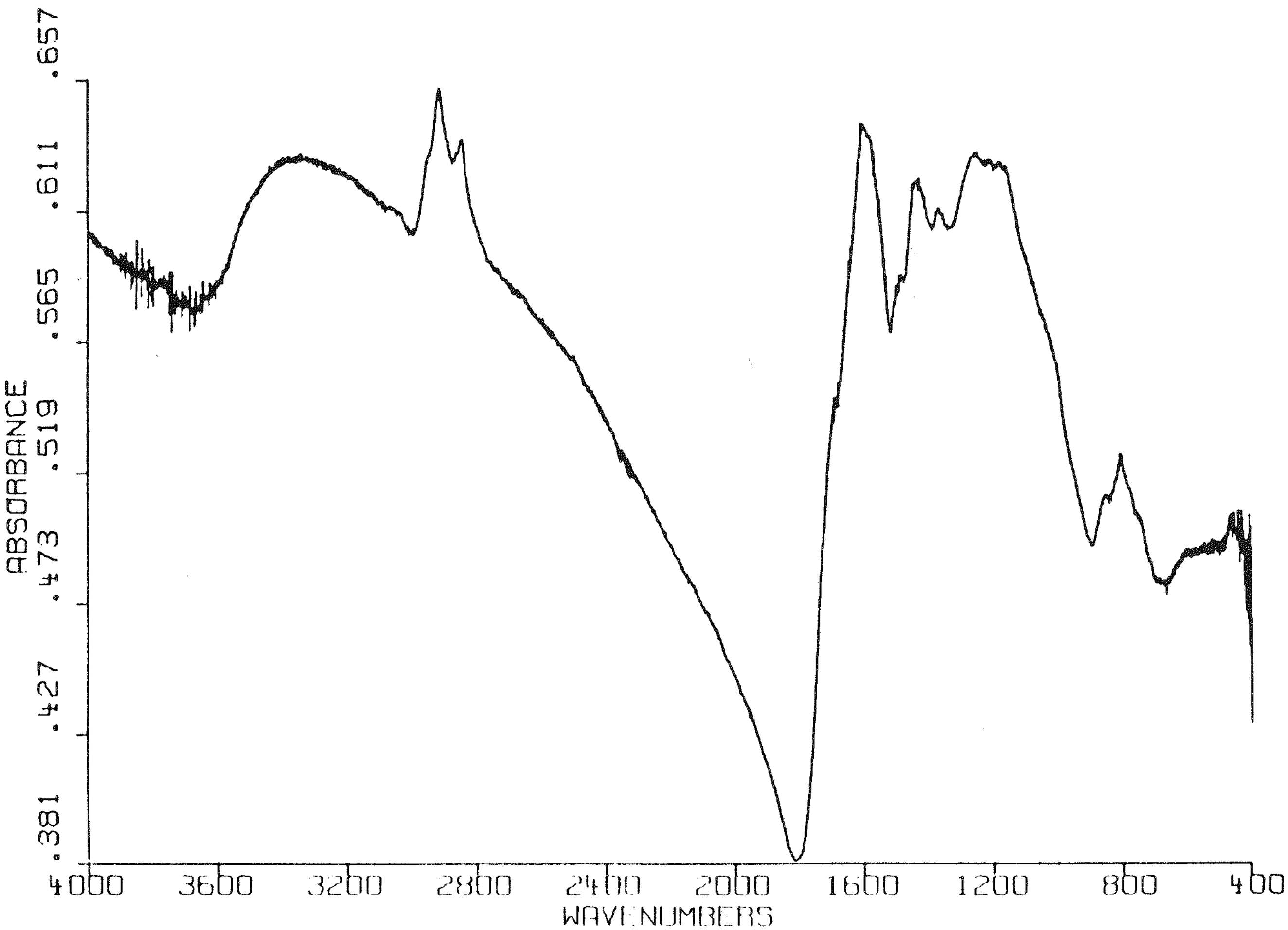


FIGURE 5-12 FTIR OF SOMEWHAT SEVERELY DRIED/OXIDIZED BELLE AYR COAL
(MG @ 100°C)

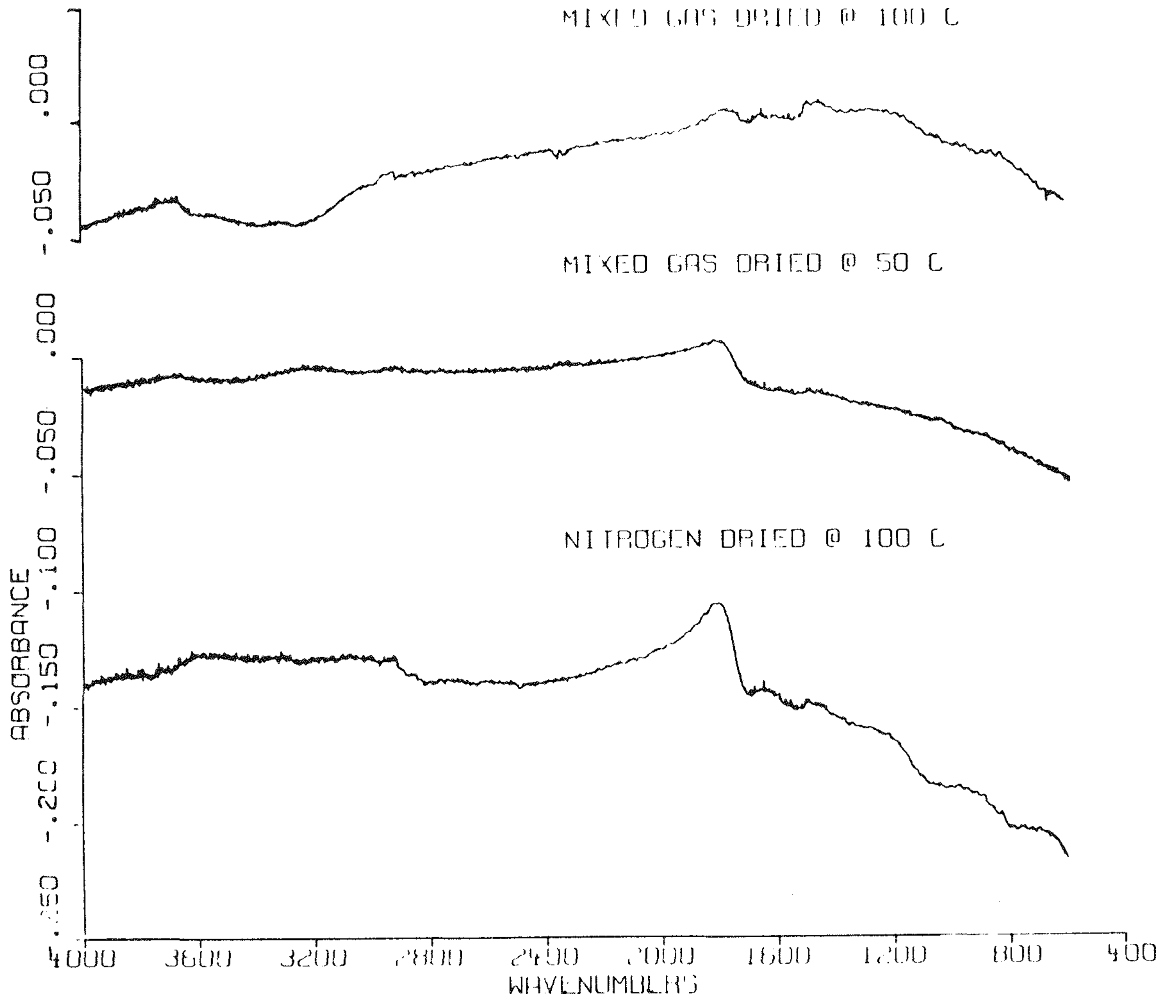


FIGURE 5-13 FTIR DIFFERENCE SCAN VS RAW COAL (MIN. SPECT. FEATURES AROMATICS) (1570 - 1650 WAVENUMBERS)

Table 5-5

LIQUEFACTION RESIDUES FOR OPTICAL CHARACTERIZATION

Sample No.	Feed Material	Liquefaction Reactor Conditions: Nominal Space Time (min.)	Atmosphere
EP-201	Pulverized As	20	H_2
EP-202	Received Belle	12	H_2
EP-203	Ayr Coal	5	H_2
EP-211	85x230 mesh Belle Ayr	20	H_2
EP-212	Dried for 1 hour at 100	20	$H_2 + H_2O$
EP-213	100°C in N_2	20	$H_2 + N_2$
EP-218	Belle Ayr Coal (Dried)	20	H_2
EP-219	1 hour at 100°C in	12	H_2
EP-220	50/50 mixture N_2/Air	5	H_2
EP-223	Belle Ayr Coal (Dried 1/4 hour at 100°C N_2 3/4 hour at 100°C 50/50 Mixture N_2 Air	20	H_2

Table 5-6

SUMMARY OF CHARACTERIZATION OF LIQUEFACTION RESIDUES

Sample No.	Major Constituents (>25%)	Minor Constituents (5 to 25%)	Trace Constituents (<5%)
EP-201	A. Granular Residue (~40%) B. Vitroplast (~25%)	A. Unconverted Huminite (~12%) B. Unconverted Fusinite Plus Semifusinite (~10%)	A. Quartz B. Semicoke (<29%)
EP-202	A. Unconverted Huminite (~40%) B. Vitroplast (~25)	A. Granular Residue (~12%) B. Unconverted Fusinite plus Semifusinite (~10%) C. Pyrrhotite	A. Quartz B. Semicoke (<10%)
EP-203	A. Unconverted Huminite (~70%)	A. Vitroplast (~20%)	A. Unconverted Fusinite + Semifusinite B. Pyrrhotite C. Quartz D. Granular Residue
EP-211 f	A. Unconverted Huminite (~50%) B. Vitroplast (~25%)	A. Semicoke (~10%)	A. Granular Residue B. Unconverted Fusinite + Semifusinite C. Pyrrhotite D. Quartz
EP-212	A. Unconverted Huminite (~40%) B. Vitroplast (~30%)	A. Semicoke ((~70%)	A. Granular Residue B. Unconverted Fusinite + Semifusinite C. Pyrrhotite D. Quartz

Table 5-6 (continued)

Sample No.	Major Constituents (>25%)	Minor Constituents (5 to 25%)	Trace Constitutents (<5%)
EP-213	A. Vitroplast (~40%) B. Unconverted Huminite (~25%)	A. Semicoke (~24%)	A. Granular Residue B. Unconverted Fusinite + Semifusinite C. Pyrrhotite D. Quartz
EP-218	A. Vitroplast (~35%) B. Unconverted Huminite (~35%)	A. Semicoke (~20%)	A. Granular Residue B. Unconverted Fusinite + Semifusinite C. Pyrrhotite D. Quartz
EP-219	A. Vitroplast (~45%) B. Unconverted Huminite (~35%)	A. Semicoke (~6%)	A. Granular Residue B. Unconverted Fusinite + Semifusinite C. Pyrrhotite D. Quartz
EP-220	A. Unconverted Huminite (~65%) B. Vitroplast (~25%)		A. Unconverted Fusinite + Semifusinite B. Pyrrhotite C. Granular Residue D. Semicoke (<1%) E. Quartz
EP-223	A. Vitroplast (~35%) B. Unconverted Huminite (~35%)	A. Semicoke (~20%)	A. Granular Residue B. Unconverted Fusinite + Semifusinite C. Pyrrhotite D. Quartz

General Comments:
1. Residues were predominantly aggregates >200 μm in diameter
2. Pyrrhotite was generally very fine grained (i.e., 90% <10 μm)
3. Semicoke was predominantly of mesophase origin

Samples EP-201 and EP-202 comprised group A. Both samples had significant amounts of granular residue and only trace amounts of semicoke (see Figure 5-14a). Sample EP-202 had higher amounts of vitroplast and uncoverted huminite, indicating that the shorter residence time of 10 minutes for EP-202 was insufficient for total dissolution, while the 15 minute space-time of EP-201 resulted in higher conversion. From the appearance of the residues, it was generally concluded that the as-received coal, liquified under hydrogen, showed the best conversion.

Two residues, EP-203 and EP-220, comprise the group B samples, which had poor conversion but showed little semicoke formation (see Figure 5-14b). These samples both contained greater than 50 percent of unconverted huminite and vitroplast. The similarity of the two residues indicates that the short space time of these samples (5 min) was probably the controlling factor in their incomplete dissolution. The greater abundance of vitroplast in sample EP-220, however, shows that the oxidation treatment of this coal may have affected its dissolution behavior.

The remaining samples, comprising group 3, all had significant amounts of semicoke formation. Within this group, the residues varied substantially in their characteristics and will be discussed in individual subgroups.

Samples EP-211, EP-212, and EP-213 were the residues from processing the dried coal for equal residence times in different atmospheres. The three samples showed decreasing amounts of combined unconverted huminite plus vitroplast, proceeding from sample EP-211 to EP-213. Sample EP-213 (processed under hydrogen and nitrogen atmosphere), had by far the greatest amount of semicoke formation (24%). (See Figure 5-14c). Sample EP-212, (processed under hydrogen and water atmosphere) had the smallest amount of semicoke formation (7%). From the appearance of the residues, none of these three samples should have shown high conversion; this is consistent with the results discussed in Section 6.

Samples EP-218 and EP-219 were residues from the dried coal which had been subjected to moderate oxidation. Both these samples showed high amounts of combined vitroplast and unconverted huminite, indicating poor conversion. Sample EP-218, the longer residence time sample, showed high (20%) amounts of semicoke formation (Figure 5-14d). The semicoke which did form in sample EP-219 was very fine grained (80% < 1 μm).

The final sample, EP-223, was also dried but subjected to only mild oxidation treatment (see Table 5-5). It contained a light amount of combined vitroplast and huminite, indicating poor conversion. It also contained a large quantity of

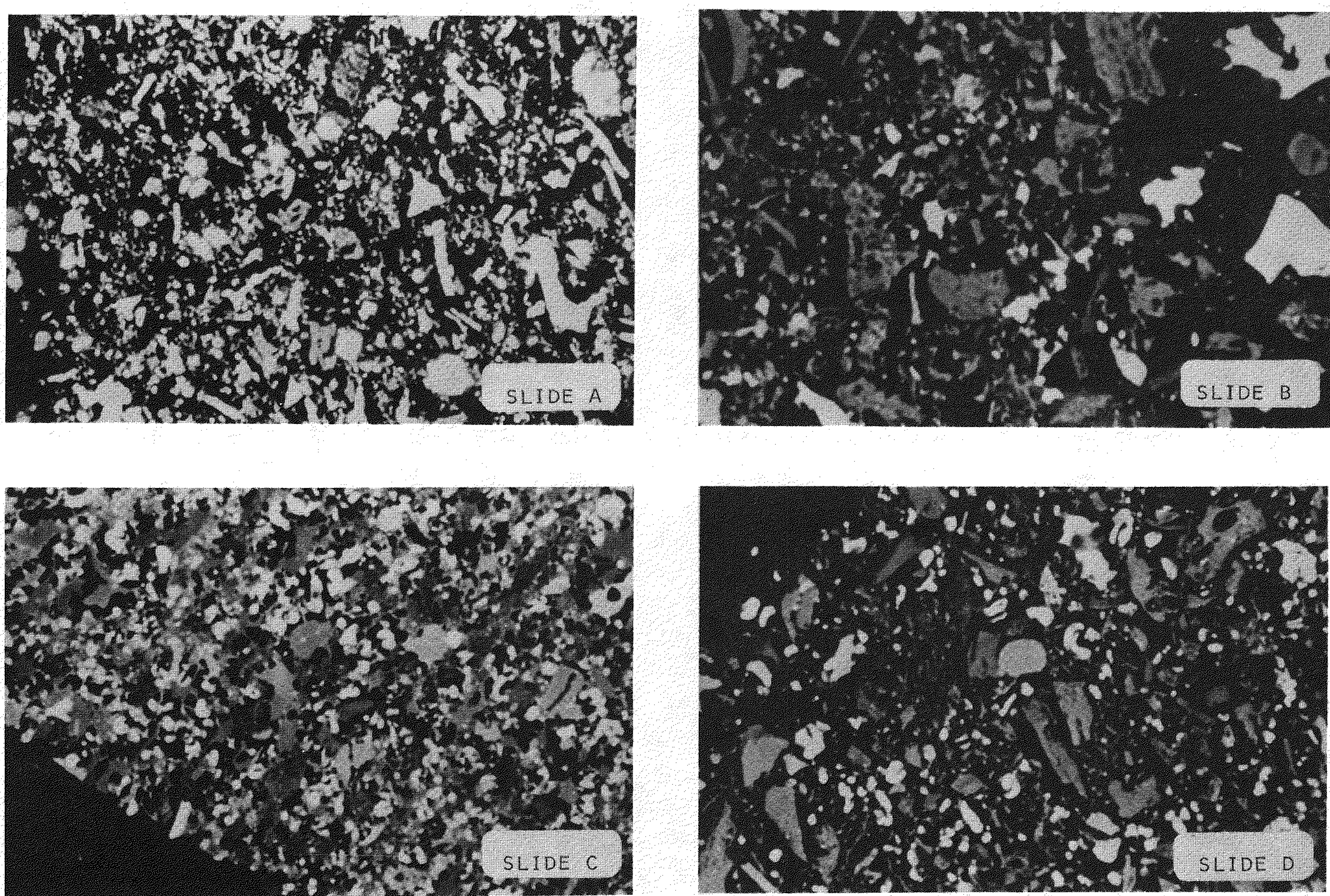


FIGURE 5-14 PHOTOMICROGRAPHS OF COAL LIQUEFACTION RESIDUES

semicoke (20%) and in overall appearance looked much like sample EP-218.

In summary, the dried, or dried and oxidized coals, showed evidence of poor conversion by containing large amounts of vitroplast, unconverted huminite and semicoke. Mild and moderate oxidation following the drying treatment tended to increase the propensity of the sample for semicoke formation. Samples processed in atmospheres of hydrogen, or hydrogen plus water, formed less semicoke than samples processed in hydrogen plus nitrogen.

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Section 6

LIQUEFACTION OF DRIED AND PARTIALLY OXIDIZED COAL

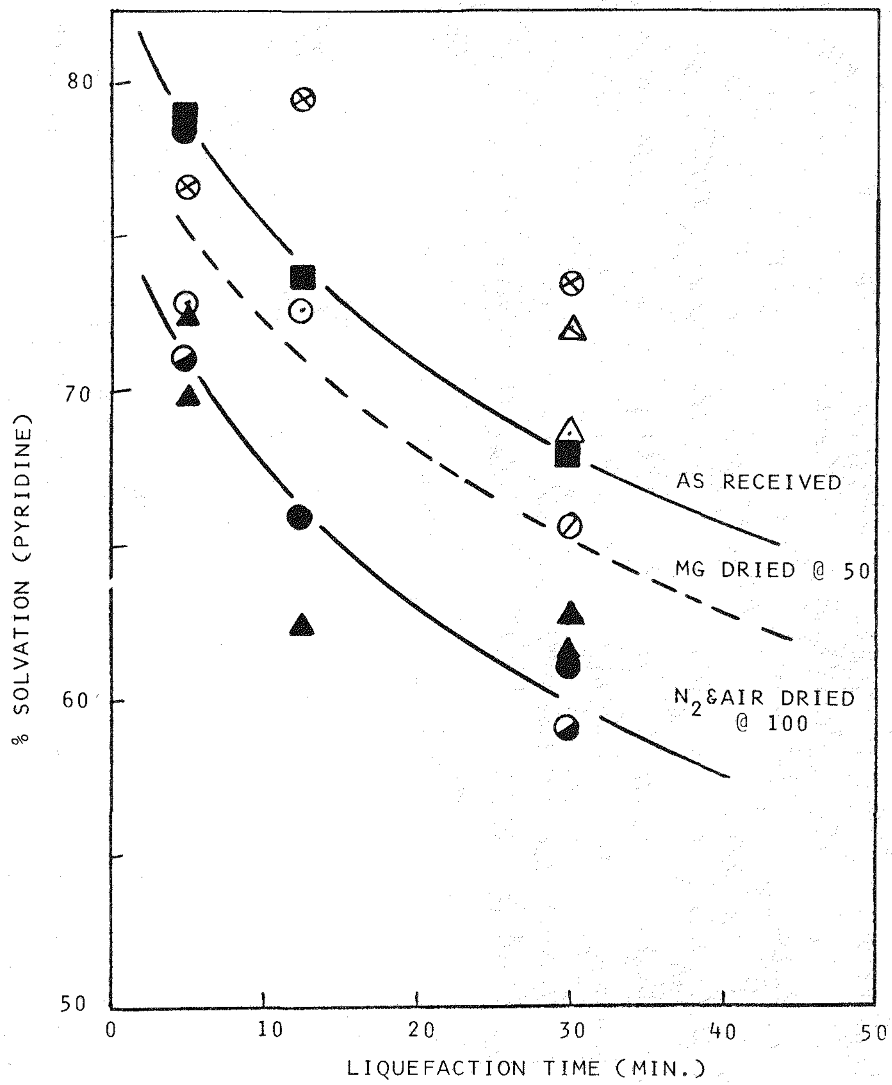
Two scales of equipment were used to observe the effect of predrying and partial oxidation on the donor solvent liquefaction of coal. Micro-scale runs were made to obtain an initial screening of overall effects while subsequent bench-scale runs were made in greater depth to obtain better material balances and provide data that should more closely approach that of a commercial facility. The coals used in these runs were recovered from the drying/oxidation runs described in the previous section. The solvent was SRC-II heavy distillate which is described in Section 7. For reference, a discussion of the units is given in Section 8.

Micro-Unit Liquefaction Experiments

A rather extensive series of runs were made using as-received, predried, and partially oxidized coals in the micro unit. The feed charge consisted of 2.0 g coal plus 3.0 g SRC-II heavy distillate solvent. The runs were made at times of 5, 12, or 30 minutes at 450°C. These run times were relatively realistic in that the heat-up period of the reactor in the sand bath averaged 2.3 minutes (time started after the reactor reached 450°C).

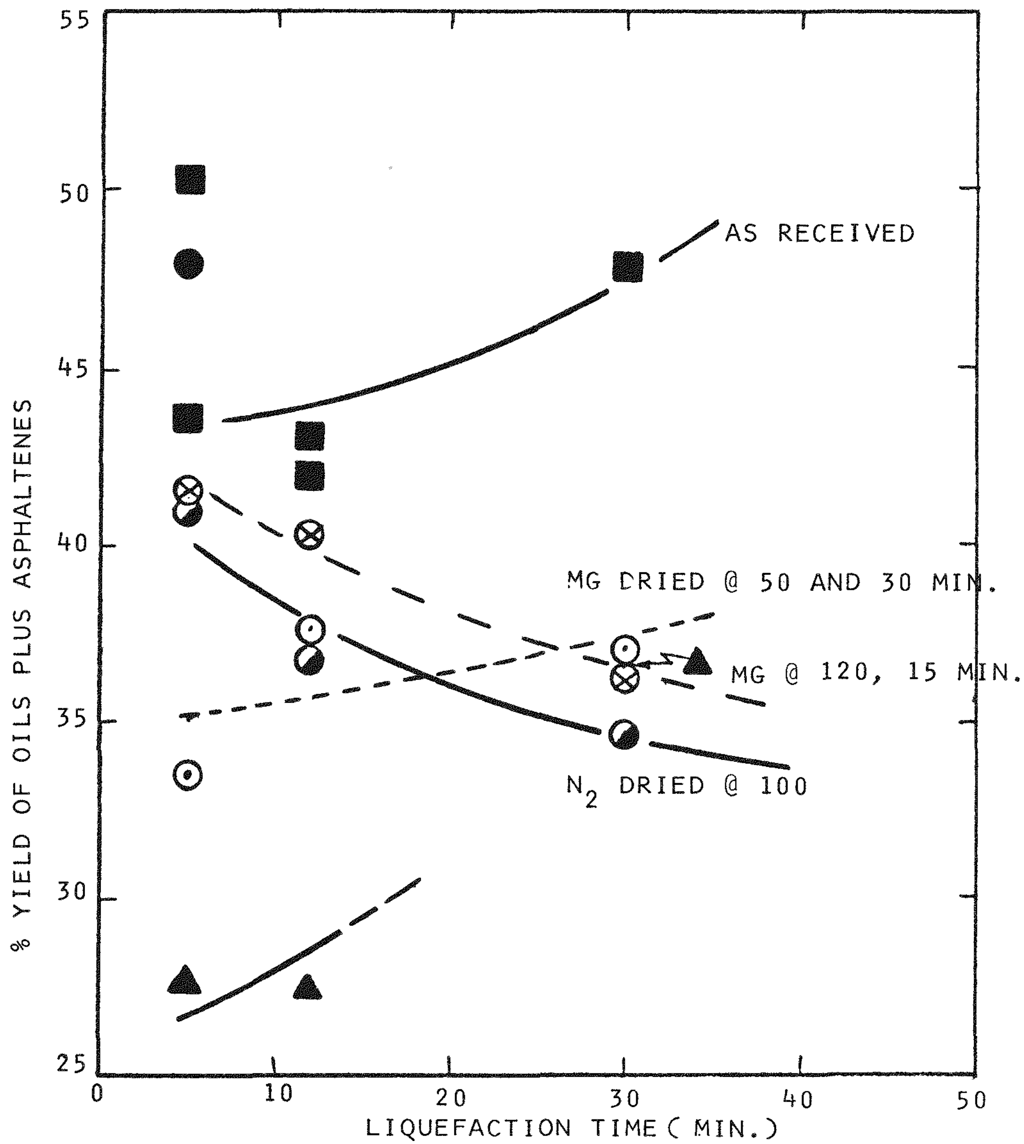
The slurry reaction products were originally to have been only extracted with pyridine to measure conversion. However, additional runs were made in which toluene was used as an extraction solvent to obtain a measurement of coal conversion into oils plus asphaltenes. The results of the liquefaction runs are summarized in Figures 6-1 and -2. (See also Tables 6-1 and -2).

The obvious first point is that the best liquefaction yields in terms of conversion of coal to toluene and pyridine solubles were observed for raw, as-received coal. This occurred in spite of the fact that the raw coal contained about 27 wt% moisture, while dried coals contained as low as 1 to 2 wt% moisture. It would be anticipated that the moisture would be in the vapor phase, thereby reducing hydrogen partial pressure and therefore coal conversion. It is obvious that changes brought about in coal characteristics by pretreatment were of more importance than the change in hydrogen partial pressure.



SYMBOL	GAS	TEMP($^{\circ}$C)	TIME	MOISTURE
■	AS RECEIVED			29.0
●	N_2	100/125	60	0.1
◐	N_2	120	30	0.6
⊙	MG	50	30	15.2
⊘	MG	50	60	14.4
⊗	MG	120	15	11.8
▲	AIR	160	60/120	0.6
△	AIR	100	5	0.3
◬	AIR	50	30	14.8

FIGURE 6-1 CONVERSION OF COAL TO PYRIDINE SOLUBLES IN THE MICRO UNIT



SYMBOL	GAS	TEMP(^{O}C)	TIME	MOISTURE
■	AS RECEIVED			29.0
●	N_2	100/125	60	0.1
◐	N_2	120	30	0.6
⊙	MG	50	30	15.2
⊘	MG	50	60	14.4
⊗	MG	120	15	11.8
▲	AIR	160	60/120	0.6

FIGURE 6-2 CONVERSION OF COAL TO TOLUENE SOLUBLES IN THE MICRO UNIT

Table 6-1

CONVERSION TO TOLUENE SOLUBLES: MICRO-UNIT RUNS

Run No.	Coal[a]	Moisture (wt%)	Run Time (Min)	Insolubles (g)	Conversion (MAF, wt%)
10	As Received	29.0	5	.7415	50.3
44	"	"	5	.8320	43.6
11	"	"	12	.8547	41.9
39	"	"	12	.8386	43.1
12	"	"	30	.7759	47.8
2	#77	0.1	5	1.0880	48.0
45	#60	0.6	5	1.2158	40.9
30	"	"	12	1.2961	36.7
40	"	"	30	1.3331	34.7
6	#68	15.2	5	1.1570	33.5
31	"	"	12	1.0909	37.6
13	"	"	30	1.0994	37.1
21	42/49 Blend	11.8	5	1.0687	41.5
23	"	"	12	1.0918	40.2
25	"	"	30	1.1600	36.1
15	72/73 Blend	0.6	5	1.4646	27.7
17	"	"	12	1.4717	27.4
46	"	"	30	1.3027	36.3

[a] Numbers refer to LFD drying/oxidation runs.

Table 6-2

CONVERSION TO PYRIDINE SOLUBLES: MICRO-UNIT RUNS

Run No.	Coal[a]	Moisture (wt%)	Run Time (Min)	Insolubles (g)	Conversion (MAF, wt%)
34	As Received	29.0	5	.3552	79.0
35	"	"	12	.4268	73.7
36	"	"	30	.5046	67.9
1	#77	0.1	5	.5081	71.6
3	"	"	12	.7483	65.9
37	#60	0.6	5	.6476	71.0
33	"	"	30	.8745	59.0
28	#126	5.4( )	30	.7970	61.0
5	#68	15.2	5	.5221	72.9
7	"	"	12	.5266	72.7
32	#57	14.4	30	.6481	65.5
20	42/49 Blend	11.8	5	.4806	76.7
22	"	"	12	.4332	79.5
24	"	"	30	.5339	73.5
14	72/73 Blend	0.6	5	.6145	72.8
38	"	"	5	.6701	69.9
16	"	"	12	.8111	62.4
18	"	"	30	.8228	61.8
27	#54	0.6	30	.8060	62.7
26	#86	14.8	30	.5396	72.0
29	#127	0.3	30	.7000	68.4

(a) Numbers refer to LFD drying/oxidation runs.

From the standpoint of converting coal into pyridine solubles in the micro unit, it appears that drying the coal in nitrogen at 100°C for 30 minutes or longer was essentially as detrimental as drying the coal in air. Specifically, conversion was reduced by 7 to 8%. When coal was dried in either mixed gas (10.8% oxygen, 89.2% nitrogen) or air at less severe conditions, intermediate liquefaction results were normally observed. For example, the drying of coal in mixed gas at 50°C and hold-times of 30 or 60 minutes resulted in a pyridine yield decrease of 2 to 3% from as-received coal. It is interesting (or perhaps anomalous) that the yield of pyridine solubles of coals dried in air at short time and low temperature or in mixed gas at short time resulted in yields above those of the as-received coal.

Another interesting point is that a tendency toward retrogressive reactions occurred with this SRC-II solvent, specifically lower conversion to pyridine solubles was observed with increased run time. It is noted that these runs were made with only a 500 psig initial nitrogen charge, and a major portion of the donatable hydrogen of the solvent may have been depleted in the longer runs, thereby leading to repolymerization. (For reference, this specific sample of SRC-II solvent contained about 0.44 wt% donatable hydrogen as determined by the method of K.S. Seshadri, et al., Fuel 57, 549 (1978). Therefore, it is not surprising that retrogressive reactions occurred.)

The effect of coal predrying and partial oxidation was similar to the above when considering coal conversion to toluene solubles. In particular, as-received coal was best and air-dried coal poorest as liquefaction feedstocks. Both nitrogen-dried and mixed gas-dried coal samples gave yields of toluene solubles which were intermediate.

A second aspect of this portion of the project was observing the effect of aging upon liquefaction characteristics. In this regard, a 40 g sample of 85x230 mesh Belle Ayr coal was placed in a flat pan with a porous cloth cover. The pan was left in the room during the somewhat dry months of October to December. Samples were periodically taken, their moisture content measured and liquefaction runs made. The following is a summary of the results:

Age (days)	Fresh	12	21	25	38	53
Moisture(wt%)	29.0	3.3	2.9	2.9	3.1	3.1
% Conversion to Toluene Solubles	43.1	47.5	48.8	44.9	43.9	39.3

Several points can be made with a degree of reservation. It first appears that the initial drying due to exposure to dry air may have been beneficial for conversion to toluene solubles. We see no obvious reason for this, considering that a nitrogen atmosphere was used in these liquefaction runs. Secondly, there is a tendency toward decreasing yields with increasing time. This may indicate a low degree of oxidation. This is consistent with similar low temperature aging studies of Dr. Schlyer in which only low levels of oxidation were observed (reference is made to Section 9 for a discussion of pertinent research). It appears that oxidation only penetrates into the coal particle to a limited extent at ambient temperature; therefore, overall liquefaction yields may only show a limited decrease with time. It is also interesting to note that the initial stages of coal surface oxidation occur rapidly, but then subsequent changes occur slowly. Again, the coal surface is only a minor portion of the coal available for liquefaction.

Another aspect of this series of experiments was to liquefy a sample of coal dried in simulated flue gas. A gas composition of 20 vol% CO_2 in nitrogen was used at drying conditions of 100°C and 1 hour time. Samples were prepared at both PSU and GR&DC. The following is a summary of the results:

Liquefaction of Flue Gas Dried Samples

Coal Samples	GR&DC-1	GR&DC-2	PSU-1	#60*
Liquefaction Time (Min)	12	30	12	12
% Conversion Toluene Solubles	34.3	38.1	37.8	36.7

*Reference sample dried in N_2 at 110°C for 30 min, see also Figure 6-2.

In summary, the drying of coal in simulated flue gas gives the same results upon liquefaction as that of drying in nitrogen. Of course, this conclusion is drawn using a limited number of experiments and a micro-scale unit, so larger scale experimentation is recommended for confirmation.

Bench-Scale Liquefaction Experiments

A series of liquefaction experiments was undertaken in a bench-scale unit which utilizes a stirred tank reactor and is capable of coninuous feed and withdrawal of coal slurry. The use of this unit provides a reasonable picture of coal liquefaction that would occur in a pilot or commercial unit. It also provides data that can be used in modelling of the kinetics of coal liquefaction (this aspect was not called for in this short series of experiments). A description of the unit, operating procedures, etc., is given in Section 8; the material balances are given in the Appendix.

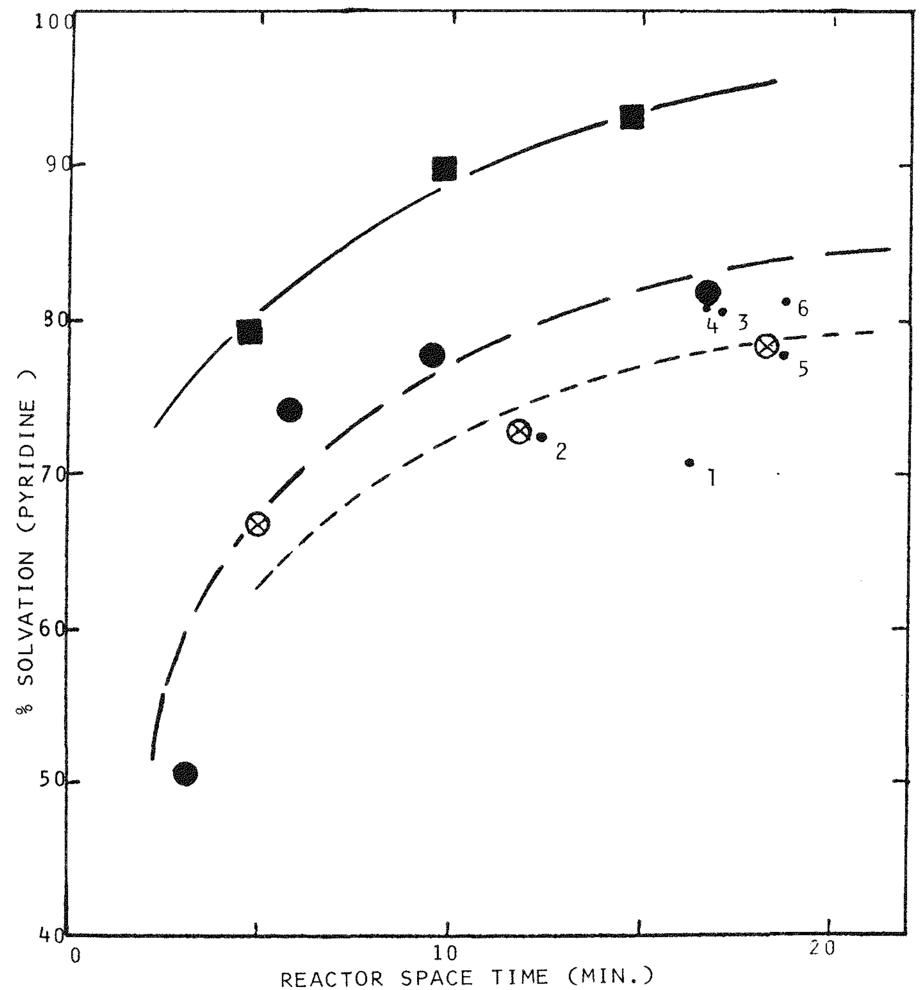
The bench-scale unit was operated at fixed conditions of temperature (450°C/842°F) and pressure (12.9 MPa/1750 psig). The coal:SRC-II solvent ratio was held constant at 1:1.5 on a basis of the pulverized raw coal having a moisture content of 27.5 wt% moisture. Namely, a correction was made to maintain a fixed ratio of moisture-free (MF) coal to solvent. The reactor space-times were varied between 4 and 20 minutes for the different coal samples.

Consistent with the micro-unit results, the highest levels of coal conversion were observed with as-received, pulverized coal. As noted in Figure 6-3, coal solvation (pyridine solubles) increased from 79 to 93% with the use of as-received coal and reactor space-times increased from 5 to 15 minutes. As shown in Figure 6-4, conversion of coal to toluene solubles (oils plus asphaltenes) followed a similar trend with an increase from 34 to 57%.

Coal samples which had been dried in nitrogen at 100°C for 1.0 hour (after heat-up) showed a reduced level of conversion to both pyridine and toluene solubles, namely average percentage point reductions of 12 and 10%, respectively.

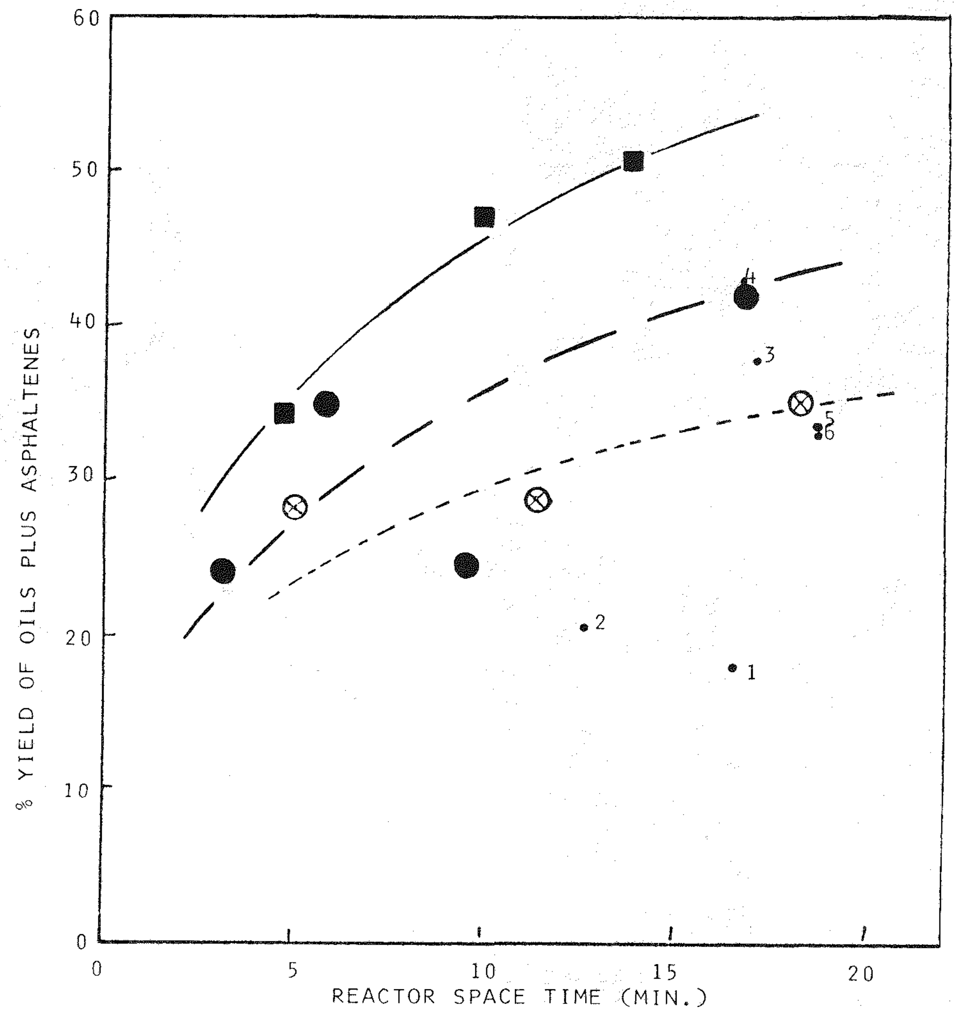
With the drying of coal samples in a mixed gas of air and nitrogen containing 10.8 vol% oxygen, a further drop in conversion from those with nitrogen drying was observed. This added decrease averaged about 5 percentage points.

Several additional runs were made to test the above results. First, when a coal sample was partially dried in nitrogen at a low temperature of 35°C for an extended period of about 9 hours, a coal moisture content of 17.8 wt% was obtained. Somewhat surprisingly, the levels of conversion of this coal sample were essentially the same as those of samples dried at a higher temperature and shorter time (see data point 4 in Figures 6-3 and -4). Secondly, the conversion of a coal sample dried in mixed gas at low temperature (50°C) was intermediate between that of samples dried in nitrogen and mixed gas at higher temperatures. Thirdly, two samples of coal were first dried in nitrogen for 30 or 15 minutes with subsequent treatment in mixed gas for a total period of 1 hour at 100°C. The levels of conversion of these coals to pyridine and toluene solubles were essentially the same as equivalent samples dried in mixed gas alone (data points 5 and 6).



SYMBOL	GAS	TEMP(^{O}C)	TIME	MOISTURE
■	AS RECEIVED			27.5
●	N_2	100	60	5.7
⊗	MG	100	60	5.6
•1	(N_2 /100/60 REACTOR GAS $H_2 \cdot N_2$)			
•2	(N_2 /100/60 REACTOR GAS $H_2 \cdot H_2O$)			
•3	MG	50	60	26.7
•4	N_2	35	540	17.8
•5	N_2	100	30	5.2
	MG	100	30	
•6	N_2	100	15	7.5
	MG	100	45	

FIGURE 6-3 % SOLVATION IN THE BENCH SCALE LIQUEFACTION RUNS



SYMBOL	GAS	TEMP(^{O}C)	TIME	MOISTURE
■	AS RECEIVED			27.5
●	N_2	100	60	5.7
⊗	MG	100	60	5.6
·1	(N_2 /100/60 REACTOR GAS $H_2 \cdot N_2$)			
·2	(N_2 /100/60 REACTOR GAS $H_2 \cdot H_2O$)			
·3	MG	50	60	26.7
·4	N_2	35	540	17.8
·5	N_2	100	30	5.2
	MG	100	30	
·6	N_2	100	15	7.5
	MG	100	45	

FIGURE 6-4 COMBINED YIELDS OF OILS PLUS ASPHALTENES IN THE BENCH SCALE LIQUEFACTION RUNS

At the request of EPRI, two additional bench-scale liquefaction runs were made using coal samples which had been dried in nitrogen, but the gas feed to the liquefaction reactor was varied. In the first run (data point 1 of the figures), the feed gas was made up of a mixture of hydrogen with sufficient nitrogen to be equivalent to moisture contained in wet as-received coal, assuming that the moisture of the feed coal vaporizes. As observed in Figures 6-3 and -4, the presence of this nitrogen was very detrimental, presumably as a result of reduced hydrogen partial pressure. In the second run (data point 2), sufficient water was added to the feed slurry to bring the moisture content of the feed coal to the equivalent of 27.5%. Again, this water addition was detrimental to coal conversion.

In addition to measuring coal conversion in terms of product extraction with various solvents, samples of coal slurry product were distilled to a fixed end-point. (The distillation technique is discussed in Section 8.) The recovery of residue was then calculated on a basis of MAF feed and product streams using the following equation:

$$\% \text{ Recovery} = \left\{\frac{(\text{FR} \times \text{WC}) - \text{CAF}}{\text{MAF Coal}}\right\} \times 100\%$$

where: FR = fraction of distillation residue in product slurry

WC = slurry product (wet cake in material balance)

CAF = ash feed in the coal (g/hr)

MAF Coal = moisture ash free coal feed rate (g/hr)

For reference, a recovery of 100% indicates no <u>net</u> yield of distillate and gases.

The recoveries of distillation residue are given as functions of space-time in Figure 6-5. In general, the residue recoveries follow trends anticipated from the extraction results. Specifically, liquefaction with as-received coal results in the least formation of residue while that with mixed gas-dried coal has generally the highest yield of residues. The results with nitrogen-dried coal appear to be intermediate but with an obvious tendency to equal those of as-received coal at short space-times. In spite of some degree of scatter of the data, it is obvious that the drying of coal in the mixed gas stream is very detrimental.

Two additional points should be made in considering the product yields from the above liquefaction runs. First, there is no pronounced retrogressive reaction

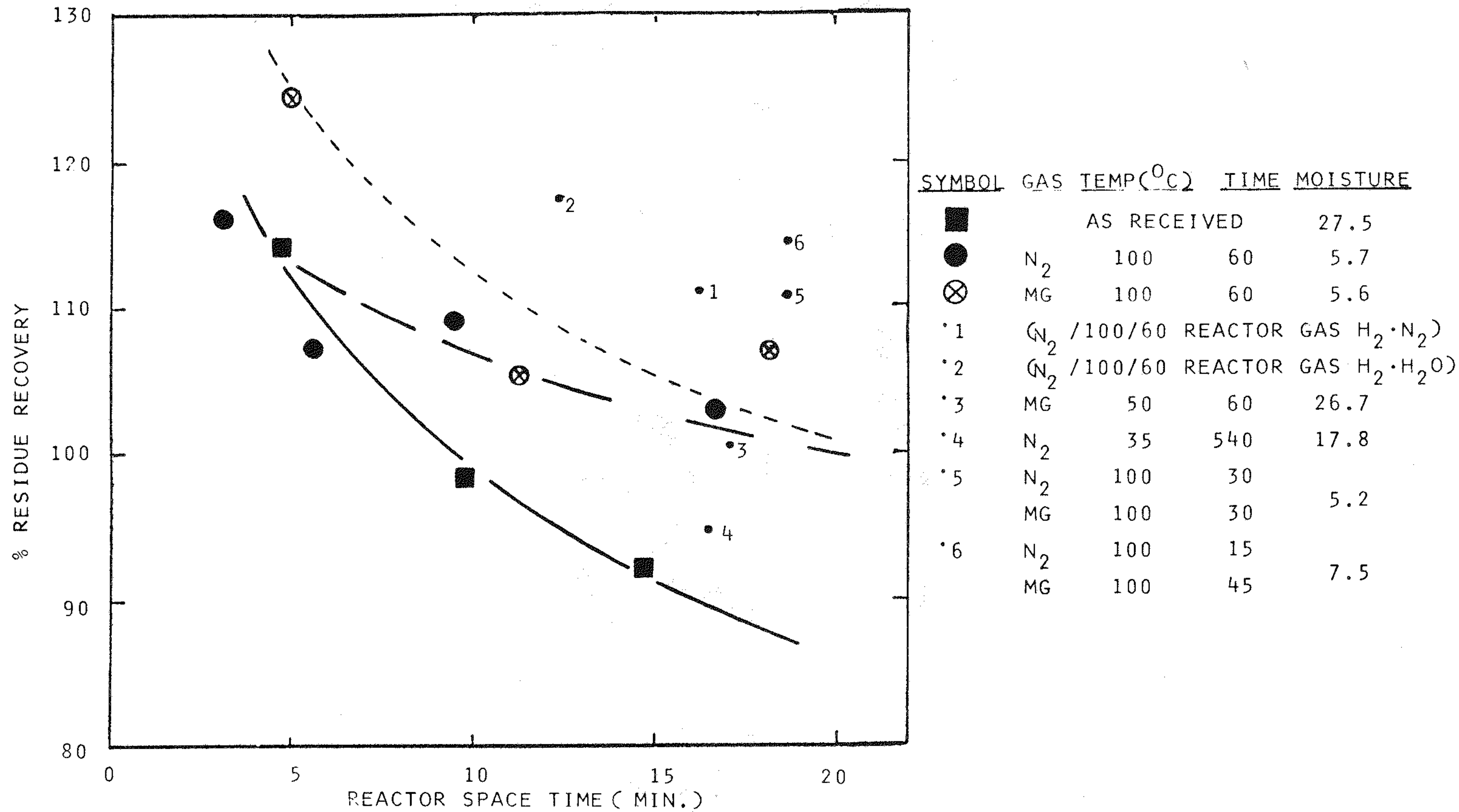


FIGURE 6-5 % RESIDUE RECOVERY FROM THE BENCH SCALE RUNS occurring at 450°C and space-times up to 20 minutes in the bench-scale unit. This is consistent with previous liquefaction experiments made with Belle Ayr coal and reported in AF-913. However, when micro-unit runs were made with this coal and SRC-II solvent in a nitrogen atmosphere, the retrogressive reaction was obvious. Apparently the SRC-II solvent can be somewhat depleted of donatable hydrogen under these reaction conditions. This results in repolymerization of coal free radicals and reduced conversion with time.

The second point concerns the yields of CO and CO_2. The following table provides a comparison of the yields of light gases from the three primary series of liquefaction runs.

Feed Coal Treatment	Liquefaction Space-Times(min.)	Average Gas Yields (g/100g)		
		CO	CO_2	C_1-C_3
As-Received	4.7, 9.8, 14.7	0.18*	1.82	3.28
N_2-Dried	3.1, 5.8, 9.5, 16.7	0.38	3.99	2.95
MG-Dried	5.0, 11.3, 18.2	0.59	4.07	4.04

*not including EP-202 @ 0.03

It is obvious that less CO and CO_2 is generated in liquefying as-received coal than either of the dried coals. While these yields are lower for the nitrogen-dried coal than those for the MG-dried coal, the difference is not as great as expected. It is our guess that the nitrogen-dried coal underwent some surface oxidation during handling, in spite of precautions taken. For reference, the yields of C_1 to C_3 gases are reasonable for these reaction conditions. The C_1 to C_3 yields increase with increasing average space-times.

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Section 7

CHARACTERIZATION OF COALS AND SOLVENTS

Two samples of coal and one sample of SRC-II solvent were used in the drying/partial oxidation/liquefaction studies of this project. The following is a summary of how these samples were obtained and their handling steps at GR&DC. Particular care was taken to insure a minimum of exposure to air, particularly after the coal samples had been pulverized.

Subbituminous Coal: Belle Ayr Mine (LR26104)

A four drum sample of subbituminous coal was obtained from the Belle Ayr Mine of Amax Coal Company through the help of Mr. George Land.

To obtain as fresh a coal sample as possible, arrangements were made to sample from a clean face of the mine. The coal was first stripped from the surface. Subsequently, a sample was scraped from the newly exposed surface, placed in drums with plastic liners, and the drums sealed. When received in GR&DC, the coal was wet and the liners were in good shape.

> Note: Based on a conversation with Mr. Land, the coal was mined about 1/2 mile west of the coal tipple. The sample was taken from the bottom 14 to 15 feet of the coal seam. At this point, the seam is designated Smith-Roland in that the two seams exist on top of each other without a parting. The total seam height is about 50 to 60 feet. In some cases, the seam is designated the Wyodak seam in that it was first mined at the Wyodak mine.

As requested by EPRI, the total Belle Ayr coal sample was crushed, coned-and-quartered, and sampled. The crushing was done in a mill with 1/4 inch slots (Jacobson Machines Works, Inc., Minneapolis, Minn., Model 1-1614-B). The bulk of the sample was placed in drums with liners, purged with nitrogen, and stored.

An analysis of the coal sample was determined at Pennsylvania State University under the auspices of Drs. Jenkins and Davis. It is given in Table 7-1.

The coal was pulverized using a Bantam Micro-Pulverizer unit of the Pulverizing Machinery Co., Summit, NH. It was fitted with a 0.125 inch screen. The pulverized coal fell directly into a Sweco Vibro-Energy Separator (30 inch diameter screener) and was typically separated into sample sizes of 84 mesh,

Table 7-1

RESULTS OF THE ANALYSIS OF BELLE AYRE COAL SAMPLE
(Prepared at PSU under the auspices of Drs. Jenkins and Davis)

1. Proximate Analysis

	As Rec'd	Dry
% Moisture	6.23	--
% Ash	4.51	4.81
% Volatile	43.50	46.39
% Fixed Carbon	45.76	48.80

2. Calorific Value

	As Rec'd	Dry
Btu	11217	11962

3. Ultimate Analysis

	Dry
% Moisture	--
% Carbon	65.56
% Hydrogen	4.74
% Nitrogen	1.16
% Chlorine	0.01
% Sulfur	.41
% Oxygen (diff)	23.31
% Ash	4.81

4. Petrographic Data (Volume Percent)

Ulminite	81.6
Humodetrinite	2.4
Porigelinite	1.4
Levigelinite	1.4
Fusinite	3.3
Semifusinite	5.9
Macrinite	1.6
Corpohuminite	0.6
Textinite	0.0
Exinite	1.7
Resinite	0.1
Alginite	0.0
Sclerotinite	0.0
Other	0.0

5. Reflectance Data

Mean Maximum Ulminite Reflectance (Oil, %) 0.360
Ulminite Rank
Subbituminous

84x145 mesh, 145x230 mesh, and through 230 mesh. It is noted that the pulverizer, screener and product receiving drums were all directly connected with a flow-through purge of nitrogen to minimize contact with air.

Pittsburgh Seam Coal: Powhatan No. 5 Mine (LR-26061)

Because of GR&DC's interest in studying a representative Pittsburgh seam coal, five drums of Powhatan No. 5 Mine coal were obtained through Mr. Jack Clements (216-752-1000) of the North American Coal Co.

To obtain a fresh sample, a truck was sent from GR&DC to the mine, along with a technician familiar with coal liquefaction experimentation. At the mine, coal is continually mined, jig washed, and piled. After about five truckloads of coal were removed from a pile prepared during the previous night shift, a sample of wet coal was transferred to 55 gallon drums with plastic bag liners. The drums and liners were sealed and returned to GR&DC. The drums were subsequently coned-and-quartered. Samples were separated for storage and analysis, and sufficient coal was crushed and pulverized for experimentation. All samples of coal were purged with nitrogen prior to sealing.

An analysis of the Powhatan coal sample as reported by Drs. Jenkins and Davis of PSU is given in Table 7-2.

SRC-II Solvent (LR-26080)

Several drums of SRC-II "heavy distillate" were obtained from the Fort Lewis facility (Tacoma, WA) of Pittsburg & Midway Coal Mining Company. The solvent, designated Sample Shipment No. 1504, has a nominal boiling range of 550-850°F. It was prepared by the SRC-II liquefaction of Powhatan Mine coal. An analysis of this solvent is given in Table 7-3.

Table 7-2

RESULTS OF THE ANALYSIS OF POWHATAN NO. 5 COAL SAMPLE
(Prepared at PSU under the auspices of Drs. Jenkins and Davis)

1. Proximate Analysis

	As Rec'd	Dry
% Moisture	1.06	--
% Ash	9.58	9.68
% Volatile	39.65	40.07
% Fixed Carbon	49.71	50.25

2. Calorific Value

	As Rec'd	Dry
Btu	13174	13315

3. Ultimate Analysis

	Dry
% Moisture	--
% Carbon	72.28
% Hydrogen	5.07
% Nitrogen	1.47
% Chlorine	0.03
% Sulfur	3.60
% Oxygen (diff)	7.86
% Ash	9.68

4. Petrographic Data (Volume Percent)

Vitrinite	89.2
Pseudovitrinite	0.0
Fusinite	1.7
Semifusinite	3.8
Macrinite	0.6
Micrinite	2.1
Exinite	2.6
Resinite	0.0
Alginite	0.0
Sclerotinite	0.0
Other	0.0

5. Reflectance Data

Mean Max. Vit. Refl.	0.655%
Vitrinite Rank	HVB

6. Gieseler Plastometer Data (Sample Run, 5/14/79)

Maximum Fluidity (D.D.P.M.)	18,619
°C @ Max Fluidity	430.5
°C @ Start Fluidity	385.0
°C @ Final Fluidity	472.0
°C Range Fluidity	87.0

Table 7-3

ANALYSIS OF SRC-II HD SOLVENT (LR26080)

1) Elemental Analysis

Carbon	88.47 wt %
Hydrogen	8.07
Nitrogen	1.10
Oxygen	1.93
Sulfur	0.43
	100.00 wt%

2. Physical Properties

Specific Gravity, 60/60°F	1.056
Kinematic Viscosity, 210°F	2.99

Distillation (D1160 @ 2.0 mm Hg, K ≈ 9.6)

5%	239°F
10%	250°
20%	259°
30%	266°
40%	275°
50%	284°
60%	300°
70%	320°
80%	347°
90%	383°
95%	414°
EP	471°

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Section 8

LIQUEFACTION AND DRYING UNITS WITH EXPERIMENTAL PROCEDURE

Coal liquefaction experiments were made in either a micro-scale batch reactor or in a bench-scale unit having a continuous feed stirred tank reactor (CFSTR) configuration. The coal for these experiments was prepared in either a laboratory-scale fluid-bed drier with a capability of a 30g charge or in a pilot-scale drier that could effectively handle a charge of 7 kg of pulverized coal. The following is a description of these units along with a summary of the operating procedures.

Bench-Scale Liquefaction Unit

A flow diagram of the coal liquefaction unit designated F-11 is shown in Figure 8-1. In summary, the system consists of feed tanks, Moyno recirculating pump, a Milton-Roy high-pressure slurry pump, preheater, one-liter stirred autoclave, let-down valves, receivers and a gas chromatograph. An auxiliary feed tank and a Braun and Luebbe pump are also available to provide high pressure flush solvent in the event of operating problems. It is noted that the autoclave was fitted with a "dead man" to reduce its effective volume to 293 cc.

The reactor was equipped with a single large (5.2 cm) diameter impeller. The agitator was run at sufficient speed (1000 rpm) to insure a minimum of mass transfer effects. There were four baffles in the reactor. As a check of the degree of agitation, a glass model was built. Visual observations of mixtures of water, air, and quartz chips were made at agitation speeds between 500 and 1000 rpm. The degree of mixing was effective and essentially no change was observed over this range. In addition, liquefaction runs have been made at 1000 and 1500 rpm; no effect of agitation on conversion was observed.

Liquefaction runs were started by first bringing the unit to operating pressure and adjusting the hydrogen flow. The reactor was heated to within 50°C of operating temperature and the feed slurry was started. The runs were of 4-6 hours in length. At least six reactor volumes passed through the unit prior to sampling.

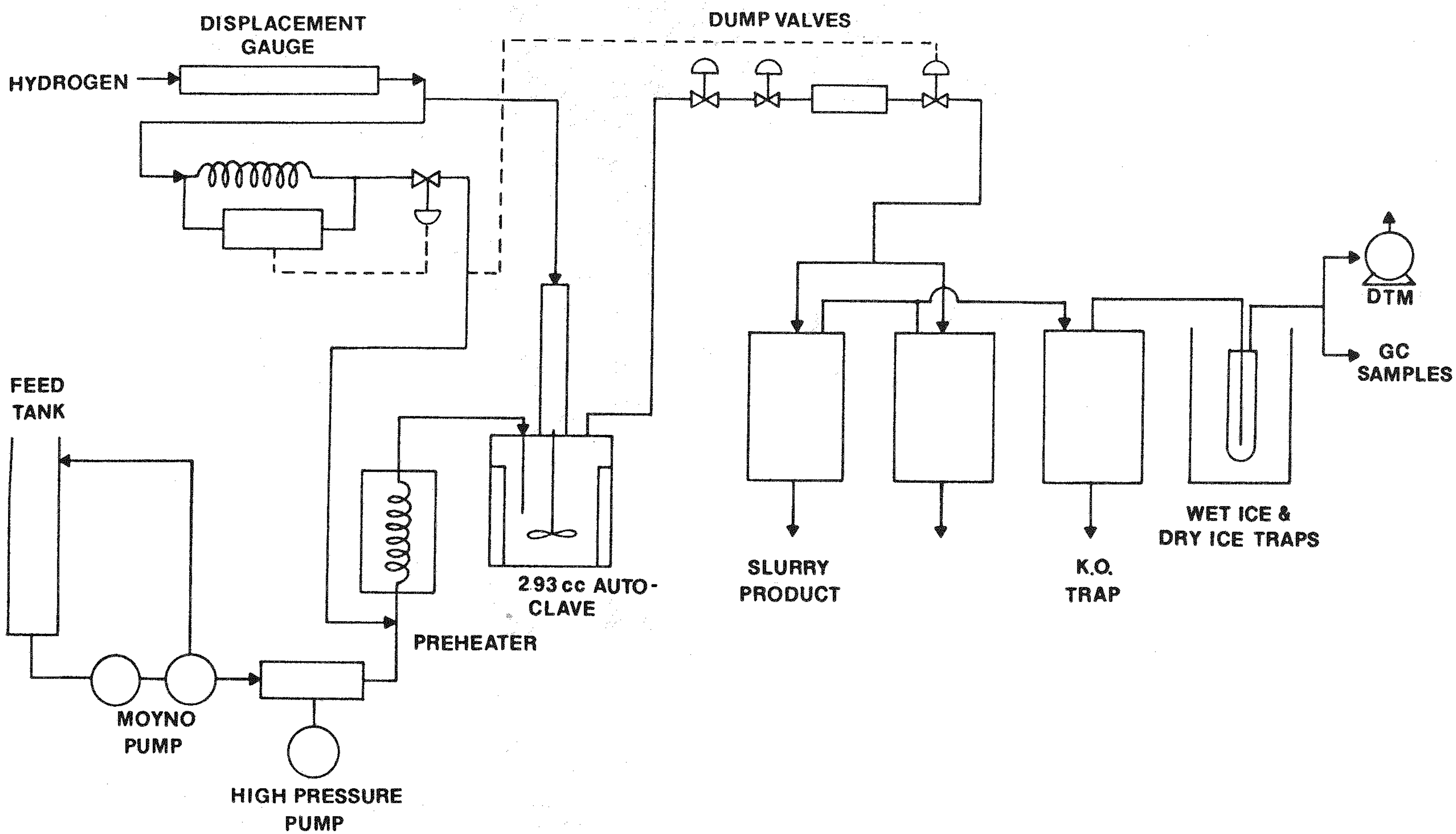

FIGURE 8-1 SCHEMATIC OF THE BENCH SCALE COAL LIQUEFACTION UNIT

The reactor was periodically opened to insure that no build-up of sediment occurred. This was the case in all runs except those of EP-206 through 208. In retrospect, an operating problem during EP-205 had resulted in some sediment forming thereby reducing the reactor volume during the next three runs. This was taken into account in Runs EP-206 through 208. A subsequent run (EP-211) confirmed the results.

Samples were taken during the latter two 60 minutes periods. These products consisted of slurry, aqueous and oil phases from the traps, and gases. The oil phase was added back to the slurry and blended. The slurry was then sampled for analysis. In addition to elemental analyses, the slurry samples were both extracted and distilled.

A detailed discussion of extracting coal slurry products has been given in our previous report (EPRI AF-913). It is sufficient herein to comment that a "Direct Extraction" technique was used in which individual samples of slurry were separately Soxhlet extracted with n-pentane, toluene and pyridine. The insoluble portions were dried in a nitrogen-swept vacuum oven and weighed. The level of the various fractions was calculated to conform with the following defintions.

Oils = n-pentane solubles
Asphaltenes = n-pentane insolubles-toluene solubles
Preasphaltenes = toluene insolubles-pyridine solubles
Pyridine Insolubles = Unreacted coal

This is an extension of petroleum chemistry terminology since petroleum contains only oils and asphaltenes. It is also noted that the use of the term "preashpaltenes" does not necessarily imply that this fraction is formed before asphaltenes, but may actually form from a polymerization of asphaltenes and/or oils.

Distillations of samples of slurry were performed in a batch still capable of operation at 1 mm pressure. About 500 g of slurry were charged to a 1 liter still pot and distilled at atmospheric pressure to 204°C (400°F). The distillation was continued at reduced pressure to 188°C (370°F) vapor temperature and 1.0 mm pressure. The average still pot temperature was about 265°C (510°F). (Note: An extensive discussion of distillation and its effect on repolymerizaton of coal liquid products is given in EPRI Report No. AF-913. In addition, it was originally proposed to stop the distillation at a pot temperature of 277°C (530°F). However, experience with other distillations indicated that the pot would have a tendency to heat up further and not have a reproducible end-point, so the above conditions were used.)

The material balances for the liquefaction runs in the F-11 unit were calculated using a computer program written for a time-sharing terminal tied to Gulf's IBM 370 computer system. A complete description of the program along with a print-out of the program in computer language "APL has been given in report AF-913.

Sand Bath Heated Micro-Reactor

A micro-reactor of about 10 cc volume was used for coal liquefaction screening runs. The reactor consisted of a 0.5 inch O.D. by about 7 inch length tube with Swagelock fittings and provisions for a gas sampling line and thermocouple. Up-and-down agitation was provided by driving a drill press head with an air motor; the unit was operated at about 800 strokes per minute. By lowering the reactor into a preheated sand bath, its contents rapidly came to temperature (i.e., 2.3 minutes for a rise from ambient to 450°C). A schematic of the unit is given in Figure 8-2.

The charge to the reactor consisted of 2.0 g coal plus 3.0 g SRC-II heavy distillate. The reactor was pressurized to 500 psig with nitrogen, sealed, and agitated for 30 minutes prior to lowering into the sand bath. After the reaction and cooling, the reactor was vented. The reactor contents were carefully removed and the unit typically flushed with the solvent to be used for Soxhlet extraction (toluene or pyridine). A subsequent flush with methylene chloride was also done, and this latter solvent was stripped off. The sample bottles were placed in the Soxhlet thimble and crushed prior to the extraction. Routine Soxhlet procedures were then used.

Fluid Bed Drying/Oxidation

The simultaneous drying/oxidation runs were performed in two scales of equipment; a glass laboratory fluid-bed dryer (LFD) and a glass and steel pilot fluid-bed dryer (PFD). Figures 8-3 and 4 depict the LFD, while Figure 8-5 shows the PFD.

Over one-hundred runs in the LFD were nominally isothermal, i.e., the cold coal charge was rapidly heated (1-5 min) to a target temperature, then held for a specified time period. All runs in the PFD were nominally isothermal, but the heating time was much longer, i.e., 30-60 min. Six non-isothermal runs were made in the LFD where the coal was heated constantly at a rate of 10°C/min from 50°C to 210°C, then cooled at 5°C/min back to about 70°C. All runs in the LFD used a 30 g (wet) coal charge and a gas rate of 5 SCFH, sufficient to give good coal fluidization without excessive entrainment. Gas superficial velocity in the LFD

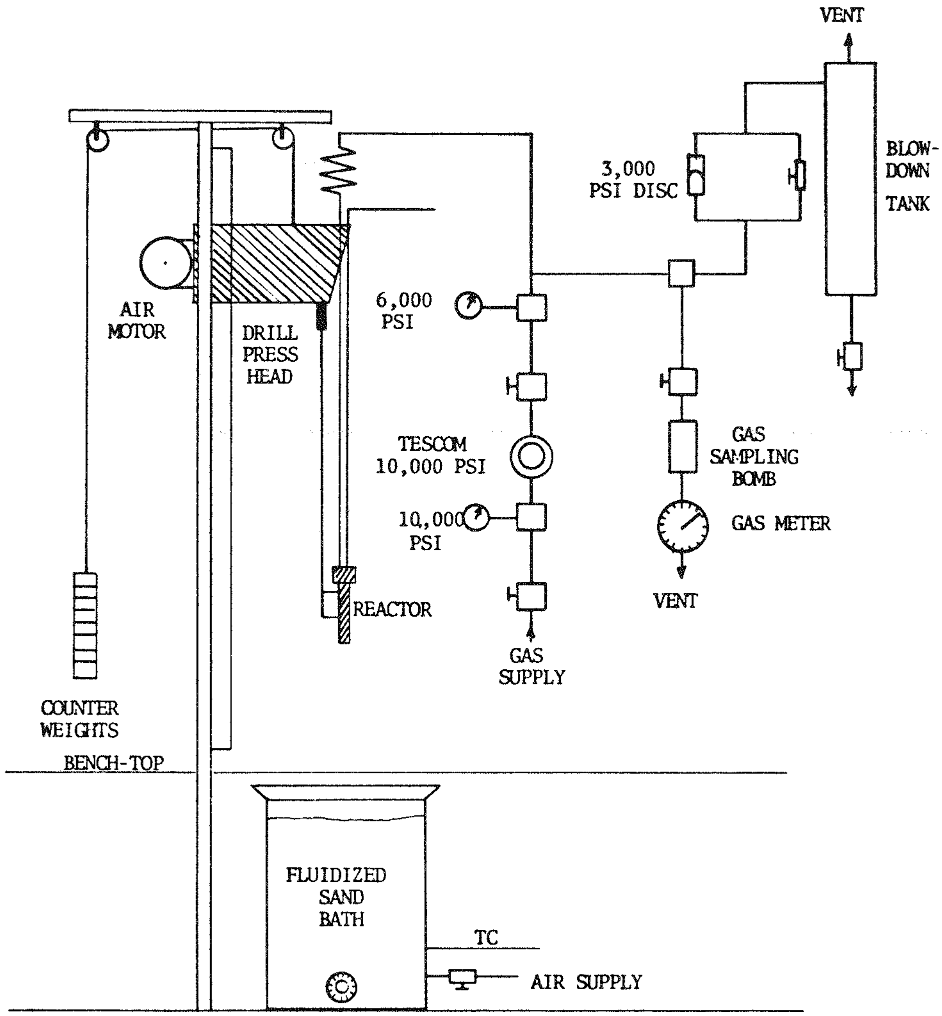

FIGURE 8-2 SCHEMATIC OF THE MICRO SCALE LIQUEFACTION UNIT

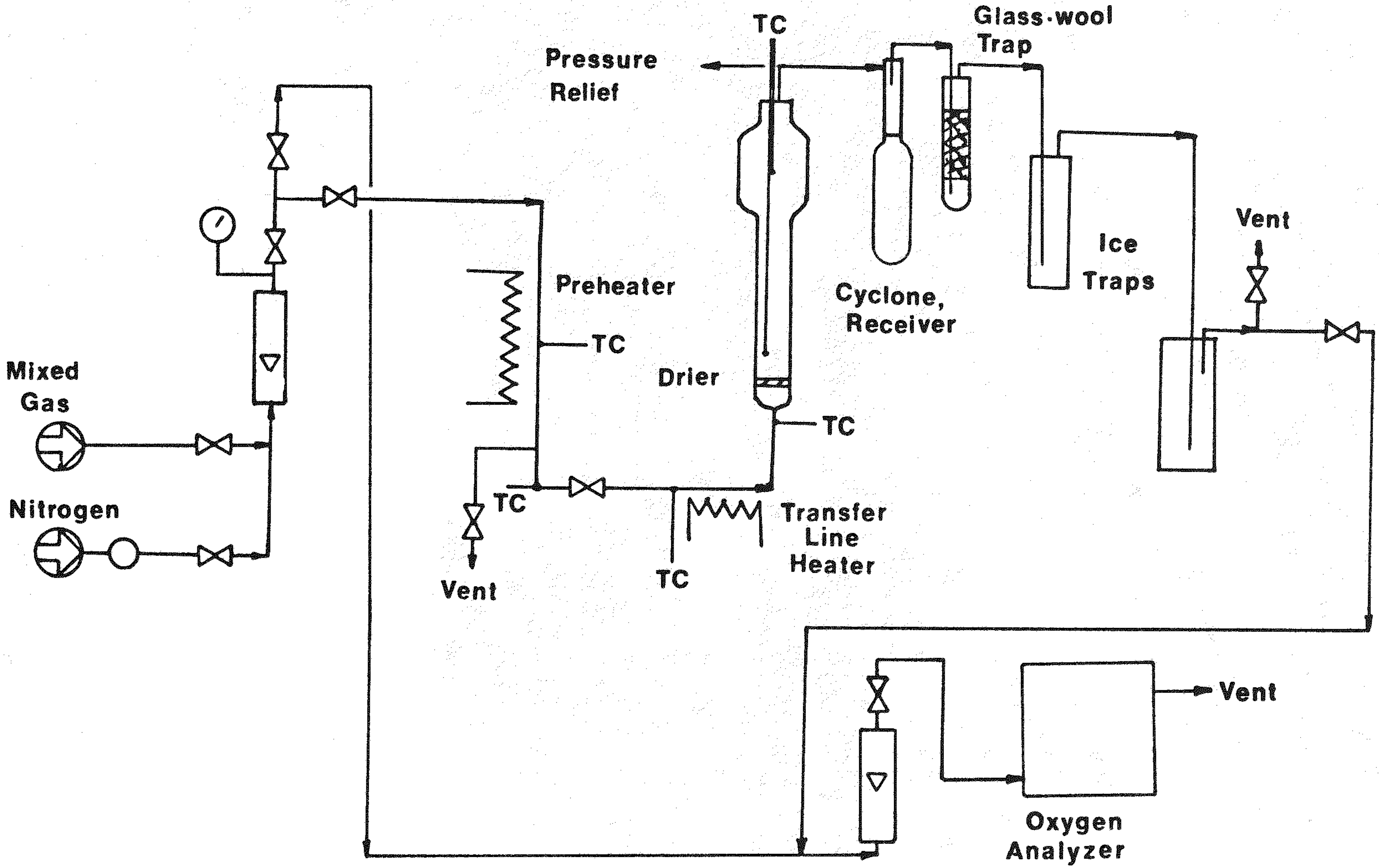

FIGURE 8-3 SCHEMATIC OF THE LABORATORY SCALE FLUID BED DRIER
(PILOT SCALE UNIT IS SIMILAR, BUT LARGER IN SIZE.)

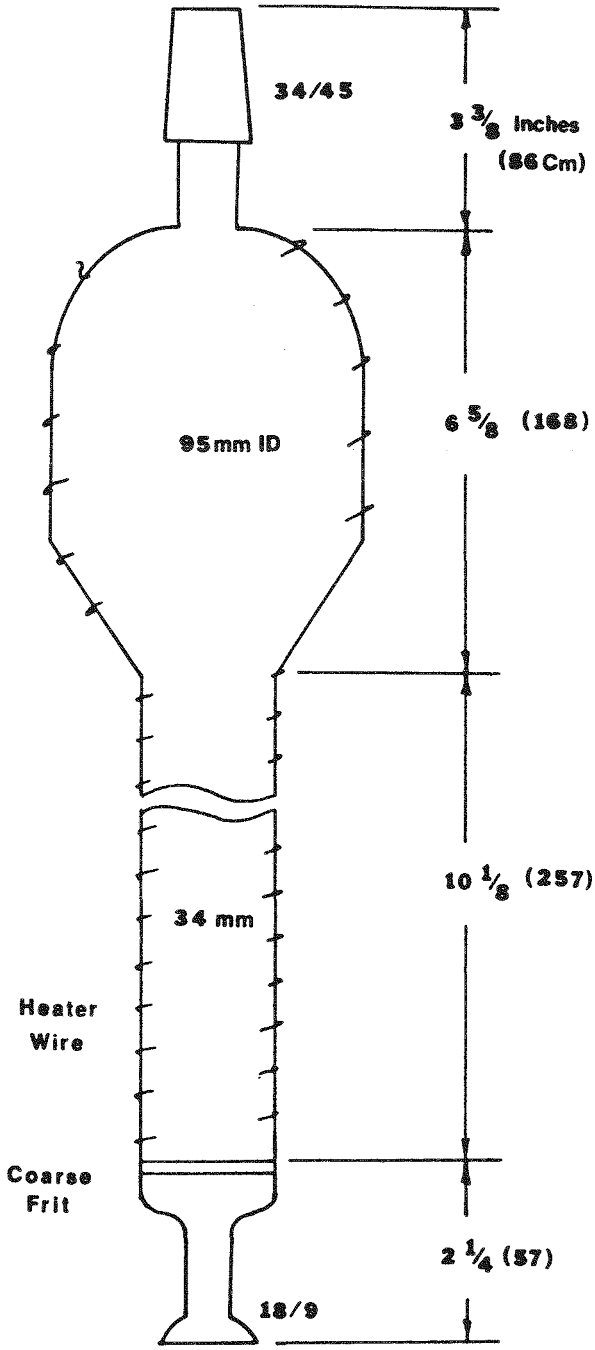

FIGURE 8-4 DRYING TUBE OF THE LABORATORY SCALE FLUID BED DRIER

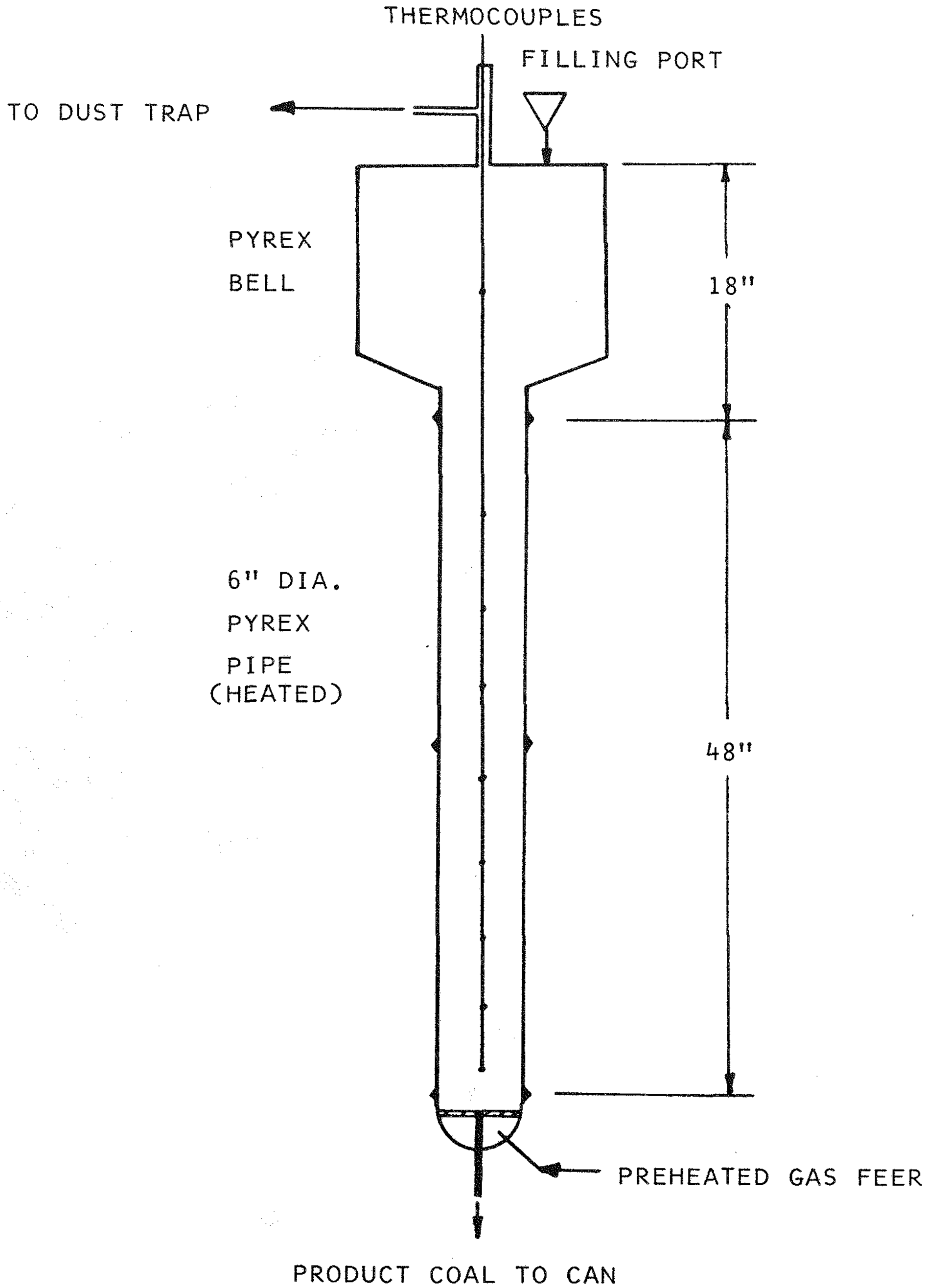

FIGURE 8-5 DRYING TUBE OF THE PILOT SCALE FLUID BED DRIER

was 4.03 cm/sec and the gas to solid ratio was 7.97 g gas/g dry coal-hr. In the PFD 7 kg (wet) coal charge was used in all but one run in which 10 kg was used. A typical gas feed rate was 31.6 SCFH, giving a superficial gas velocity of 1.36 cm/sec and a gas to solid ratio of 0.216 g gas/g dry coal-hr. The gas feed was either nitrogen, instrument air (dry), or a mixed gas ("MG") composed of nitrogen plus air and containing approximately 10.8 vol % oxygen.

Coal for all runs was crushed, pulverized and sieved under a nitrogen blanket. The coal powder was loaded into a nitrogen-purged reactor and care was taken to avoid any pre-exposure to oxygen. For reference, a variety of coal mesh sizes were prepared for the experiments. These are given in the discussion of results.

Figure 8-3 shows the operating flowsheet for the LFD. The gas feed was preheated in a 1 kW furnace, then conveyed through a traced line to the fluid bed inlet. By a valve manifold, the preheated gas was vented until the start of a run (SOR), then instantaneously directed to a fritted glass grid at the base of the bed. The reactor was electrically heated and insulated to maintain bed temperature. Bed temperature was automatically controlled with a West temperature controller using a control couple slightly above the surface of the fluidized coal. The controller was turned on immediately before gas flow, i.e., both the reactor and coal were cold at SOR. The temperature readout couple was immersed in the fluidized coal. A vertical trace within the bed showed temperature to be uniform (±3°C).

The exit gas from the reactor was passed into a wet scrubber containing surfactant solution to remove entrained fine coal particles. A cyclone collector and fiberglass filter were initially used but gave poor performance, the former due to insufficient velocity and the latter due to wetting. The particle-free gas air passed through a chilled trap to remove condensate and a slipstream was continuously analyzed for oxygen content using a Beckman polarographic instrument, calibrated with air and nitrogen. Bomb samples of gas were taken for carbon oxide analysis.

Operation of the PFD was similar to that of the LFD, with the major exception being due to the long heatup time. Nitrogen was fed during this period, then the gas feed was switched to MG for the isothermal hold period. The gas feed was preheated in a steam-jacketed exchanger and the reactor was electrically heated and insulated to maintain bed temperature. The precision of temperature control was not as good as that of the LFD. A wet scrubber was also used to remove entrained coal prior to gas analysis.

Oxygen Consumption and Analysis

The oxygen consumption rate (OCR) was determined by obtaining good analyses of the feed and product gas streams and calculating the value by material balance. The approach was used for selected runs with the LFD and PFD units.

Coal oxygen content is generally reported "by difference", i.e., total mass minus non-oxygen components. This is because actual oxygen content in coal is difficult to determine analytically. The oxygen is contained in several forms, both organic and inorganic. The inorganic oxygen comprises moisture (both free and bound), relatively volatile mineral oxygen such as carbonates and sulfates, and relatively non-volatile mineral oxygen such as silicates. The organic oxygen is contained in several functional groups, as previously reported in EPRI Project 713-1.

Three analytical methods, performed by three different labs, were evaluated for direct determination of oxygen in the raw and oxidized coals. Fast Neutron Activation Analysis (FNAA) was done by Uranium-West Labs.* In this method, oxygen in the coal reacts with a fast neutron and releases high-energy radiation which is counted. This method nominally detects total oxygen, including organic, inorganic, and moisture. Fluoride may interfere if present at high levels.

Analysis by the Unterzaucher method[5] was performed by Microanalysis Labs (P.O. Box 5088, Wilmington, DE 19808). In this method the coal is flushed briefly with nitrogen, then pyrolyzed to an "oxygen-free" char. The volatile oxygen-containing products are catalytically decomposed to CO_2 which is measured. This method nominally measures organic O and some mineral O (e.g., carbonates). Moisture should also appear as oxygen.

Dr. David Schlier performed analysis by "Charged Particle Activation Analysis" (CPAA). Briefly, the sample is irradiated in a cyclotron and then released γ (positron annihilation gamma rays) are counted

$$^{16}O\ (^{3}He,p)\ ^{18}F \text{ ------> } 5keV\ \gamma\ (\lambda=110 \text{ min})$$

Since carbon is also activated, the method yields the ratio (O/C), which is considered to be of higher accuracy than the determination of absolute oxygen. The method determines total oxygen, including organic, inorganic and moisture.

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The following is an outline of the specific procedure used.

The samples as received from Gulf were transferred to a dry box with an argon atmosphere. One mil aluminum packets were filled with coal powder (approximately 1 gram) and sealed. These were placed in a polyethylene bag along with standards in identical aluminum packets. This bag was then removed from the dry box and taken to the 60" Brookhaven cyclotron. The samples and standards were removed from the bag and placed on the sample wheel for irradiation.[6] The rotating wheel was irradiated with a 10 MeV helium-3 beam for 300 seconds at a beam current of 03 microamperes. The samples and standards in the aluminum packets were removed from the wheel, placed in counting vials and counted every 20 minutes for 5 hours utilizing a Searle model 1185 automatic gamma counter. The isotopes produces are fluorine-18 from oxygen and carbon-11 from carbon-12. The count data were punched onto paper tape which could be analyzed using a PDP 11/34 minicomputer. The computer program CLSQ was used to determine the quantities of carbon-11 and fluorine-18 present at the end of bombardment. The standards were analyzed using a linear regression and the oxygen to carbon ratio for the coal sample was computed from this equation.

Oxygen depth profiles were done on two samples from the EP series. The sample was placed in the inert atmosphere box and sieved using 60, 100, 200, 325, and 400 mesh sieves. A sample from each of these fractions was placed in the 1 mil aluminum packets and irradiated and analyzed as previously described.

Section 9

APPENDICES

APPENDIX A

Material Balance Calculations

RUN NUMBER	EP 201	
(EPRI PROJECT)		
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	14.68	
SOLVENT/COAL RATIO	60/40	
SOLVENT TYPE	SRC II HD	
COAL TYPE	BELLE AYR	
RESULTS:		
PERCENT SOLVATION	92.94	
OIL YIELD (GM/100GMAF)	26.00	

FEED:	G/HR	G/100G MAF
COAL MAF	324.30	100.00
COAL ASH	27.04	8.34
COAL MOISTURE	133.27	41.09
COAL TOTAL	484.61	149.43
SOLVENT TOTAL	724.61	223.44
HYDROGEN CONSUMPTION	3.97	1.23
TOTAL	1213.19	374.09
PRODUCTS:		
CO	.64	.20
CO2	5.47	1.69
C1-C3	15.84	4.88
C4-C6	3.27	1.01
H2O(OUT)	170.11	52.45
NH3 PLUS H2S	3.41	1.05
OILS	808.93	249.44
ASPHALTENES	80.49	24.82
PR-ASPHALTENES	75.39	23.25
PYRIDINE INSOLUBLES	49.92	15.39
TOTAL	1213.45	374.17

	C	H	N	O	S	METAL	TOTAL
COAL	246.63	32.59	4.38	182.21	1.54	17.27	484.61
SOLV	641.06	58.48	7.97	13.98	3.12		724.61
H2 CONS.		3.97					3.97
TOTAL	887.69	95.04	12.35	196.20	4.66	17.27	1213.19
GAS	16.76	4.12		4.34			25.21
FILT	.00	.00	.00	.00	.00		.00
WC	870.97	71.42	9.54	40.65	4.87	17.27	1014.72
NH3		.60	2.81				3.41
H2S		.00			.00		.00
H2O(OUT)		18.90		151.21			1,70.11
TOTAL	887.73	95.04	12.35	196.20	4.87	17.27	1213.45

RUN NUMBER (EPRI PROJECT)	EP 202
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	9.77
SOLVENT/COAL RATIO	60/40
SOLVENT TYPE	SRC II HD
COAL TYPE	BELLE AYR
RESULTS:	
PERCENT SOLVATION	89.87
OIL YIELD (GM/100GMAF)	14.41

FEED:	G/HR	G/100G MAF
COAL MAF	487.59	100.00
COAL ASH	40.66	8.34
COAL MOISTURE	200.37	41.09
COAL TOTAL	728.62	149.43
SOLVENT TOTAL	1089.45	223.44
HYDROGEN CONSUMPTION	8.36	1.71
TOTAL	1826.43	374.58
PRODUCTS:		
CO	.12	.03
CO2	8.30	1.70
C1-C3	18.43	3.78
C4-C6	3.66	.75
H2O(OUT)	251.74	51.63
NH3 PLUS H2S	5.06	1.04
OILS	1159.72	237.85
ASPHALTENES	159.91	32.80
PR-ASPHALTENES	130.41	26.75
PYRIDINE INSOLUBLES	90.04	18.47
TOTAL	1827.39	374.78

	C	H	N	O	S	METAL	TOTAL
COAL	370.81	48.99	6.58	273.96	2.32	25.96	728.62
SOLV	963.84	87.92	11.98	21.03	4.68		1089.45
H2 CONS.		8.36					8.36
TOTAL	1334.65	145.27	18.57	294.98	7.00	25.96	1826.43
GAS	19.63	4.77		6.10			30.51
FILT	.00	.00	.00	.00	.00		.00
WC	1315.15	111.64	14.40	65.11	7.83	25.96	1540.08
NH3		.89	4.16				5.06
H2S		.00			.00		.00
H2O(OUT)		27.97		223.77			251.74
TOTAL	1334.78	145.27	18.57	294.98	7.83	25.96	1827.39

RUN NUMBER	EP 203	
(EPRI PROJECT)		
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	4.65	
SOLVENT/COAL RATIO	60/40	
SOLVENT TYPE	SRC II HD	
COAL TYPE	BELLE AYR	

RESULTS:	
PERCENT SOLVATION	79.18
OIL YIELD (GM/100GMAF)	5.93

FEED:	G/HR	G/100G MAF
COAL MAF	1023.74	100.00
COAL ASH	85.36	8.34
COAL MOISTURE	420.69	41.09
COAL TOTAL	1529.79	149.43
SOLVENT TOTAL	2287.39	223.44
HYDROGEN CONSUMPTION	-6.71	-.66
TOTAL	3810.47	372.21
PRODUCTS:		
CO	1.74	.17
CO2	21.16	2.07
C1-C3	10.96	1.07
C4-C6	2.33	.23
H2O(OUT)	487.75	47.64
NH3 PLUS H2S	3.20	.31
OILS	2348.08	229.36
ASPHALTENES	288.54	28.18
PR-ASPHALTENES	348.23	34.02
PYRIDINE INSOLUBLES	298.48	29.16
TOTAL	3810.49	372.21

	C	H	N	O	S	METAL	TOTAL
COAL	778.54	102.87	13.82	575.19	4.86	54.50	1529.79
SOLV	2023.65	184.59	25.16	44.15	9.84		2287.39
H2 CONS.		-6.71					-6.71
TOTAL	2802.20	280.75	38.98	619.34	14.70	54.50	3810.47
GAS	16.97	2.85		16.38			36.20
FILT	.00	.00	.00	.00	.00		.00
WC	2785.24	223.25	37.05	169.40	13.89	54.50	3283.33
NH3		.41	1.93				2.35
H2S		.05			.81		.86
H2O(OUT)		54.19		433.56			487.75
TOTAL	2802.21	280.75	38.98	619.34	14.70	54.50	3810.49

RUN NUMBER (EPRI PROJECT)	EP 206	
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	17.40	
SOLVENT/COAL RATIO	66/34	
SOLVENT TYPE	SRC II HD	
COAL TYPE	NITROGEN DRIED BELLE AYR	
RESULTS:		
PERCENT SOLVATION	77.64	
OIL YIELD (GM/100GMAF)	-1.72	

FEED:	G/HR	G/100G MAF
COAL MAF	322.04	100.00
COAL ASH	26.91	8.36
COAL MOISTURE	21.21	6.59
COAL TOTAL	370.16	114.94
SOLVENT TOTAL	650.40	201.96
HYDROGEN CONSUMPTION	7.15	2.22
TOTAL	1027.71	319.13
PRODUCTS:		
CO	1.57	.49
CO2	16.98	5.27
C1-C3	13.54	4.21
C4-C6	5.56	1.73
H2O(OUT)	30.57	9.49
NH3 PLUS H2S	47.15	14.64
OILS	644.86	200.24
ASPHALTENES	83.36	25.88
PR-ASPHALTENES	85.19	26.45
PYRIDINE INSOLUBLES	98.93	30.72
TOTAL	1027.70	319.12

	C	H	N	O	S	METAL	TOTAL
COAL	221.34	18.12	39.29	74.53	1.38	15.50	370.16
SOLV	575.41	52.49	7.15	12.55	2.80		650.40
H2 CONS.		7.15					7.15
TOTAL	796.75	77.76	46.44	87.09	4.18	15.50	1027.71
GAS	20.57	3.84		13.24			37.65
FILT	.00	.00	.00	.00	.00		.00
WC	776.17	62.38	8.89	46.67	2.72	15.50	912.34
NH3		8.05	37.55				45.60
H2S		.09			1.46		1.55
H2O(OUT)		3.40		27.17			30.57
TOTAL	796.74	77.76	46.44	87.09	4.18	15.50	1027.70

RUN NUMBER (EPRI PROJECT)	EP 207	
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	12.12	
SOLVENT/COAL RATIO	66/34	
SOLVENT TYPE	SRC II HD	
COAL TYPE	BELLE AYR	
RESULTS:		
PERCENT SOLVATION	74.14	
OIL YIELD (GM/100GMAF)	-.40	

FEED:	G/HR	G/100G MAF
COAL MAF	434.57	100.00
COAL ASH	36.31	8.36
COAL MOISTURE	28.62	6.59
COAL TOTAL	499.50	114.94
SOLVENT TOTAL	965.62	222.20
HYDROGEN CONSUMPTION	12.63	2.91
TOTAL	1477.75	340.05
PRODUCTS:		
CO	1.66	.38
CO2	16.60	3.82
C1-C3	12.93	2.98
C4-C6	7.85	1.81
H2O(OUT)	59.52	13.70
NH3 PLUS H2S	3.81	.88
OILS	963.90	221.81
ASPHALTENES	154.20	35.48
PR-ASPHALTENES	108.77	25.03
PYRIDINE INSOLUBLES	148.69	34.22
TOTAL	1477.93	340.09

	C	H	N	O	S	METAL	TOTAL
COAL	330.23	27.04	5.86	111.20	2.06	23.12	499.50
SOLV	854.29	77.93	10.62	18.64	4.15		965.62
H2 CONS.		12.63					12.63
TOTAL	1184.51	117.59	16.48	129.84	6.22	23.12	1477.75
GAS	21.98	4.05		13.02			39.04
FILT	.00	.00	.00	.00	.00		.00
WC	1162.71	106.43	14.57	63.91	4.81	23.12	1375.56
NH3		.41	1.91				2.32
H2S		.09			1.40		1.49
H2O(OUT)		6.61		52.91			59.52
TOTAL	1184.69	117.59	16.48	129.84	6.22	23.12	1477.93

RUN NUMBER	EP 208	
(EPRI PROJECT)		
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	6.49	
SOLVENT/COAL RATIO	66/34	
SOLVENT TYPE	SRC II HD	
COAL TYPE	BELLE AYR	
RESULTS:		
PERCENT SOLVATION	50.63	
OIL YIELD (GM/100GMAF)	.60	

FEED:	G/HR	G/100G MAF
COAL MAF	811.75	100.00
COAL ASH	67.83	8.36
COAL MOISTURE	53.46	6.59
COAL TOTAL	933.04	114.94
SOLVENT TOTAL	1803.72	222.20
HYDROGEN CONSUMPTION	-33.46	-4.12
TOTAL	2703.31	333.02
PRODUCTS:		
CO	2.10	.26
CO2	23.47	2.89
C1-C3	12.27	1.51
C4-C6	5.60	.69
H2O(OUT)	126.74	15.61
NH3 PLUS H2S	10.63	1.31
OILS	1808.56	222.80
ASPHALTENES	189.97	23.40
PR-ASPHALTENES	55.73	6.86
PYRIDINE INSOLUBLES	468.60	57.73
TOTAL	2703.68	333.07

	C	H	N	O	S	METAL	TOTAL
COAL	616.84	50.50	10.95	207.71	3.85	43.18	933.04
SOLV	1595.75	145.56	19.84	34.81	7.76		1803.72
H2 CONS.		-33.46					-33.46
TOTAL	2212.59	162.61	30.79	242.52	11.61	43.18	2703.31
GAS	21.62	3.57		18.26			43.44
FILT	.00	.00	.00	.00	.00		.00
WC	2191.35	143.41	24.35	111.60	8.97	43.18	2522.87
NH3		1.38	6.44				7.82
H2S		.16			2.64		2.80
H2O(OUT)		14.08		112.66			126.74
TOTAL	2212.96	162.61	30.79	242.52	11.61	43.18	2703.68

RUN NUMBER (EPRI PROJECT)	EP 210A
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	19.96
SOLVENT/COAL RATIO	66/34
SOLVENT TYPE	SRC II HD
COAL TYPE	NITROGEN DRIED BELLE AYR
RESULTS:	
PERCENT SOLVATION	59.65
OIL YIELD (GM/100GMAF)	8.54

FEED:	G/HR	G/100G MAF
COAL MAF	263.05	100.00
COAL ASH	21.98	8.36
COAL MOISTURE	17.33	6.59
COAL TOTAL	302.36	114.94
SOLVENT TOTAL	587.39	223.29
HYDROGEN CONSUMPTION	-2.41	-.92
TOTAL	887.34	337.32
PRODUCTS:		
CO	1.78	.68
CO2	17.68	6.72
C1-C3	13.94	5.30
C4-C6	9.72	3.69
H2O(OUT)	10.09	3.83
NH3 PLUS H2S	2.95	1.12
OILS	609.86	231.84
ASPHALTENES	74.88	28.47
PR-ASPHALTENES	18.30	6.96
PYRIDINE INSOLUBLES	128.13	48.71
TOTAL	887.34	337.32

	C	H	N	O	S	METAL	TOTAL
COAL	199.89	16.37	3.55	67.31	1.25	13.99	302.36
SOLV	519.66	47.40	6.46	11.34	2.53		587.39
H2 CONS.		-2.41					-2.41
TOTAL	719.55	61.36	10.01	78.65	3.77	13.99	887.34
GAS	24.68	4.57		13.88			43.13
FILT	.00	.00	.00	.00	.00		.00
WC	694.87	55.21	7.99	55.81	3.30	13.99	831.17
NH3		.43	2.01				2.45
H2S		.03			.48		.51
H2O(OUT)		1.12		8.97			10.09
TOTAL	719.55	61.36	10.01	78.65	3.77	13.99	887.34

RUN NUMBER (EPRI PROJECT)	EP 210B
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	10.55
SOLVENT/COAL RATIO	66/34
SOLVENT TYPE	SRC II HD
COAL TYPE	BELLE AYR

RESULTS:	
PERCENT SOLVATION	58.83
OIL YIELD (GM/100GMAF)	-5.72

FEED:	G/HR	G/100G MAF
COAL MAF	499.06	100.00
COAL ASH	41.70	8.36
COAL MOISTURE	32.87	6.59
COAL TOTAL	573.64	114.94
SOLVENT TOTAL	1108.93	222.20
HYDROGEN CONSUMPTION	1.80	.36
TOTAL	1684.37	337.51
PRODUCTS:		
CO	2.53	.51
CO2	26.79	5.37
C1-C3	19.86	3.98
C4-C6	13.36	2.68
H2O(OUT)	65.90	13.20
NH3 PLUS H2S	7.68	1.54
OILS	1080.38	216.48
ASPHALTENES	160.11	32.08
PR-ASPHALTENES	60.63	12.15
PYRIDINE INSOLUBLES	247.17	49.53
TOTAL	1684.41	337.51

	C	H	N	O	S	METAL	TOTAL
COAL	379.24	31.05	6.73	127.70	2.37	26.55	573.64
SOLV	981.07	89.49	12.20	21.40	4.77		1108.93
H2 CONS.		1.80					1.80
TOTAL	1360.31	122.34	18.93	149.10	7.14	26.55	1684.37
GAS	35.20	6.42		20.92			62.54
FILT	.00	.00	.00	.00	.00		.00
WC	1325.14	107.81	16.59	69.60	2.59	26.55	1548.28
NH3		.50	2.34				2.85
H2S		.28			4.55		4.84
H2O(OUT)		7.32		58.58			65.90
TOTAL	1360.35	122.34	18.93	149.10	7.14	26.55	1684.41

RUN NUMBER	EP 211
(EPRI PROJECT)	
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	16.73
SOLVENT/COAL RATIO	66/34
SOLVENT TYPE	SRC II HD
COAL TYPE	BELLE AYR

RESULTS:	
PERCENT SOLVATION	81.91
OIL YIELD (GM/100GMAF)	21.72

FEED:	G/HR	G/100G MAF
COAL MAF	314.75	100.00
COAL ASH	26.30	8.36
COAL MOISTURE	20.73	6.59
COAL TOTAL	361.78	114.94
SOLVENT TOTAL	699.38	222.20
HYDROGEN CONSUMPTION	1.42	.45
TOTAL	1062.59	337.60

PRODUCTS:		
CO	1.24	.39
CO2	12.41	3.94
C1-C3	9.69	3.08
C4-C6	6.76	2.15
H2O(OUT)	50.82	16.15
NH3 PLUS H2S	4.35	1.38
OILS	767.73	243.92
ASPHALTENES	55.82	17.73
PR-ASPHALTENES	70.51	22.40
PYRIDINE INSOLUBLES	83.24	26.45
TOTAL	1062.55	337.59

	C	H	N	O	S	METAL	TOTAL
COAL	239.18	19.58	4.25	80.54	1.49	16.74	361.78
SOLV	618.74	56.44	7.69	13.50	3.01		699.38
H2 CONS.		1.42					1.42
TOTAL	857.92	77.45	11.94	94.04	4.50	16.74	1062.59
GAS	17.19	3.17		9.73			30.09
FILT	.00	.00	.00	.00	.00		.00
WC	840.70	68.19	10.69	39.14	1.83	16.74	977.29
NH3		.27	1.24				1.51
H2S		.17			2.67		2.84
H2O(OUT)		5.65		45.17			50.82
TOTAL	857.88	77.45	11.94	94.04	4.50	16.74	1062.55

RUN NUMBER (EPRI PROJECT)	EP 212	
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	16.27	
SOLVENT/COAL RATIO	66/34	
SOLVENT TYPE	SRC II HD	
COAL TYPE	NITROGEN DRIED BELLE EYR	

RESULTS:	
PERCENT SOLVATION	70.94
OIL YIELD (GM/100GMAF)	-1.27

FEED:	G/HR	G/100G MAF
COAL MAF	322.67	100.00
COAL ASH	26.96	8.36
COAL MOISTURE	21.25	6.59
COAL TOTAL	370.89	114.94
SOLVENT TOTAL	720.58	223.32
HYDROGEN CONSUMPTION	-.56	-.17
TOTAL	1090.90	338.09
PRODUCTS:		
CO	2.27	.70
CO2	18.31	5.67
C1-C3	21.65	6.71
C4-C6	14.72	4.56
H2O(OUT)	35.34	10.95
NH3 PLUS H2S	2.70	.84
OILS	716.49	222.05
ASPHALTENES	61.87	19.17
PR-ASPHALTENES	96.80	30.00
PYRIDINE INSOLUBLES	120.75	37.42
TOTAL	1090.90	338.08

	C	H	N	O	S	METAL	TOTAL
COAL	245.20	20.07	4.35	82.57	1.53	17.17	370.89
SOLV	637.43	58.15	8.00	13.91	3.10		720.58
H2 CONS.		-.56					-.56
TOTAL	882.63	77.66	12.35	96.47	4.63	17.17	1090.90
GAS	35.30	7.05		14.61			56.95
FILT	.00	.00	.00	.00	.00		.00
WC	847.32	66.33	10.98	50.45	3.66	17.17	995.90
NH3		.29	1.37				1.67
H2S		.06			.97		1.03
H2O(OUT)		3.93		31.41			35.34
TOTAL	882.62	77.66	12.35	96.47	4.63	17.17	1090.90

RUN NUMBER (EPRI PROJECT)	EP 213	
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	12.38	
SOLVENT/COAL RATIO	66/34	
SOLVENT TYPE	SRC II HD	
COAL TYPE	NITROGEN DRIED BELLE AYR	

RESULTS:	
PERCENT SOLVATION	72.43
OIL YIELD (GM/100GMAF)	-1.94

FEED:	G/HR	G/100G MAF
COAL MAF	423.93	100.00
COAL ASH	35.42	8.36
COAL MOISTURE	27.92	6.59
COAL TOTAL	487.27	114.94
SOLVENT TOTAL	946.61	223.29
HYDROGEN CONSUMPTION	-4.17	-.98
TOTAL	1429.72	337.25
PRODUCTS:		
CO	2.43	.57
CO2	26.83	6.33
C1-C3	11.56	2.73
C4-C6	7.83	1.85
H2O(OUT)	46.36	10.94
NH3 PLUS H2S	1.89	.45
OILS	938.40	221.36
ASPHALTENES	95.20	22.46
PR-ASPHALTENES	146.88	34.65
PYRIDINE INSOLUBLES	152.32	35.93
TOTAL	1429.71	337.25

	C	H	N	O	S	METAL	TOTAL
COAL	322.14	26.37	5.72	108.48	2.01	22.55	487.27
SOLV	837.47	76.39	10.41	18.27	4.07		946.61
H2 CONS.		-4.17					-4.17
TOTAL	1159.61	98.60	16.13	126.75	6.08	22.55	1429.72
GAS	24.02	3.74		20.89			48.65
FILT	.00	.00	.00	.00	.00		.00
WC	1135.57	89.40	14.70	64.64	5.93	22.55	1332.80
NH3		.31	1.43				1.73
H2S		.01			.15		.16
H2O(OUT)		5.15		41.21			46.36
TOTAL	1159.60	98.60	16.13	126.75	6.08	22.55	1429.71

RUN NUMBER (EPRI PROJECT)	EP 214
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	17.09
SOLVENT/COAL RATIO	60/40
SOLVENT TYPE	SRC II HD
COAL TYPE	MILDLY DRY/OXIDIZED (PFD5) BELLE

RESULTS:	
PERCENT SOLVATION	80.62
OIL YIELD (GM/100GMAF)	12.50

FEED:	G/HR	G/100G MAF
COAL MAF	280.02	100.00
COAL ASH	23.39	8.35
COAL MOISTURE	110.52	39.47
COAL TOTAL	413.92	147.82
SOLVENT TOTAL	625.11	223.24
HYDROGEN CONSUMPTION	.18	.06
TOTAL	1039.21	371.12
PRODUCTS:		
CO	1.77	.63
CO2	16.24	5.80
C1-C3	15.12	5.40
C4-C6	9.23	3.30
H2O(OUT)	125.02	44.65
NH3 PLUS H2S	3.39	1.21
OILS	660.11	235.74
ASPHALTENES	70.60	25.21
PR-ASPHALTENES	60.01	21.43
PYRIDINE INSOLUBLES	77.66	27.73
TOTAL	1039.15	371.10

	C	H	N	O	S	METAL	TOTAL
COAL	212.97	27.63	3.78	153.30	1.33	14.91	413.92
SOLV	553.03	50.45	6.88	12.06	2.69		625.11
H2 CONS.		.18					.18
TOTAL	766.00	78.26	10.66	165.36	4.02	14.91	1039.21
GAS	24.79	4.75		12.82			42.36
FILT	.00	.00	.00	.00	.00		.00
WC	741.16	59.19	9.09	41.41	2.62	14.91	868.38
NH3		.34	1.57				1.90
H2S		.09			1.40		1.48
H2O(OUT)		13.89		111.13			125.02
TOTAL	765.95	78.26	10.66	165.36	4.02	14.91	1039.15

RUN NUMBER (EPRI PROJECT)	EP 215	
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	16.67	
SOLVENT/COAL RATIO	63/37	
SOLVENT TYPE	SRC II HD	
COAL TYPE	MILDLY DRY/OXIDIZED (PFD 2) BELLE	

RESULTS:	
PERCENT SOLVATION	81.03
OIL YIELD (GM/100GMAF)	13.83

FEED:	G/HR	G/100G MAF
COAL MAF	299.92	100.00
COAL ASH	25.02	8.34
COAL MOISTURE	70.36	23.46
COAL TOTAL	395.30	131.80
SOLVENT TOTAL	669.79	223.33
HYDROGEN CONSUMPTION	1.50	.50
TOTAL	1066.59	355.63
PRODUCTS:		
CO	1.42	.47
CO2	13.98	4.66
C1-C3	13.51	4.51
C4-C6	8.75	2.92
H2O(OUT)	98.33	32.79
NH3 PLUS H2S	3.25	1.08
OILS	711.28	237.16
ASPHALTENES	86.58	28.87
PR-ASPHALTENES	47.48	15.83
PYRIDINE INSOLUBLES	81.93	27.32
TOTAL	1066.52	355.61

	C	H	N	O	S	METAL	TOTAL
COAL	228.00	24.27	4.05	121.60	1.42	15.96	395.30
SOLV	592.57	54.05	7.37	12.93	2.88		669.79
H2 CONS.		1.50					1.50
TOTAL	820.56	79.82	11.41	134.53	4.30	15.96	1066.59
GAS	22.38	4.31		10.97			37.66
FILT	.00	.00	.00	.00	.00		.00
WC	798.11	64.16	9.76	36.15	3.13	15.96	927.28
NH3		.35	1.65				2.01
H2S		.07			1.17		1.25
H2O(OUT)		10.93		87.41			98.33
TOTAL	820.50	79.82	11.41	134.53	4.30	15.96	1066.52

RUN NUMBER (EPRI PROJECT)	EP 218
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	18.19
SOLVENT/COAL RATIO	66/34
SOLVENT TYPE	SRC II HD
COAL TYPE	DRY/OXIDIZED(PFD14-16) BELLE
RESULTS:	
PERCENT SOLVATION	78.34
OIL YIELD (GM/100GMAF)	8.39

FEED:	G/HR	G/100G MAF
COAL MAF	289.86	100.00
COAL ASH	24.18	8.34
COAL MOISTURE	15.83	5.46
COAL TOTAL	329.87	113.80
SOLVENT TOTAL	646.27	222.96
HYDROGEN CONSUMPTION	.36	.13
TOTAL	976.51	336.89
PRODUCTS:		
CO	1.40	.48
CO2	11.04	3.81
C1-C3	14.91	5.14
C4-C6	9.72	3.35
H2O(OUT)	39.41	13.60
NH3 PLUS H2S	3.54	1.22
OILS	670.58	231.35
ASPHALTENES	77.10	26.60
PR-ASPHALTENES	61.86	21.34
PYRIDINE INSOLUBLES	86.96	30.00
TOTAL	976.53	336.90

	C	H	N	O	S	METAL	TOTAL
COAL	220.22	17.67	3.91	71.28	1.38	15.42	329.87
SOLV	571.76	52.15	7.11	12.47	2.78		646.27
H2 CONS.		.36					.36
TOTAL	791.98	70.19	11.02	83.75	4.15	15.42	976.51
GAS	23.48	4.77		8.83			37.08
FILT	.00	.00	.00	.00	.00		.00
WC	768.53	60.50	8.71	39.89	3.46	15.42	896.50
NH3		.50	2.31				2.81
H2S		.04			.69		.73
H2O(OUT)		4.38		35.03			39.41
TOTAL	792.01	70.19	11.02	83.75	4.15	15.42	976.53

RUN NUMBER (EPRI PROJECT)	EP 219
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	11.29
SOLVENT/COAL RATIO	66.3/33.7
SOLVENT TYPE	SRC II HD
COAL TYPE	DRIED/OXIDIZED BELLE AYR

RESULTS:	
PERCENT SOLVATION	72.81
OIL YIELD (GM/100GMAF)	-4.15

FEED:	G/HR	G/100G MAF
COAL MAF	467.11	100.00
COAL ASH	38.97	8.34
COAL MOISTURE	25.52	5.46
COAL TOTAL	531.60	113.80
SOLVENT TOTAL	1041.48	222.96
HYDROGEN CONSUMPTION	2.66	.57
TOTAL	1575.74	337.33
PRODUCTS:		
CO	3.46	.74
CO2	19.62	4.20
C1-C3	20.14	4.31
C4-C6	6.86	1.47
H2O(OUT)	63.43	13.58
NH3 PLUS H2S	6.22	1.33
OILS	1022.11	218.81
ASPHALTENES	154.34	33.04
PR-ASPHALTENES	113.57	24.31
PYRIDINE INSOLUBLES	165.98	35.53
TOTAL	1575.73	337.33

	C	H	N	O	S	METAL	TOTAL
COAL	354.89	28.48	6.30	114.86	2.22	24.85	531.60
SOLV	921.40	84.05	11.46	20.10	4.48		1041.48
H2 CONS.		2.66					2.66
TOTAL	1276.29	115.18	17.76	134.96	6.70	24.85	1575.74
GAS	28.41	5.43		16.24			50.08
FILT	.00	.00	.00	.00	.00		.00
WC	1247.88	101.74	13.57	62.34	5.63	24.85	1456.00
NH3		.90	4.19				5.09
H2S		.07			1.07		1.13
H2O(OUT)		7.05		56.38			63.43
TOTAL	1276.28	115.18	17.76	134.96	6.70	24.85	1575.73

RUN NUMBER	EP 220	
(EPRI PROJECT)		
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	5.03	
SOLVENT/COAL RATIO	66.3/33.7	
SOLVENT TYPE	SRC II HD	
COAL TYPE	BELLE AYR	
RESULTS:		
PERCENT SOLVATION	66.63	
OIL YIELD (GM/100GMAF)	.46	

FEED:	G/HR	G/100G MAF
COAL MAF	1048.06	100.00
COAL ASH	87.43	8.34
COAL MOISTURE	57.25	5.46
COAL TOTAL	1192.73	113.80
SOLVENT TOTAL	2336.75	222.96
HYDROGEN CONSUMPTION	-3.86	-.37
TOTAL	3525.63	336.40
PRODUCTS:		
CO	5.80	.55
CO2	43.94	4.19
C1-C3	27.92	2.66
C4-C6	18.36	1.75
H2O(OUT)	112.62	10.75
NH3 PLUS H2S	8.23	.79
OILS	2341.58	223.42
ASPHALTENES	291.46	27.81
PR-ASPHALTENES	238.46	22.75
PYRIDINE INSOLUBLES	437.18	41.71
TOTAL	3525.55	336.39

	C	H	N	O	S	METAL	TOTAL
COAL	796.27	63.89	14.13	257.72	4.98	55.75	1192.73
SOLV	2067.32	188.58	25.70	45.10	10.05		2336.75
H2 CONS.		-3.86					-3.86
TOTAL	2863.59	248.61	39.84	302.82	15.02	55.75	3525.63
GAS	51.78	8.98		35.25			96.01
FILT	.00	.00	.00	.00	.00		.00
WC	2811.73	225.82	34.14	167.46	13.79	55.75	3308.69
NH3		1.22	5.70				6.92
H2S		.08			1.24		1.31
H2O(OUT)		12.51		100.10			112.62
TOTAL	2863.52	248.61	39.84	302.82	15.02	55.75	3525.55

RUN NUMBER (EPRI PROJECT)	EP 221
REACTION CONDITIONS:	
TEMPERATURE -C/-F	450.0 842.0
PRESSURE PSIG	1750
SPACE TIME (MIN)	18.70
SOLVENT/COAL RATIO	66/34
SOLVENT TYPE	SRC II HD
COAL TYPE	DRY/OXIDIZED (PFD17) BELLE

RESULTS:	
PERCENT SOLVATION	77.67
OIL YIELD (GM/100GMAF)	7.52

	G/HR	G/100G MAF
FEED:		
COAL MAF	280.82	100.00
COAL ASH	23.43	8.34
COAL MOISTURE	16.69	5.94
COAL TOTAL	320.93	114.29
SOLVENT TOTAL	628.77	223.91
HYDROGEN CONSUMPTION	.96	.34
TOTAL	950.67	338.53
PRODUCTS:		
CO	2.16	.77
CO2	15.35	5.47
C1-C3	20.26	7.21
C4-C6	4.05	1.44
H2O(OUT)	34.29	12.21
NH3 PLUS H2S	4.55	1.62
OILS	649.89	231.43
ASPHALTENES	69.60	24.78
PR-ASPHALTENES	64.38	22.93
PYRIDINE INSOLUBLES	86.13	30.67
TOTAL	950.66	338.53

	C	H	N	O	S	METAL	TOTAL
COAL	213.36	17.27	3.79	70.25	1.33	14.94	320.93
SOLV	556.27	50.74	6.92	12.14	2.70		628.77
H2 CONS.		.96					.96
TOTAL	769.63	68.97	10.70	82.38	4.04	14.94	950.67
GAS	24.44	4.99		12.39			41.82
FILT	.00	.00	.00	.00	.00		.00
WC	745.19	59.48	7.69	39.51	3.20	14.94	870.00
NH3		.64	3.01				3.65
H2S		.05			.84		.89
H2O(OUT)		3.81		30.48			34.29
TOTAL	769.63	68.97	10.70	82.38	4.04	14.94	950.66

RUN NUMBER (EPRI PROJECT)	EP 223	
REACTION CONDITIONS:		
TEMPERATURE -C/-F	450.0	842.0
PRESSURE PSIG	1750	
SPACE TIME (MIN)	18.74	
SOLVENT/COAL RATIO	66/34	
SOLVENT TYPE	SRC II HD	
COAL TYPE	DRY/OXIDIZED (PFD18) BELLE	

RESULTS:	
PERCENT SOLVATION	81.30
OIL YIELD (GM/100GMAF)	7.05

FEED:	G/HR	G/100G MAF
COAL MAF	278.33	100.00
COAL ASH	23.21	8.34
COAL MOISTURE	24.45	8.78
COAL TOTAL	325.99	117.12
SOLVENT TOTAL	621.33	223.24
HYDROGEN CONSUMPTION	1.57	.56
TOTAL	948.89	340.92
PRODUCTS:		
CO	1.58	.57
CO2	12.65	4.54
C1-C3	14.49	5.21
C4-C6	3.21	1.15
H2O(OUT)	51.47	18.49
NH3 PLUS H2S	3.94	1.41
OILS	640.96	230.29
ASPHALTENES	72.66	26.11
PR-ASPHALTENES	72.66	26.11
PYRIDINE INSOLUBLES	75.25	27.04
TOTAL	948.88	340.92

	C	H	N	O	S	METAL	TOTAL
COAL	211.48	17.99	3.75	76.63	1.32	14.81	325.99
SOLV	549.69	50.14	6.83	11.99	2.67		621.33
H2 CONS.		1.57					1.57
TOTAL	761.18	69.70	10.59	88.62	3.99	14.81	948.89
GAS	18.22	3.62		10.10			31.94
FILT	.00	.00	.00	.00	.00		.00
WC	742.95	59.70	7.54	32.78	3.77	14.81	861.54
NH3		.65	3.05				3.70
H2S		.01			.22		.24
H2O(OUT)		5.72		45.75			51.47
TOTAL	761.17	69.70	10.59	88.62	3.99	14.81	948.88

APPENDIX B

Task 2 Report: Literature Survey

INVESTIGATION OF THE LIQUEFACTION OF PARTIALLY DRIED AND OXIDIZED COALS

TASK 2 REPORT: LITERATURE SURVEY

Research Project RP779-25

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Clean Liquid and Solid Fuels

Section 1

ABSTRACT

One of the first tasks in the project "Investigation of the Liquefaction of Partially Dried and Oxidized Coals" is that of carrying out a detailed literature survey. The goals are to establish a range of safe operating conditions to dry and partially oxidize coal and to survey available information on the rates and mechanism of oxidizing coal. The survey has been limited to reaction conditions considered likely for the primary project; namely, temperatures from ambient to 200°C and oxygen concentrations from 0 to 21% (air).

Pulverized coal can be safely dried and partially oxidized in a fluid-bed unit over a fairly wide range of conditions. Based on Bureau of Mines publications and data, coal will not form an explosive mixture with a gas stream containing nitrogen with less than 10% oxygen. Generally, flammability decreases as rank, moisture content and coal particle size increases. With increasing temperature, dust clouds are much less likely to ignite than piles of dust; therefore, care must be taken to insure that the coal is well fluidized.

The initial step of the surface oxidation of coal appears to be the formation of peroxides through the oxidation of alkyl groups. This is followed by series and parallel reactions leading to carbonyl, carboxyl and hydroxyl functional groups. The loss of CO, CO_2 and H_2O also occurs at this time.

Numerous emperical equations have been used to correlate the reaction rate data. The most popular approach is the use of the Elovich equation of the following form:

$$\frac{dq}{dt} = Ae^{-\alpha q}$$

where q is the quantity of oxygen adsorbed, t is time, and A and α are constants. This approach apparently correlates fairly well with a diffusion controlled process. Reported activation energies also indicate that the reactions are diffusion controlled.

Section 2

INTRODUCTION AND SCOPE OF THE LITERATURE SURVEY

The literature survey was undertaken to determine the following:

1. A range of safe coal drying conditions for both the subbituminous coal (Belle Ayr) and the bituminous Pittsburgh seam coal.
2. Available data on fluid bed drying and surface oxidation of coal.

In surveying the literature, it is evident that there are a wide variety of reasons for studying the mechanism of coal oxidation. Some researchers are interested in understanding the spontaneous ignition of coal to develop an "early warning" technique which will be helpful in the coal mines. Others are interested in the coking properties of the coals and how they are influenced by weathering. In addition, caking coals can be partially oxidized to avoid agglomeration during the initial stages of gasification. It is also noted that pre-oxidation of coal affects flotation efficiency by changing surface characteristics, thereby allowing surfactants to be more effective.

Others also study coal partial oxidation as a purely academic interest to study the surface changes of coal when exposed to oxidation and their work is not directed by any stated practical application.

Coal can obviously be oxidized by liquid-solid and by gas-solid reactions. Due to the specific goal of this project, this survey has been limited to the gas-solid reactions, with an upper temperature limit of 250°C.

This survey will be divided in four segments. In the first the available literature will be described, emphasizing the reasons for carrying out the work, the analytical and experimental techniques used (if unusual) and the results. In the second segment the mechanism of coal oxidations proposed by the various authors will be illustrated. Rate data will be discussed in the third segment, and explosion limits of coal/gas mixtures will be given in the fourth.

This search was made using both the open literature and the computer search files of DOE and API. The specific descriptors were coal, coal/oxidation, oxidation, drying, surface properties, coal fines, dusts, powders, flammability, combustion properties, and ignition. Emphasis was placed upon the period of 1964 through April 1979; however, prior literature was checked using Chemical Abstracts and references cited in appropriate publications. Copies of most of the pertinent articles were obtained.

It is also strongly noted that Drs. Hertzberg and Cashdollar of the Bureau of Mines (Pittsburgh Energy Technical Center) were contacted concerning explosion limits. These discussions greatly helped the literature search.

Section 3

GENERAL DESCRIPTION OF THE OXIDATION OF COAL IN THE GAS-SOLID SYSTEM

SPONTANEOUS COMBUSTION OF COAL

The concentration of CO in the air of coal mines is monitored continuously to detect incipient heating of coal which can result in spontaneous combustion and explosion. Chamberlain, et al[1] studied the oxidation of coal and gases produced at low temperature on the assumption that there may be other gases (e.g., those contributing to gob-stink) that may serve as a better indicator of incipient heating than CO. They heated three English coals, size 425-250μ (40-60 mesh), from -40° to 120°C at 0.5°C/minute in ten O_2/N_2 mixtures ranging from 0.5% to 100.0% O_2. The gases produced were dried and analyzed for ketones, aldehydes, alcohols, aliphatic acids and CO. The authors concluded that:

1. No gases are detected when coal is oxidized below 10°C. However, the oxidation reaction could start at lower temperatures, but its detection depends on the limits of the gas analysis equipment.
2. CO is still the best indicator for possible spontaneous combustion.
3. All gases produced could be the results of the decomposition of peroxides, and CO could be a by-product from the decomposition of these gases.

These authors also propose a series of reaction mechanisms to explain the origin of the various compounds they detected. These mechanisms are discussed in another section of this report.

Marinov[6] studied the onset of coal self-ignition by investigating the mechanism of coal-oxygen interaction at low temperatures. He heated thin layers ($0.1 g/cm^2$) of powdered (<0.2mm [70 mesh] sieve), Eastern Europe coals[3] from ambient temperature to a final temperature ranging from 60° to 250°C. The heating rate was maintained constant at 1°C/min and the coal was flushed with vapor saturated air at a rate of $0.5 m^3/hr$.

Marinov examined the raw and oxidized coals for changes in weight, elemental analysis, oxygen functional groups, IR spectra, and spin concentration. In addition the yields of CO, CO_2 and H_2O were measured and the light hydrocarbons evolved before self-ignition were analyzed by GLC. He identified a region in which an interaction of coal with water was taking place in the presence of oxygen. The hydrogen content remained invariable or increased. Above this region there is another region called "oxidative dehydrogenation" which results in the formation of hydrogen-rich gaseous products. This region precedes the self-ignition region. He concludes that coal oxidation can be considered a redox process in which the aromatic part of coal acts as an oxidizing agent, where hydrogen is affected only slightly by molecular oxygen.

In a following report[7] Marinov illustrates the results of experiments to study the rate of reaction between porous substances and gases, in which the layer thickness is a variable. He used the same coals as in reference 6 and a "DERIVATOGRAPH", a device for simultaneous TG and DTA work. All runs were made with coals ground to pass a 0.2mm sieve (70 mesh) in air saturated with water vapors at room temperature. The heating rates were 1.3 and 3.5°C/min and the final temperatures were about 300°C.

Marinov found that self-ignition occurs only in thick layers and that it is prevented by oxidative reduction, hydrolysis and dehydrogenation reactions. The self-ignition of coal is probably caused by the burning of low molecular weight products, initiated by aldehyde oxidation. In thin layers self-ignition is impossible probably because the concentration of reactive organic matter is small.

In the third report of the series[8] Marinov examines the induction period and its effects on coal self-ignition. He heated samples of the above coals at 60° $\pm$ 2°C for up to 1200 hours. Samples were withdrawn regularly and analyzed. After about 300 hours, the onset of oxidative dehydrogenation was detected. This was accompanied by an increase in alcoholic content for all three coals studied, a two-fold increase of carboxylic groups in lignite and brown coal, and the appearance of carboxylic groups in the black coal. The temperature of self-ignition was not changed for the lignite and the brown coal, but was decreased for the black coal.

Since these changes occur well beyond the length of time that we are planning to use in our work, these and other findings reported by Marinov are of very limited importance to us and the interested reader is referred to the original article.

EFFECT OF OXIDATION ON THE COKING PROPERTIES OF COAL

Two methods are typically used to follow the degree of oxidation of coking coal. In one the Ruhr dilatometer is used and the change of coking properties is related to the reduction of the dilatation of the coal on heating. This test is simple and very sensitive with coals having high dilatation properties. However, it does not reveal the extent of oxidation, particularly when the coking properties have been largely destroyed.

The other method, based on the change in the flotation characteristics of coal, uses the changes in the surface properties of the coal particles upon oxidation. This method requires considerable care and is limited by the fact that it gives information only on the surface oxidation.

Ignasiak et al[2] developed a new and fast method to measure the oxidation levels of coal. This method is based on the volume of the gas produced by the pyrolysis of coal and on its composition.

Starting from the knowledge that oxidation of coal below 200°C causes an increase in the oxygen functional groups of the coal, these authors reasoned that pyrolysis of the oxidized coals should decompose these groups and produce more CO and CO_2 than nonoxidized or fresh coals. They established that the content of CO and CO_2 in the pyrolysis gas is a sensitive indicator of the extent of oxidation. A better indication of the oxidation levels was given by the ratio of the gas volumes produced by the pyrolysis of an equal amount of oxidized and fresh coal.

Ignasiak et al[3] used this technique to study the pyrolysis products of coals oxidized with $^{18}O_2$ at 100°C for 72 hours. They analyzed the products by mass spectrometry and found that 60% of the $^{18}O_2$ had been incorporated in the tar + water fraction, indicating that upon mild weathering oxygen crosslinks are introduced.

Wachowska et al[5] continued to study the coking and weathering properties of coals. To clarify the types of oxygen or carbon linkages that are responsible for the loss of the fluid character of vitrinites, they removed or blocked all of the oxygen functional groups resulting from the oxidation of coal, and measured the dilatation of the products. They concluded that the swelling properties were largely destroyed by ether bond formation occurring at low temperature. The presence of other oxygen functional groups did not affect the fluid character of the sample and no polymerization through C-C bond formation occurred. These authors do not address themselves to the mechanism of coal oxidation, but the types of oxygen functional groups that they found in oxidized coal were in agreement with the mechanisms illustrated elsewhere[1,5,12,16,19].

It is pointed out that the various techniques used by these authors to eliminate or block the oxygen functional groups are well established and have been reviewed in our previous literature survey[20] to EPRI.

Volborth and co-workers have made extensive studies on the stoichiometry of coals. Volborth[11] distinguishes four fundamental types of water in coal:

- surface moisture (adsorbed water)
- inherent moisture (equilibrium water)
- decomposition moisture (organic moisture)
- mineral matter moisture (water of hydration).

Experimentally these moistures cannot be readily differentiated and, in practice, four classes of water are determined:

- as-received or total water
- air-dried moisture (what is left after drying in air)
- air-dry moisture (water lost by an as-received sample by air drying)
- equilibrium moisture

In determining the moisture by heating coal at 105°C, we are not entirely correct in assuming that only water is liberated[9]. In fact, varying amounts of CH_4, C_2H_6, N_2, NH_3, CO_2 and other volatile components are evolved below 100°C[11]. These authors suggest that a better analysis is obtained if oxygen is not obtained by difference but is determined by fast neutron activation (FNA), on the as-received and dried coal and on the ash. They point out that a better coal material balance can be calculated, and the inaccuracy of the other analyses can be evaluated. They applied this technique to six Wyoming coals[9] and to 33 other coals[10].

Section 4

MECHANISM OF THE CHEMICAL OXIDATION OF COAL (IN THE GAS-SOLID SYSTEM)

It is generally thought that the first step in the oxidation of coal is the formation of peroxides by the oxidation of the aliphatic and olefinic structures present in coal[1,5,12,16,19].

For a saturated hydrocarbon the basic mechanism is believed to be as follows:

$$R\text{-}H + O_2 \rightarrow R\cdot + HO_2^{\cdot}$$
$$R\cdot + O_2 \rightarrow R\text{-}O\text{-}O\cdot$$
$$R\text{-}O\text{-}O\cdot + HO_2^{\cdot} \rightarrow R\text{-}O\text{-}O\text{-}H + O_2$$
$$R\text{-}O\text{-}O\cdot + R\text{-}H \rightarrow R\text{-}O\text{-}O\text{-}H + R^{\cdot}$$

R is aliphatic or aromatic. If it is aromatic, the oxidation probably occurs at the carbon α to the aromatic ring[16]. The decomposition of the peroxide R-O-OH can produce various oxygen containing compounds such as acids, peracids, alcohols, ketones, aldehydes, esters and ethers.

The mechanism for these decomposition reactions are not completely agreed upon. Thus deVries et al[19] propose the following mechanism for the decomposition of the peroxides and the formation of the various compounds listed above.

$$R\text{-}CH_2\text{-}\underset{}{\overset{O\text{-}OH}{CH}}\text{-}CH_2\text{-}R \xrightarrow{-1/2\ O_2} R\text{-}CH_2\text{-}\overset{OH}{CH}\text{-}CH_2\text{-}R \text{ (Alcohol)}$$

$$\downarrow -H_2O$$

$$R\text{-}CH_2\text{-}\overset{O}{\overset{\|}{C}}\text{-}CH_2\text{-}R \text{ (Ketone)}$$

$$\downarrow \text{(additional oxidation)}$$

$$R\text{-}CH\ \text{-}\overset{O}{\overset{\|}{C}}\text{-}\underset{OOH}{CH}\text{-}R \text{ (Peroxyketone)}$$

$$\downarrow +O_2$$

$$R\text{-}CO + HOOC\text{-}CH_2\text{-}R$$

Aldehyde (carboxylic acid)

$$\downarrow +O_2$$

$$HOOC\text{-}CH_2\text{-}R \text{ (Carboxylic acid)}$$

$$\downarrow +$$

$$R\text{-}\underset{O}{\underset{\|}{C}}\text{-}OOH \text{ (Peracid)}$$

$$+$$

$$R\text{-}\underset{O}{\underset{\|}{C}}\text{-}CH_2R \text{ Ketone}$$

$$R\text{-}COOH + R\text{-}\underset{O}{\underset{\|}{C}}\text{-}O\text{-}CH_2\text{-}R \text{ (Ester)}$$

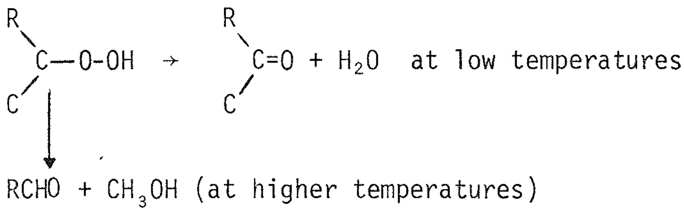
(acid)

Chamberlain et al[1] propose instead the following mechanisms, depending on the type of peroxides.

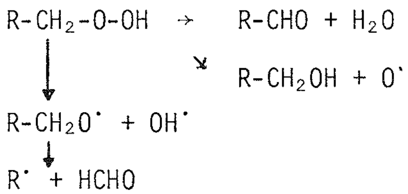
(1) Tertiary alkyl peroxides

$$R\text{-}\underset{C}{\overset{C}{C}}\text{-}O\text{-}OH \rightarrow \begin{matrix} R \\ CH_3 \end{matrix}\!\!>C{=}O + CH_3OH$$

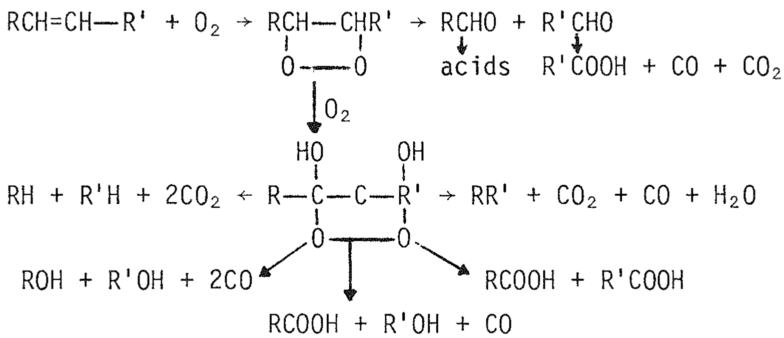
(2) Secondary alkyl peroxides



(3) Primary alkyl peroxides



Chamberlain et al[1] also propose that in one case the olefin is oxidized via the formation of a cyclic peroxide to produce the following products:



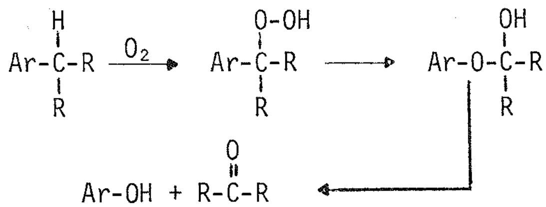
In a second case the other mechanism involves the formation of the peroxide through a radical-propagated chain mechanism (not shown); the overall reaction is

$$R\text{-}CH_2\text{-}CH{=}CH\text{-}R' + O_2 \rightarrow R\text{-}\underset{\displaystyle O\text{-}O\text{-}H}{\underset{|}{C}H}\text{-}CH{=}CH\text{-}R'$$

The olefinic peroxide can then decompose to the saturated peroxide and go through the reactions illustrated earlier.

It is pointed out that according to deVries et al[19] ketones, aldehydes, alcohols, acids and esters can all be formed from the aliphatic peroxides. However, according to Chamberlain et al[1] only alcohols, ketones, and aldehydes are formed from the aliphatic hydrocarbons. All of the compounds are produced from the olefinic hydrocarbons. In addition, Chamberlain et al show that CO and CO_2 are produced from the decomposition of the aldehydes which are themselves produced by the decomposition of the olefinic peroxide.

It is also noted that when a peroxide is formed on the carbon α to an aromatic ring, phenols are produced as illustrated by Albers et al[16]



Ar = Aromatic
R - Aliphatic

The phenolic structures can be further oxidized to quinonic structures. In addition all the hydroxylic structures can condense with radicals and produce etheric structures[5,16].

Volborth[12] also addressed the mechanism of coal oxidation. He appears to agree with the above authors in that the formation of peroxide is one of the initial steps of coal oxidation, resulting in the formation of the other oxygen-containing groups. According to Volborth[12], this is especially true of saturates and aromatic hydrocarbons. In fact oxygen-compounds are readily oxidized, while the corresponding hydrocarbons have to first be activated through the formation of peroxides.

It is interesting to note that deVries et al and Chamberlain et al do not discuss the oxidation of aromatic- and of oxygen-containing structures.

Mechanistic studies of coal oxidation are also discussed by Butuzova and Kucher[25] and Kukharenko[26]. The first work was undertaken to compare the adsorption of an ionic and nonionic surface active agent on coals with increasing amounts of oxygen-containing groups generated by medium-temperature (150°C) oxidation. The levels of the various oxygen-containing groups were followed as a function of time for the oxidation of Type Zh "fat" coal up to 10 days. It appears that the following stages occurred:

- Paramagnetic centers - maximum at 1 day
- Peroxides - maximum at about 3.5 days
- Hydroxyls - maximum at about 5.5 days
- Carbonyls - maximum at 7 days
- Carboxyls - continually rising with leveling off at 10 days.

It is noted that while successive interconversion may have taken place, it is also possible that simultaneous parallel reactions may have occurred. The adsorption curves went through successive peaks and valleys depending upon surface functionality. It was also noted that vitrinite underwent much more functionality change than fusinite. Optical density of aqueous extracts appeared to continuously rise with oxidation time indicating that the number of hydrolysable bonds and amount of hydrolysis product increased.

Kukharenko[26] concluded that the reactivity of coals is determined:

1. by the amount and forms of oxygen,
2. by the amount of condensed and substituted aromatic nuclei, and
3. by the concentration of free radicals, the nature of which differs and which is connected with the first two indices.

A detailed mechanism of coal partial oxidation is also proposed by Marinov[6-8]. A portion of the mechanism "bears a resemblance to the one electron transfer process in redox polymers". The overall mechanism is somewhat lengthy and beyond the scope of this paper, so its review is left to the reader.

Olpinski et al[18] suggest that at the beginning of the low temperature oxidation of coal, oxygen is attached to coal both physically and chemically. As the temperature increases, the physically held oxygen decreases, while the chemisorbed oxygen increases. More oxygen is taken up from the air as the temperature is increased. This oxygen is either chemisorbed or released as CO and CO_2. Above 100-150°C the consumed oxygen appears only as CO and CO_2, and their evolution increases until the amount of oxygen evolved equals that consumed. In this process, heat is evolved and spontaneous ignition occurs when more heat is produced than can escape from the system. The amount of heat loss depends on the content of carbon and mineral matter, physical and chemical structure of the coal surface, heat conductivity, particle size, degree of oxidation, air flow, etc. (Since only the abstract of a Polish publication is on hand no more details are available at this time.)

Section 5

KINETICS OF COAL OXIDATION AT LOW TEMPERATURE IN THE GAS-SOLID SYSTEM

The study of the kinetics of coal oxidation at low temperatures of 25°C to 200°C has been the subject of a fair amount of work. Various theories have been proposed, and a somewhat limited number of equations have been used to correlate the results. To an extent these equations tend to overlap in the primary areas of interest. Emphasis in this literature search will be placed on the more recent publications, but it is pointed out that the general consensus is that an empirical approach may well be satisfactory if not best.

Coal oxidation at low temperatures (to be defined later) appears to occur in two stages. The first is a "rapid reversible equilibrium process"[28] in which oxygen adsorbs on the coal sample and can be reversibly desorbed by evacuation at the same temperature. A "slow irreversible process" occurs when oxygen is chemisorbed on the coal surface. Some of this combined oxygen may be released as CO, CO_2 and H_2O.

The subject of physically adsorbed oxygen on coal has had little recent emphasis. With char degassed under vacuum at 1000°C for 10 hours, the data of Allardice[28] indicate that about 3g oxygen/100g char will be reversibly adsorbed from air at 25°C. At temperatures of 65° and 110°C, the levels are reduced by one-half and three-quarters. Char was used to isolate physical from chemical adsorption; extrapolation to coal is questionable.

Allardice also referred to Hayward and Trapnell[29] quoting figures of 1.6 to 5.0 kcal/mole as reasonable heats of physical absorption of oxygen. Schlyer[14] has indicated that the amount of physically adsorbed oxygen increases with decreasing rank. At ambient temperature in air, the O/C ratio of a New Mexico subbituminous coal increased from 0.21 to about 0.27. This indicates an oxygen adsorption of about 4.6g/100g coal. Much of this absorption occurred within 2-4 minutes.

The kinetics of the chemisorption of oxygen on coal has been the subject of fairly extensive research. Data have been correlated by a variety of equations including the following which were cited by Wood[30]:

$$\frac{dq}{dt} = Ae^{k_1 t} + Be^{k_2 t}$$

$$\frac{dq}{dt} = Ct^{k_3}$$

$$\frac{dq}{dt} = Dt^{-1/2}$$

$$\frac{dq}{dt} = Ee^{k_4 q}$$

where q is the quantity of oxygen absorbed, t is time and A,B,C,D, k_1, k_2, k_3 and k_4 are constants. Wood then effectively fit his data to the so-called Elovich equation:

$$\frac{dq}{dt} + Ae^{-\alpha q}$$

or upon integration:

$$q = \frac{2.3}{\alpha} \log (A\alpha t + 1)$$

or $$q = \frac{2.3}{\alpha} [\log(t+t_0)-\log t_0]$$

where: A, α are constants and $t_0 = 1/A\alpha$

The Elovich equation is basically an empirical relationship which provides a convenient means of graphing and interpolating the data. As pointed out by Allardice[28], it also provides a means of extrapolating the data to determine the initial rate of irreversible adsorption which cannot be measured directly due to the simultaneous reversible adsorption process. In this method, the amount of oxygen (preferably chemisorbed) is plotted against the log of time. A linear relationship is typically observed. α is calculated from the slope and is substituted in the integrated equation with a value of q and t to calculate the initial rate A.

The effectiveness of the Elovich equation is particularly well discussed by Allardice[28] and by Harris and Evans[24]. Numerous references are cited, and it is pointed out that attempts to justify the equation in terms of a physical model have not been successful. However, over a significant range, diffusion data show an excellent agreement with the equation. Allardice indicated that the constant A is the initial rate of chemisorption when the rate of physical adsorption is 0. His data indicated that A was independent of pressure and that the initial adsorption was a zero order reaction. An Arrhenius plot of A at different temperatures gave an activation energy of 11.1kcal/mole. He therefore concluded that the rate of chemisorption is controlled by the migration of oxygen across the surface to active sites where chemical reaction occurs.

Other researchers[31] have also found low activation energies and there is a general consensus that diffusion is a controlling factor in coal oxidation. Some of the better work in this area is that of Sommers and Peters[32]. Using as a basis that the consumption of oxygen is independent of decomposition reactions, the rate of oxygen consumption is therefore equal to the reaction rate at the boundary surface:

$$\frac{dq}{dt} = kO$$

where O = oxygen concentration. Then taking into account diffusion:

$$\frac{dq}{dt} = D_s\left(\frac{d^2 Q}{dx^2}\right)$$

where x = distance; the equations were solved for various limiting conditions. The data were correlated using this diffusion model. The activation energy of the diffusion term was somewhat high though with a range of 26kcal/mole for anthracite to 16kcal/mole for a low ranked coal (76.6% C). The factor $\sqrt{D/k}$ showed a direct correlation with rank.

The implication that diffusion is a controlling factor is confirmed by the low activation energies found by others including Jones and Townend[34], Oreshko[35] and Kreulen[36], as discussed by Kreulen, et al[37].

On a more empirical approach, the oxidation data of Schmidt and Elder[33] should be cited. They studied the oxidation of a range of bituminous coals including Pittsburgh, Lower Kittanning, and Pocahontas 3 and 4 at temperatures of 30°, 50° and 99°C. The rate of oxidation increased by a factor of 3 with a volatile matter increase from 18 to 38%. The rate of oxidation decreased rapidly with increased oxygen consumption; it was inversely proportional to the one-fourth power of oxygen consumed. The rate of oxidation was proportional to the 0.6 power of oxygen concentration in the gas phase and the 0.8 power of run time. The value of the activation energy therefore was of the order of 11kcal/mole.

In a discussion of diffusion effects and rate controlling mechanisms, reference should be made to the reviews given by van Krevelen[38] and Carpenter and Giddings[39]. A further discussion of diffusion mechanisms is brought out by Dugan and Moran[40] based upon their DTA experiments and by Albers, et al[16].

The work of Dugan and Moran also points out an interesting consideration in oxidation. After the initial diffusion controlled reaction, more direct internal burning occurs in which cavities form in the inside of the coal particles. This appears to happen at higher temperatures, namely in the region of 190°C and above. Since Dugan and Moran worked with bituminous coals (New South Wales) and coals of lower rank are typically more easily oxidized, this onset of combustion may well occur at lower temperatures for lower ranked coals. This phenomenon may be considered loosely as an ignition temperature below which air drying of coal should be done. This aspect will be looked at further.

Sweeney[41] followed the change of oxygen pressure as it was consumed by coal in the temperature range of 110° to 170°C. The preferred rate equation was

$$-\frac{dPi}{dt} = K_s Pi/(B_W - Pi)^{1/2}$$

where: Pi = oxygen pressure at time t
K_S = rate constant
$B_W = Po + b_W$ = the calculated "effective initial pressure".

After solving the equations, it was shown that oxygen pressure correlated well against time to the 0.6 power. It was stated that the process is pure chemical sorption and not diffusion controlled.

Kam, et al[42] studied coal oxidation in a packed bed at 200°-277°C. The type of equipment and reaction conditions are outside of our area of interest; the research is cited for reference only.

The next point of consideration in the study of reaction kinetics is the rate of formation of various components on the coal surface and of CO and CO_2. By-product water can also be considered, but problems arise because of the difficulty of separating coal oxidation product from retained moisture that is released as a function of temperature and perhaps surface chemical properties.

Albers et al[16] have studied the structural changes occurring during the low temperature oxidation of coal (see also Lenart and Oelert[43]). Three German coals were ground to less than 0.2mm (70 mesh) and oxidized under a hot oxygen stream of 2.5 l/hour. The temperatures varied from 110° to 170°C and the times from 0 to 100 hours. The water vapors, CO and CO_2 were determined gravimetrically. The oxidized coals were examined by elemental analysis and IR spectroscopy.

The authors followed changes in the levels of various carbons (aromatic, methyl, etc.), oxygens (hydroxyl, carbonyl and total), and hydrogens as functions of time and temperature. The data were fitted to a first-order equation:

$$y = y_0 + c_0 \, [1-\exp(-kt)]$$

where y = % of group content at time t, y_0 = % at t_0, and c_0 is an extrapolated % at t = ∞. From an Arrheneous plot it was concluded that the oxidation reactions were independent of diffusion processes. (It is noted that the activation energies ranged between 7 and 22kcal/mole). It was found that reactivity toward oxygen decreases with increasing rank and decreasing reaction temperature. It is also noted that hydroxyl concentration remained essentially flat with time (0 to 100h), carbonyl rose to a high level (∿10%) then remained constant, and the remaining oxygen, while less than carbonyl, continued to rise during the whole reaction period.

The postulated mechanism was that of initial autoxidation of aliphatic α-groups to hydroperoxide followed by subsequent oxidation of the hydroperoxide to ketone. Their oxidation mechanism is similar to that of deVries et al[5] and of Chamberlain et al[1] which have been previously discussed.

Swann and Evans[44] reported the results of oxidizing Yallourn brown coal (Australian) at mild temperatures of 35° and 70°C for a long period of 45 days. In spite of a sizable loss of CO, CO_2 and H_2O, a weight gain of 1.5g/100g dry was observed. At 35°C primarily phenolic groups were formed, while at 70°C the formation of carboxyl groups was predominant. The yields of CO and CO_2 were reported as follows (in units of g/100g dry coal):

Temperature (°C)	CO	CO_2
35	0.1	1.3
70	0.5	5.0

Consistent with the above discussed mechanisms, Swann and Evans reported a reduction of aliphatic infrared peaks and an increase of carbonyl peaks on oxidation. Changes also were observed for hydroxyl and ether region peaks, but the results were too diffuse to draw conclusions.

Also consistent with the above, Volborth[45] indicates that compounds with oxygen functional groups attached are more susceptible to further oxidation than their progenitors and that peroxides appear to be the first species formed. Further oxidation steps are then described.

The loss of CO and CO_2 upon drying and oxidation is cited by a number of researchers, but the kinetics of the reactions have not been reported. Therefore, only relative yields will be discussed herein.

Averaging the data of Schmidt and Elder[33], the following yields of gaseous products from bituminous coals were recovered after "the highest state of oxidation attained":

Temperature (°C)	Yield (g/100g of Dry Coal) CO_2	CO	SO_2	Total O_2 Consumed
99.3	1.0	0.08	0.03	2.2
50.0	0.4	0.02	-	-
30.0	0.13	-	-	1.2

Calculated on a similar basis for four bituminous coals oxidized for 145 hours, the gaseous yields of deVries et al[19] are summarized below:

Temperature (°C, avg.)	Yield (g/100g Feed Coal)	
	CO_2	CO
100	2.4	0.3
80	0.6	0.1
60	0.1	0.02

The yields were strongly influenced by the coal rank in that the yields of CO and CO_2 increased by a factor of about 2.5-3.0 with a coal carbon content decrease of 88.7 to 84.2%.

Less of an influence of rank was reported by Sommers and Peters[32], but their work was done in the temperature range of 190°-260°C. Typically less than 5% of the consumed oxygen was released as CO and on the order of 10 to 40% as CO_2. Three-quarters of these gases were released in the first 30 minutes of the four hour run periods.

CO_2 can be driven off low ranked coals even when heated in a stream of nitrogen. Allardice and Evans[46] observed the following recoveries of CO_2 when Yallourn brown coal was heated and evacuated for a period of 20 hours:

Temperature (°C)	CO_2 (g/100g Dry Coal)
30	0.03
60	0.03
90	0.19
120	0.31

It was pointed out that the weight loss of CO_2 would show up as "moisture" for the typical analytical procedure of drying at 110°C for several hours. Furthermore, the measurement of moisture in coal is not a direct process. Specifically, isotherms were determined for the drying of this coal. Isosteric heats of sorption increased from the latent heat of vaporization (10.4kcal/mole) in the bulk phase region to 14kcal/mole in the monolayer region. This implies that enhanced hydrogen bonding of the water to functional groups on the coal surface occurs. References cited by the authors indicated that much higher levels had been observed, but interference may have happened due to side reactions. Schlyer[14] observed a value of 11.8kcal/mole for the heat of desorption of H_2O at 153°-220°C (7.7kcal/mole for CO_2, O_2 at 81°-140°C).

Experimentation was done by Spackman et al[47] as part of a sub-project to observe the effect of oxidation on subsequent coking properties. For the temperatures of 120° to 220°C a continual increase of weight was observed for the 4-hour run period. At 250°C a maximum occurred with "gasification" then becoming the controlling factor.

In addition to the above literature, various articles were picked up in the abstracts, but they were not reviewed due to language problems or time delays in obtaining copies. Using titles and/or abstracts for selection, the more interesting articles are cited as references 27 and 48 through 53.

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Section 6

SAFE OPERATING CONDITIONS: FLAMMABILITY LIMITS

Three actions were undertaken to insure that safe operation could be maintained in the drying/oxidation of coal in a fluid-bed unit. A computer literature survey was carried out, and discussions were held with Drs. K. Cashdollar and M. Hertzberg of the Bureau of Mines (Bruceton, PA). Finally, the larger fluid-bed dryer/oxidizer was built in an explosion cell.

The literature search brought forward numerous articles on the combustion of coal particles, but most of the emphasis was on mine safety and the design or operation of burners. We limited our study to combustion problems that may exist in a fluid-bed dryer with emphasis on recent publications. These publications were supplemented with the above mentioned discussions. In no way should this literature search be considered conclusive.

A summary of Drs. K. Cashdollar and M. Hertzberg comments[54] is given below:

1. The lower flammability limit of coal dust (bituminous) in air is about 200 mg/liter; no upper limit was stated.
2. Powdered Pitt seam coal (35% volatiles) will not explode in a gas stream containing less than 13 1/2% oxygen. The limit for Pocahontas (16% volatiles) is 18% oxygen.

3. It was <u>estimated</u> that Belle Ayr subbituminous coal would have an oxygen limit of 16%. Operation would be in the safe range for oxygen gas contents of 13% or less.
4. It was indicated that a source of ignition was very unlikely in our fluid-bed unit, even if air was used. In Bureau experimentation an ignitor-match is used to test flammability limits; a spark is typically not sufficient.
5. When an explosion occurs with coal powder <u>in air</u>, a maximum pressure increase of X6 (i.e., 6 atmospheres) is observed.

The above information was derived in part from their research reported in two recent publications[55,56]. It is also noted that they used a chemical matchhead array as the ignitor; ignitor energies in the range of 150 joules were needed.

The limiting oxygen content in a gas stream for the explosion of coal dust powder is strongly dependent upon coal rank. Hartmann, et al[57] at the Bureau of Mines reported that semianthracite and low-volatile bituminous coals could not be ignited by a spark in air. The limiting oxygen contents for spark test ignition for MV, HV_A and HV_B bituminous coals and lignite were 18.5, 17.0, 15.0 and 15.0%, respectively. With the use of a 850°C (1560°F) furnace, ignition of the bituminous coals and lignite occurred with 10.5 and 7.5% oxygen, respectively. Essentially the same results are reported later by Nagy, et al[58], "For Pittsburgh coal dust the lowest oxygen concentrations for ignition are 18, 16, and 11 percent by volume for induction spark, heated coil, and furnace (1,560°F) sources, respectively. The oxygen concentrations for a lignite dust are 16, 15, and 8 percent for these same igniting sources."

It is also noted that the moisture content of the coal affects its explosion limits. When the moisture content of Pittsburgh seam bituminous and an unspecified subbituminous coal were above 8%, flammability limits and the required ignition power increased significantly[57] with increasing moisture content.

Another point worth noting is that a layer of coal dust ignites at a much lower temperature than a dust cloud, namely 200°C vs. 300°-700°C for high volatile coals. This is of particular advantage for fluid-bed drying over tray drying.

In addition to the above, the Bureau of Mines has continued to publish reports on the flammability of coal dusts, mine gases and the likes. Several of these are cited in references 59 through 64. Particular attention is directed to the following paragraphs from Report No. IC 8258[61]:

> " A cloud of 200-mesh coal dust will ignite at 1,100°F, and 0.05 ounce of such dust per cubic foot of air will create such an explosive cloud in the presence of as little as 10 percent oxygen. Larger concentrations of dust will propagate more violent explosions until the dust cloud becomes so dense that it excludes air (or oxygen) and acts to smother the flame. Tests by the Bureau indicate that the maximum explosive dust concentration is about 12 ounces per cubic foot. The presence of methane or other combustible gases increases the flammability of coal dust, and the finer the dust the more explosive it becomes.
>
> During normal drying operations, a high volume of drying gases pass through a fluidized bed dryer system and the temperatures and distilled gases are dissipated rapidly, reducing danger of explosion. However, it is not difficult to visualize serious danger of dust explosions during startup, shutdown, and standby operations.
>
> Dust particles finer than 325-mesh are not unusual in dryer systems, and layers of encrusted dust are common on the inner surfaces of drying chambers, ductwork, and dust

> collectors. The temperatures in these areas during normal operation are below the ignition temperature of dust clouds, but they are high enough to contribute to extremely rapid spontaneous heating following shutdown. Such layers of dust oxidize quickly to critical temperature and become incandescent at 350° to 400°F in only a few minutes."

Additional articles were pulled out by the abstract services; however, the articles appeared to only be indirectly related to our operation. They are cited as references 65 through 70.

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